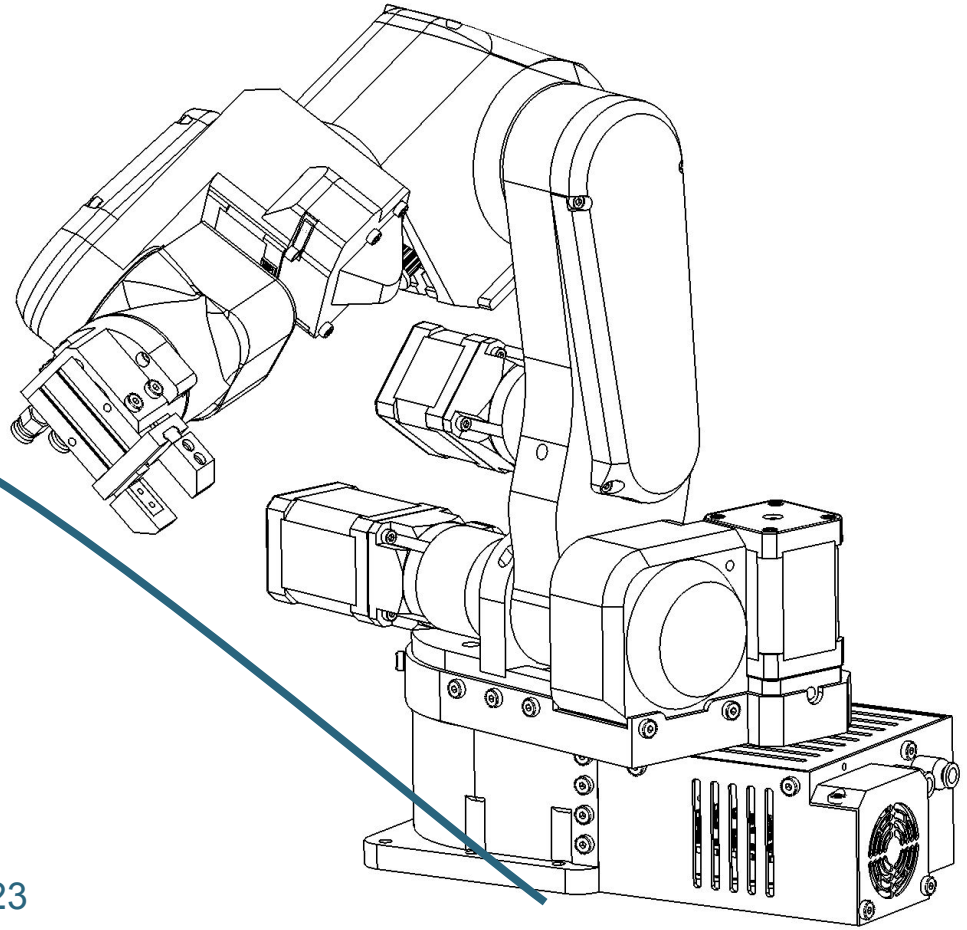
Source Robotics

PAROL6 ASSEMBLY MANUAL



Version 1.2, rev1, date 27.10.2023

PAROL6 机械臂装配手册

中文对照版 | 每个原文页面后附对应中文译页

译文说明：本文件为机器辅助翻译，供理解和装配参考。涉及安全、电气、紧固件及尺寸时，请同时核对前一页原版图文；如有冲突，以原始英文手册及实际零件为准。

PAROL6 装配手册

版本 1.2，修订版 1，日期：2023-10-27

Disclaimer important notice

Source Robotics

The PAROL6 robot arm and its associated part are specifically intended for engineering development, demonstration, or evaluation purposes. They are not considered to be a final product suitable for general consumer use. Individuals handling these products must have proper electronics and robotics training and adhere to good engineering practices. It should be noted that these goods may not fulfill all the required design, marketing, and manufacturing-related protective measures, including product safety and environmental considerations typically found in finished products. The PAROL6 robot arm is not compliant with the European Union directives concerning electromagnetic compatibility, restricted substances (RoHS), recycling (WEEE), FCC, CE, or UL. Consequently, it may not meet the technical requirements outlined in these directives or other related directives.

The user assumes full responsibility and liability for the correct and safe handling of the robot. Additionally, the user agrees to indemnify the Source Robotics team against any claims arising from the use or handling of the products. Given the open construction of the product, it is the user's responsibility to take all necessary precautions.

EXCEPT FOR THE INDEMNITY DESCRIBED ABOVE, NEITHER PARTY SHALL BE HELD LIABLE TO THE OTHER FOR ANY INDIRECT, SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES.

第 2 页中文译文

免责声明与重要须知

PAROL6 机械臂及其相关零件专为工程开发、演示或评估用途设计，不属于适合一般消费者使用的最终产品。操作这些产品的人员必须接受适当的电子和机器人技术培训，并遵循良好的工程实践。请注意，本产品可能未满足成品通常具备的所有设计、营销和制造相关保护措施，包括产品安全和环境方面的要求。PAROL6 机械臂不符合欧盟关于电磁兼容性、限制有害物质（RoHS）、电子电气设备废弃物回收（WEEE）的指令，也不符合 FCC、CE 或 UL 要求。因此，它可能不满足上述指令或其他相关指令规定的技术要求。

用户对机械臂的正确、安全操作承担全部责任和义务。此外，用户同意就因使用或操作产品而产生的任何索赔，向 Source Robotics 团队提供赔偿并使其免受损害。由于产品采用开放式结构，用户有责任采取所有必要的预防措施。

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Disclaimer important notice

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Before building this robot please read the whole manual! This machine can cause harm to you and people around you. It can burn, electrocute and physically hurt you if you are not careful. Please read these instructions carefully!



Also after reading and building this robot make sure to make yourself comfortable with robot control software and how to operate it!

第 3 页中文译文

免责声明与重要须知

组装此机械臂前，请先阅读整本手册！本设备可能伤害你及周围的人。如果操作不慎，可能造成烫伤、触电或身体伤害。请仔细阅读这些说明！

阅读手册并完成组装后，也请务必熟悉机械臂控制软件及其操作方式！

License

Source Robotics

Robotic arms commander software: Link, STL files: Link and control board software are licensed under GPLv3 license!

PAROL6 STEP files and PAROL6 control board are not open source!

第 4 页中文译文

许可证

机械臂控制软件：链接；STL 文件：链接；控制板软件：均采用 GPLv3 许可证！

PAROL6 的 STEP 文件和 PAROL6 控制板不是开源的！

Source Robotics

Welcome to the PAROL6 assembly manual. PAROL6 is 6 axis robotic arm aimed for education, research and small automation. It is made by Source Robotics, designed by Petar Crnjak

All additional information can be found here:

- Github project repository - most up to date BOM, parts, docs, manuals…
- Docs page - latest documentation on PAROL software
- Official company page - Source robotics

You can also ask help and advice from community:

- Official discord server: link
- Or send us an email at: info@source-robotics.com

And you can follow us on social media:



第 5 页中文译文

装配手册

欢迎阅读 PAROL6 装配手册。PAROL6 是一款面向教育、研究和小型自动化的六轴机械臂，由 Source Robotics 制作，设计者为 Petar Crnjak。

更多信息请访问：

- GitHub 项目仓库——最新的 BOM（物料清单）、零件、文档和手册。
- 文档页面——PAROL 软件的最新文档。
- 官方公司页面——Source Robotics。

你也可以向社区寻求帮助和建议：

- 官方 Discord 服务器：链接。
- 或发送邮件至：info@source-robotics.com。

你也可以在社交媒体上关注我们：

Assembly manual

Source Robotics

All screws shown are same size and only for reference. For correct size read the instructions stated on the page what screw to use. Some parts are not show; mostly bearings, small screws and belts but are referenced on the page of the building instructions.

This document only shows assembly manual, all other files like BOM, printing table, STL files can be found on GitHub page of the project.

All parts should be printed with PETG material. Reference printing table for optimal infills and printing settings.

Wires are not shown but detailed instructions are given on how to route the wires.

Belts are not shown but detailed instructions are given on how to attach them correctly.

This window shows what parts you will need on this page.

This window show building instructions.

This window shows tips and notes.

第 6 页中文译文

装配手册

图中显示的螺钉尺寸都相同，仅供参考。正确尺寸请阅读对应页面上的说明。部分零件没有画出，主要是轴承、小螺钉和皮带，但会在装配步骤页面中注明。

本文档只包含装配手册；BOM、打印参数表、STL 文件等其他文件可在项目 GitHub 页面找到。

所有零件都应使用 PETG 材料打印。最佳填充率和打印设置请参考打印参数表。

图中未显示电线，但手册提供了详细的布线路径说明。

图中未显示皮带，但手册提供了正确安装皮带的详细说明。

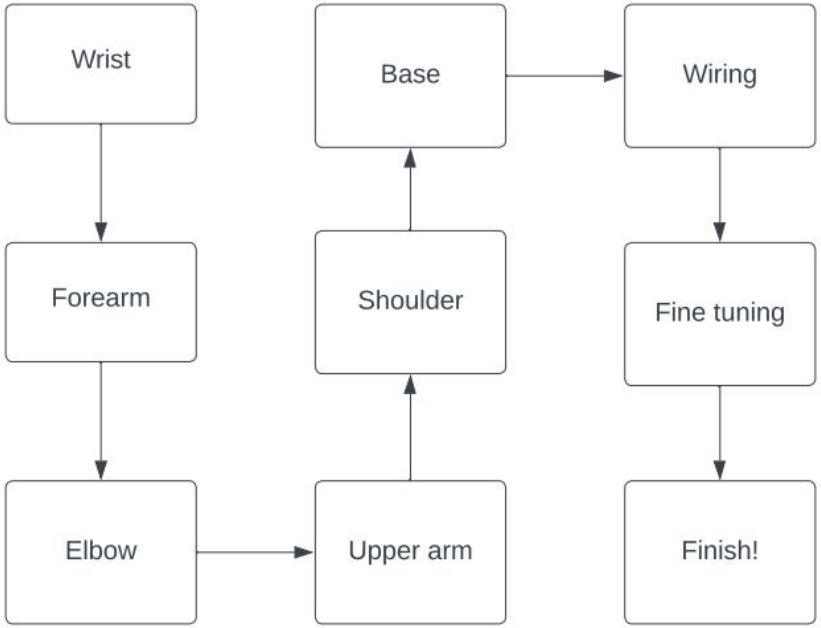
- 此区域显示本页所需零件。
- 此区域显示装配说明。
- 此区域显示提示和备注。

Assembly manual - Road map on building PAROL6 and using it!

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Robot needs to be built from wrist to the base. That means we start by building the wrist then lower arm, elbow, upper arm and base. Any other order will not work. This needs to be done this way since all wires pass thru the the robot.

After assembling the robot it needs to be wired up to the PAROL control board. After that some fine tuning and testing can be done.



第 7 页中文译文

装配手册——PAROL6 的组装与使用路线图

机械臂必须从手腕向底座方向组装。也就是说，先组装手腕，再组装前臂、肘部、上臂，最后组装底座。其他组装顺序都不可行。必须这样做，是因为所有线缆都要穿过机械臂内部。

机械臂组装完成后，需要连接 PAROL 控制板。之后即可进行微调和测试。

Assembly manual - Print preparation

Source Robotics

Some parts need to be printed with support. Before using those parts make sure to clean all support material.

Printer settings we used:

- Printer: Prusa MK2S
- Material: PETG
- Layer height:
- Support: Reference printing table
- Infill: Reference printing table

第 8 页中文译文

装配手册——打印准备

部分零件需要使用支撑打印。使用这些零件前，请确保清理干净所有支撑材料。

我们使用的打印机设置：

- 打印机：Prusa MK2S。
- 材料：PETG。
- 层高：原文未填写。
- 支撑：参考打印参数表。
- 填充率：参考打印参数表。

Assembly manual - Wire and pneumatic tube preparation

Source Robotics

Wires for gripper need to be twisted pair wires.
Try to follow these guidelines on how much wire length you need to have for every part of the robot.
Wire lengths:

- Motor joint 1 - 4 wires 1 meters
- Sensor joint 1 - 3 wires 1 meters
- Motor joint 2 - 4 wires 1 meters
- Limit switch joint 2 - 2 wires 1 meters
- Motor joint 3 - 4 wires 1 meters
- Limit switch joint 3 - 2 wires 1 meters
- Motor joint 4 - 4 wires 1 meters
- Sensor joint 4 - 3 wires 1 meters
- Motor joint 5 - 4 wires 1 meters
- Limit switch joint 5 - 2 wires 1 meters
- Motor joint 6 - 4 wires 1 meters
- Sensor joint 6 - 3 wires 1 meters
- Pneumatic tubes - 2 tubes 1 meters
- Gripper wires - 4 wire 1 meters (2 twisted wire pair)

Since robot is built from the wrist to the base of the robot you will have more and more wires as you reach the end. To keep it clean and easy to assemble before adding next link wrap group of wires with insulation tape (we used cloth fabric tape). For example after finishing lower arm assembly wrap Joint 6 motor with tape, Joint 5 motors with tape… Try to use tape of different color to mark different joints. This will help you A LOT when you will need to wire up the robot. When you assemble the robot you will have more length of the wires than you need cut them so you can wire the control board and keep everything clean. Usually after the wires exit the base of the robot they need to have extra 8 cm. For tubes leave 15 cm.

第 9 页中文译文

装配手册——电线与气动管准备

夹爪电线需要使用双绞线。

请尽量按照以下建议，为机械臂各部分准备相应长度的电线。

电线长度：

- 电机 J1：4 根线，每根 1 米。
- 传感器 J1：3 根线，每根 1 米。
- 电机 J2：4 根线，每根 1 米。
- J2 限位开关：2 根线，每根 1 米。
- 电机 J3：4 根线，每根 1 米。
- J3 限位开关：2 根线，每根 1 米。
- 电机 J4：4 根线，每根 1 米。
- 传感器 J4：3 根线，每根 1 米。
- 电机 J5：4 根线，每根 1 米。
- J5 限位开关：2 根线，每根 1 米。
- 电机 J6：4 根线，每根 1 米。
- 传感器 J6：3 根线，每根 1 米。
- 气动管：2 根，每根 1 米。
- 夹爪电线：4 根，每根 1 米（组成 2 对双绞线）。

由于机械臂从手腕向底座组装，越接近末端时，已经穿入的线缆就越多。为了保持整洁并便于组装下一段连杆，每完成一段后，用绝缘胶带（我们使用布基线束胶带）把这一组电线缠扎起来。例如完成前臂组装后，用胶带整理 J6 电机线、J5 电机线等。尽量用不同颜色的胶带标记不同关节，这会在连接机械臂电路时帮上很大忙。

组装过程中，线缆会留得比实际需要更长；可将多余部分剪短，以便连接控制板并保持整洁。通常线缆从机械臂底座穿出后，还需要额外留出 8 cm。气动管则留出 15 cm。

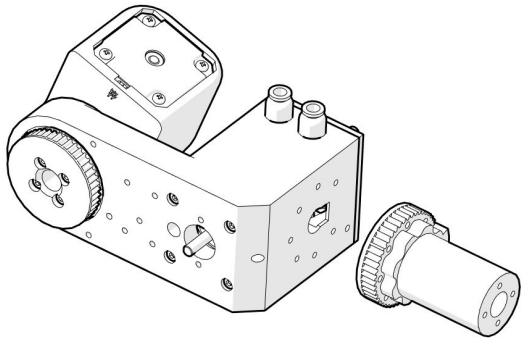
Assembly manual - Screws and attaching parts

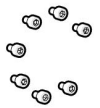
PAROL6 uses M3 -M5 screws to attach parts. Most functional print to print attachments are done with a crew and an undersized 3D printed hole.
For example on image at the right side we are attaching two 3D printed parts that will carry load and are not esthetic parts (esthetic parts are for example covers and robot will work without them). Screws are in this example M3 screws and holes are undersized to 2.7-2.8mm that means that when we screw in the screws we are tapping holes in 3D printed parts.
There are multiple benefits to this:

- Connection is strongest compared to tapping holes with a tap or using brass inserts
- It is simple and fast
- No need to prepare the hole, it can be printed undersized

Cons are that you can't disassemble it a lot of times. In case you feel screws slipping in the hole. Put some super glue in the hole and wait for it to cure. After that re tap the hole.

Most metal to metal parts require no preparation. For example attaching stepper motors or gearboxes. Those parts already have threaded holes. Attaching shaft couplers to motor shafts requires blue threadlocker. This will be discussed more, later in assembly manual. It is also recommended to apply threadlocker to any screw that pairs with metal. So any screw that connects to the motors and gearboxes





第 10 页中文译文

装配手册——螺钉与零件连接

PAROL6 使用 M3–M5 螺钉连接零件。大多数承力的打印件对打印件连接，采用螺钉配合略小于螺钉直径的 3D 打印孔。

例如右侧图片中，我们用 M3 螺钉连接两个承受载荷的 3D 打印件，而不是装饰件（例如外盖等装饰件即使不装，机械臂也能工作）。此处孔径打印为 2.7–2.8 mm，略小于螺钉直径；拧入螺钉时，螺钉会在 3D 打印件中攻出螺纹。

这种做法有以下优点：

- 与使用丝锥攻牙或黄铜热熔嵌件相比，连接强度最高。
- 简单、快速。
- 无需预先加工孔，可直接打印小孔。

缺点是无法反复拆装很多次。如果感觉螺钉在孔内打滑，可在孔内滴入一些强力胶，等待其固化后再重新攻牙。

大多数金属对金属连接无需额外准备，例如固定步进电机或减速器；这些零件本身已有螺纹孔。将轴联轴器固定到电机轴上时，需要使用蓝色螺纹锁固剂，后续装配手册会进一步说明。也建议在任何与金属配合的螺钉上使用螺纹锁固剂，例如连接电机和减速器的螺钉。

Assembly manual - Needed tools

Source Robotics

Tools you will need to assemble this robot:

- Screwdriver set
- Drill
- Soldering iron
- Imbus wrench (Allen key / HEX key) set
- Torque screwdriver
- Rubber hammer
- Hammer
- Pliers

For screwing we recommend using an electric screwdriver and then precision torque screwdriver or last few turns by hand. If you use drill or electric screwdriver for all the parts you risk destroying your threads.

第 11 页中文译文

装配手册——所需工具

组装机械臂需要以下工具：

- 螺丝刀套装。
- 电钻。
- 电烙铁。
- 内六角扳手（Allen key / HEX key）套装。
- 扭矩螺丝刀。
- 橡胶锤。
- 锤子。
- 钳子。

建议先使用电动螺丝刀拧紧，再用精密扭矩螺丝刀完成最后几圈，或最后几圈用手拧紧。如果所有零件都用电钻或电动螺丝刀拧紧，可能会损坏螺纹。

Assembly manual - Needed consumable items

Source Robotics

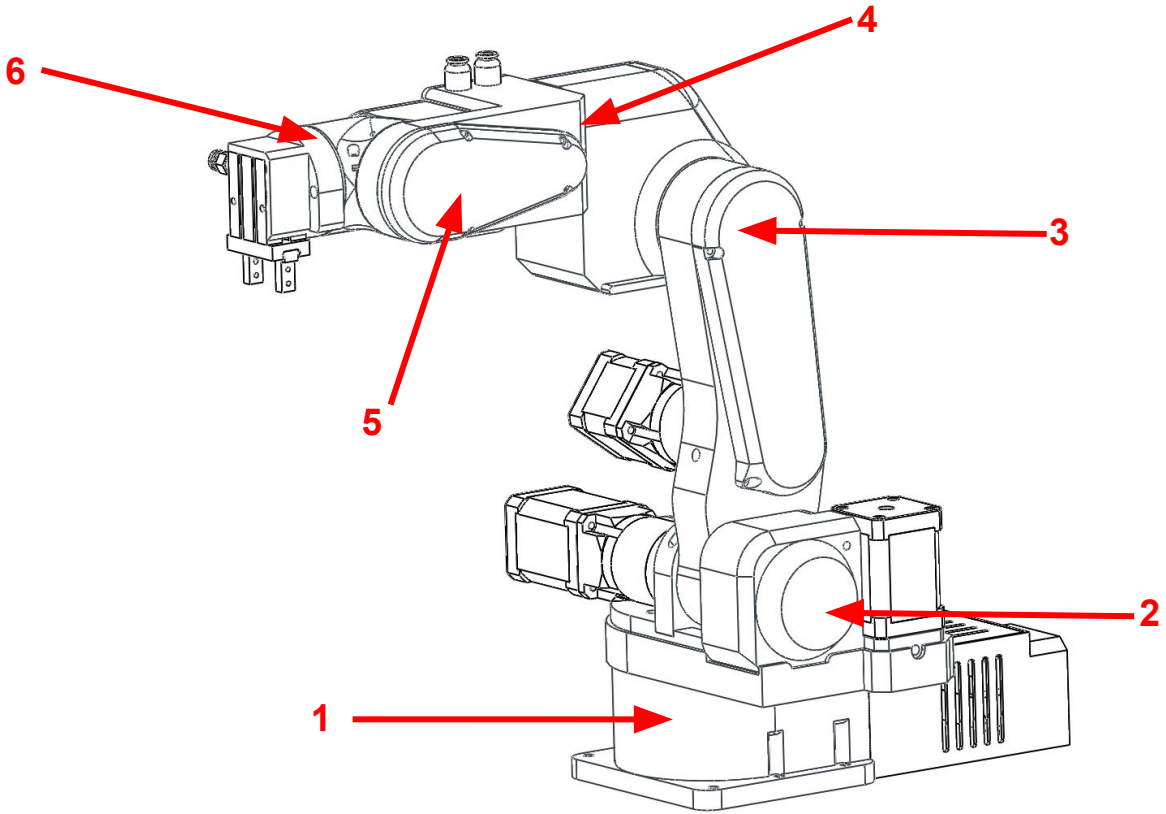
- Blue threadlocker
- Lithium grease
- Insulation tape
- Solder
- Heat shrink tube
- Wire harness
- Cloth Fabric Tape

第 12 页中文译文

装配手册——所需耗材

- 蓝色螺纹锁固剂。
- 锂基润滑脂。
- 绝缘胶带。
- 焊锡。
- 热缩管。
- 线束。
- 布基线束胶带。

Assembly manual - Joint numbering and names



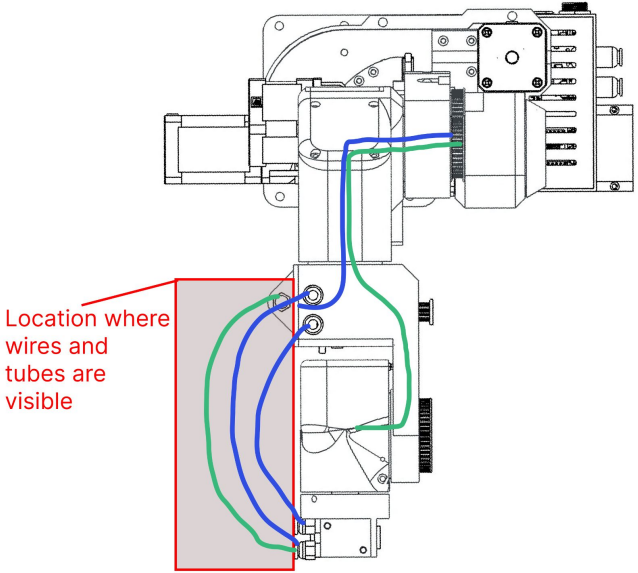
第 13 页中文译文

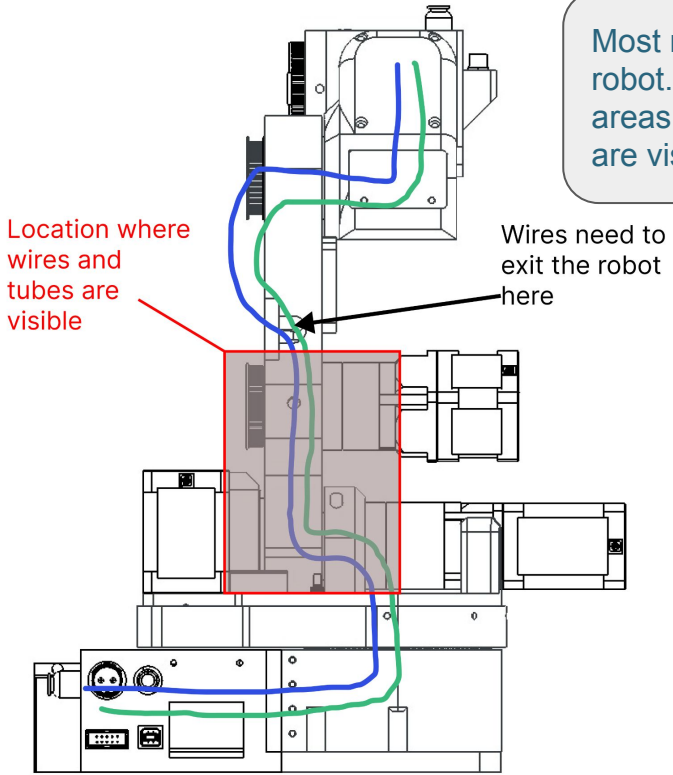
装配手册——关节编号与名称

图示关节编号：6、4、5、3、2、1。（编号位置请对照原版图示。）

Assembly manual - Wiring

Source Robotics





Most robot wires go thru the robot. Only the red marked areas are where robot wires are visible.

第 14 页中文译文

装配手册——布线

机械臂的大部分电线都从机械臂内部穿过。只有红色标出的区域会露出机械臂外部。

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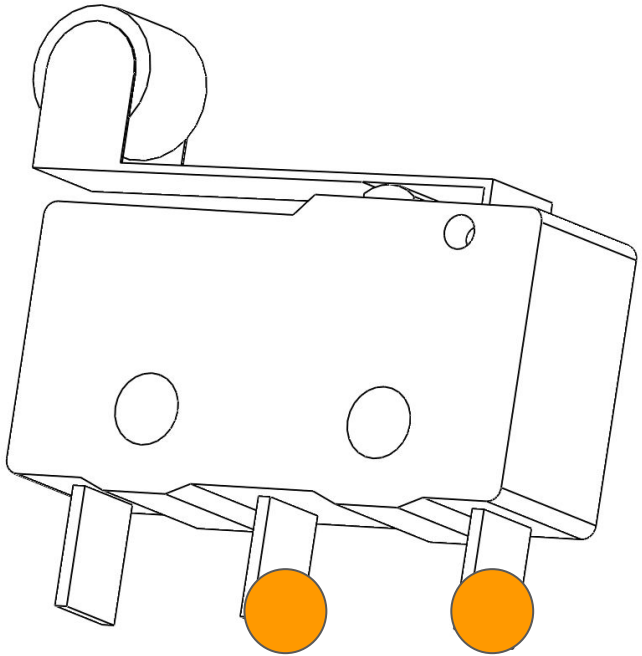
Preparing parts

第 15 页中文译文

零件准备

（本页为章节标题页。）

Limit switches



Attach limit switch wires to contacts indicated with orange. Apply heat shrink tube on the solder joints. Length of the wire needs to be 1 m.

第 16 页中文译文

限位开关

将限位开关导线焊接到橙色标示的触点上。焊点处套上热缩管。导线长度需为 1 m。

Source Robotics

Additional notes

第 17 页中文译文

附加说明

（本页为章节标题页。）

Additional notes

- In case a belt is not tight enough you can always add additional tension bearings.
- When assembling the robot there is a lot of wires. As you go wrap them in the one single wire harness with insulation tape or cloth tape.
- If during the assembly process the tape gets in the way remove it and add it after that assembly step.
- try to mark the wires so you know what is what after you built the robot.
- Belt tension is done with Ball bearing 3x8x4 and Ball bearing 3x10x4 ball bearings. depending on how much tension you want you can choose any combination. They need to be 2 in pair always on a screw of 14 - 20 mm length ——————————>
- It is recommended for screw to follow ISO 4762 specification.
- **SCREWS USED TO ATTACH GEARBOX TO MOTOR ARE THE ONES YOU GET WITH THE GEARBOX**



第 18 页中文译文

附加说明

- 如果皮带张力不够，可以增加张紧轴承。
- 装配机械臂时会有很多电线。组装过程中，将电线逐步整理成一束，并用绝缘胶带或布基胶带缠扎。
- 如果组装过程中胶带妨碍操作，可先将其拆下，完成该步骤后再重新缠好。
- 尽量给电线做标记，以便组装完成后识别每根线的用途。
- 皮带张紧使用 3×8×4 mm 和 3×10×4 mm 滚珠轴承。根据所需张力，可选择任意组合。轴承必须始终成对安装在一颗长度为 14–20 mm 的螺钉上。（原图此处有指向说明的箭头。）
- 建议使用符合 ISO 4762 规格的螺钉。
- 用于连接减速器和电机的螺钉，使用减速器随附的螺钉。

Additional notes

Parol6 robotic arm needs homing since it uses open loop-stepper drivers. Joints 1,4 and 6 are homed with inductive sensors. Homing is done with small screws that trigger the sensor when it gets close to it. You need to find “sweet” spot where the sensors trigger.

第 19 页中文译文

附加说明

PAROL6 机械臂需要执行回零，因为它使用开环步进驱动器。J1、J4 和 J6 通过电感式传感器回零。回零由小螺钉触发：螺钉靠近传感器时会触发信号。你需要找到传感器能够触发的“最佳位置”（sweet spot）。

Source Robotics

Wrist assembly

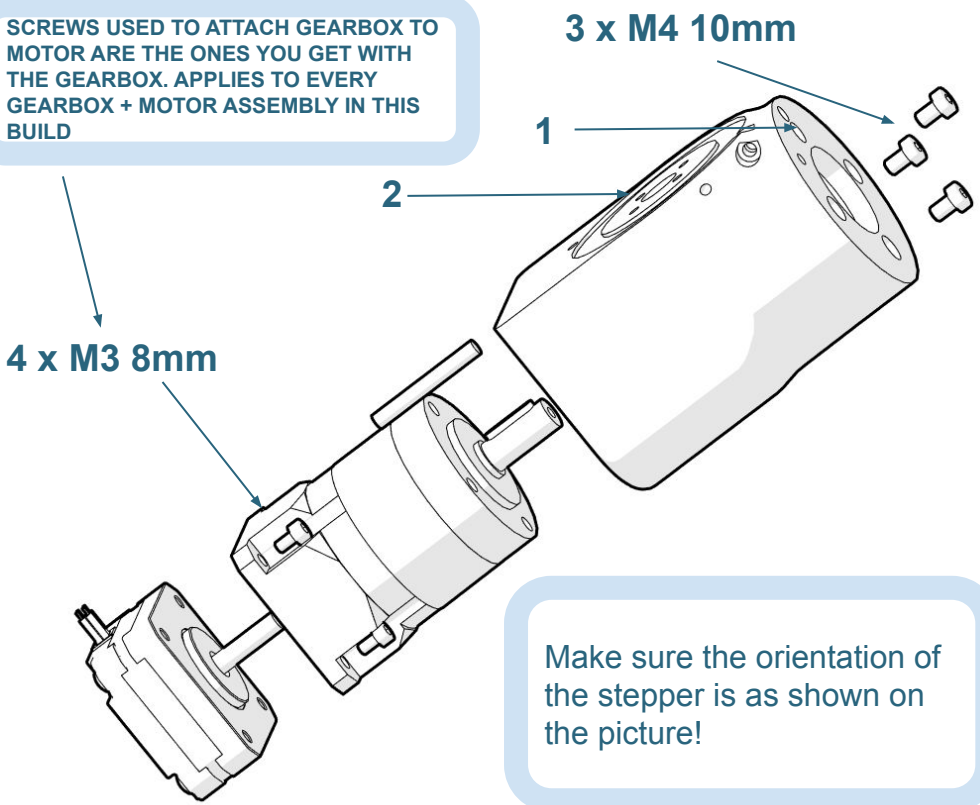
第 20 页中文译文

手腕组件

（本页为章节标题页。）

Wrist assembly

SCREWS USED TO ATTACH GEARBOX TO MOTOR ARE THE ONES YOU GET WITH THE GEARBOX. APPLIES TO EVERY GEARBOX + MOTOR ASSEMBLY IN THIS BUILD



Make sure the orientation of the stepper is as shown on the picture!

Wrist
Gearbox 10:1
Stepper1
Sensor1
3 x M4 10mm
4 x M3 8mm

First attach stepper 1 to the gearbox. The gearbox comes with its own coupler that need to be tightened with 2 set screw. Holes to tighten those screws are on the side of the gearbox.

Now press fit sensor 1 in wrist part. It needs to lay flat on top part of the wrist as indicated by arrow 1.

Wires for the motor and sensor need to exit from the hole market with arrow 2.

第 21 页中文译文

手腕组件

零件与紧固件：

- Wrist（手腕件）。
- Gearbox 10:1（10:1 减速器）。
- Stepper 1（步进电机 1）。
- Sensor 1（传感器 1）。
- 3 颗 M4×10 mm 螺钉。
- 4 颗 M3×8 mm 螺钉。

首先将步进电机 1 固定到减速器上。减速器自带联轴器，需要用 2 颗紧定螺钉锁紧。用于拧紧这两颗螺钉的孔位于减速器侧面。

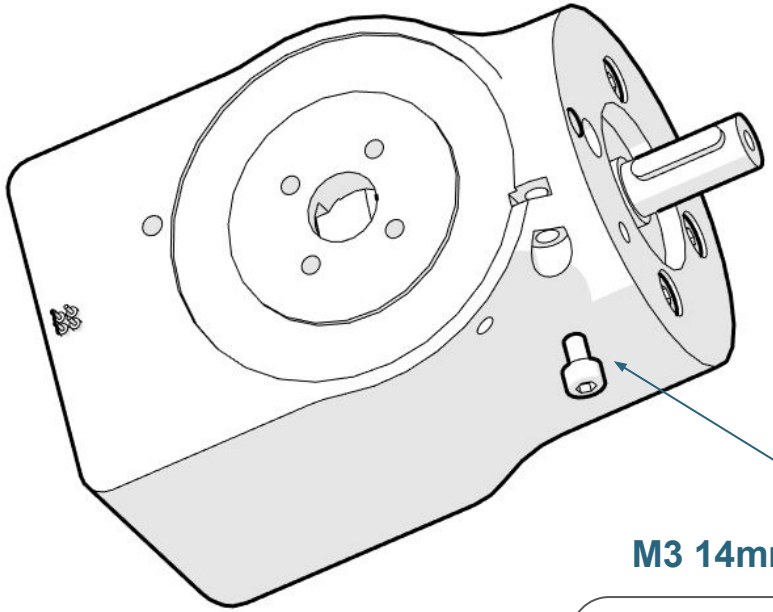
接着将传感器 1 压入 Wrist（手腕件）。传感器应平贴在手腕件顶部，位置如箭头 1 所示。

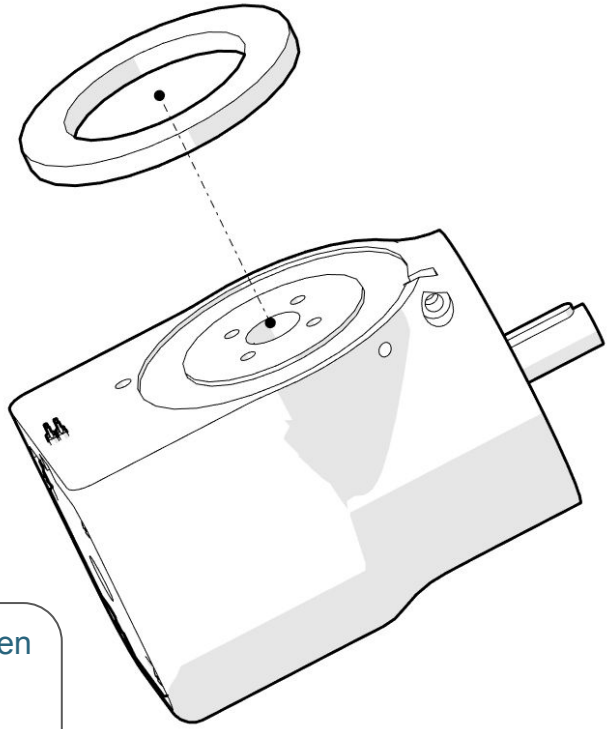
电机和传感器的导线应从箭头 2 标示的孔穿出。

确保步进电机方向与图片所示一致！

减速器和电机之间使用的螺钉为减速器随附螺钉。本装配中所有“减速器 + 电机”组件均适用此要求。

Wrist assembly





M3 14mm

Bearing 1
*M3 14mm
*M3 nut

If the sensor is loose you can tighten it with M3 nut and a screw.

Attach bearing 1 to wrist part.
Assembled part is on the next page.

第 22 页中文译文

手腕组件

零件与紧固件：

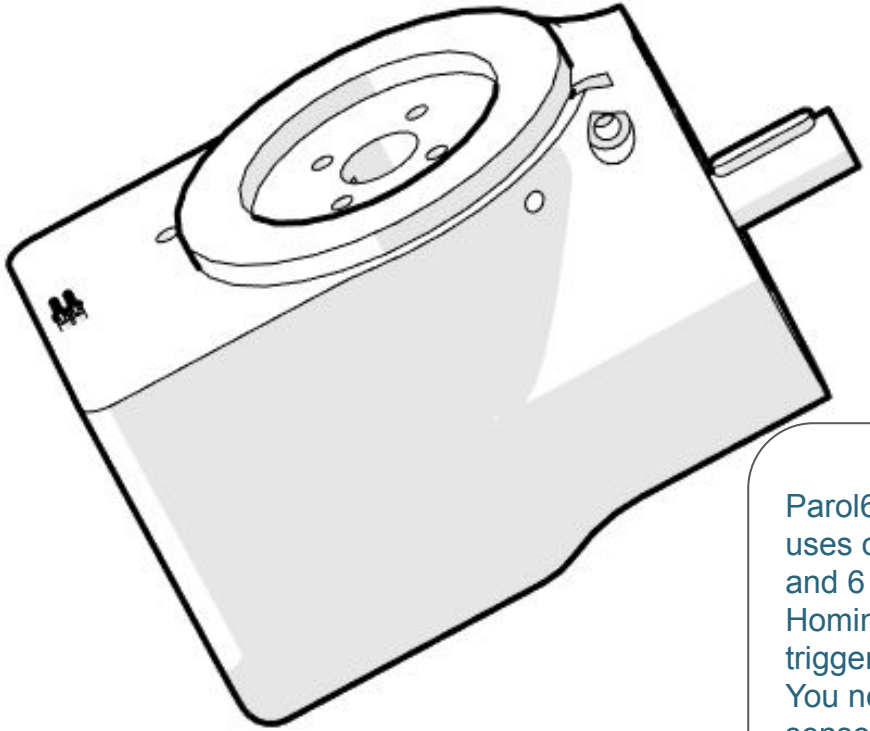
- Bearing 1（轴承 1）。
- M3×14 mm 螺钉（*）。
- M3 螺母（*）。

如果传感器松动，可使用 M3 螺母和螺钉将其固定。

将轴承 1 安装到 Wrist（手腕件）上。组装完成后的部件见下一页。

图示标注：M3×14 mm。

Wrist assembly



Parol6 robotic arm needs homing since it uses open loop-stepper drivers. Joints 1,4 and 6 are homed with inductive sensors. Homing is done with small screws that trigger the sensor when it gets close to it. You need to find “sweet” spot where the sensors trigger.

第 23 页中文译文

手腕组件

PAROL6 机械臂需要执行回零，因为它使用开环步进驱动器。J1、J4 和 J6 通过电感式传感器回零。回零由小螺钉触发：螺钉靠近传感器时会触发信号。你需要找到传感器能够触发的“最佳位置”（sweet spot）。

Source Robotics

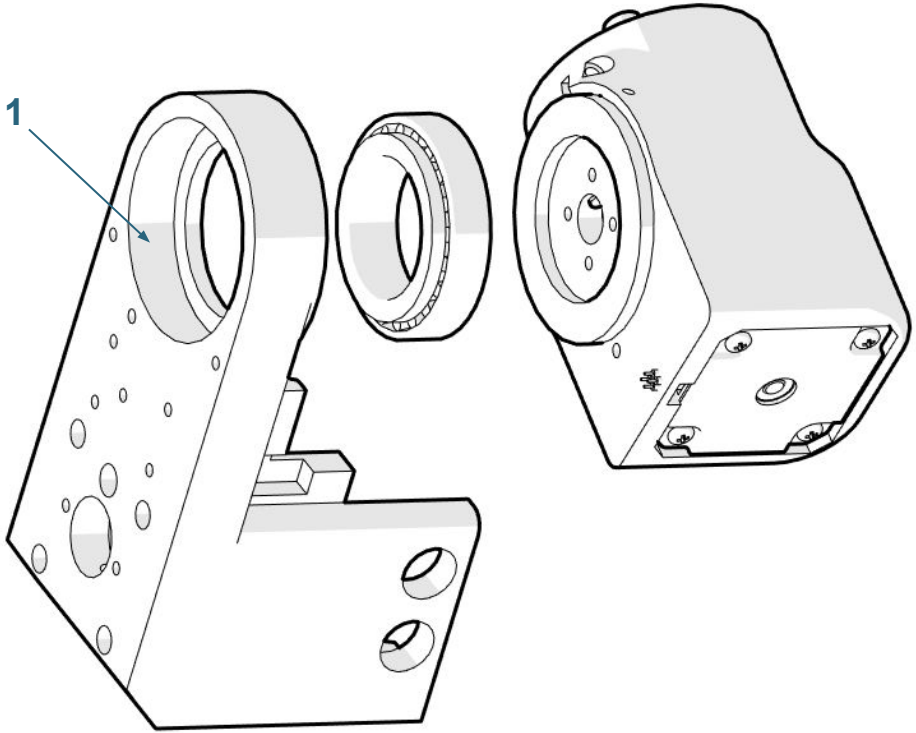
Forearm assembly

第 24 页中文译文

前臂组件

（本页为章节标题页。）

Forearm assembly



Bearing 2
J5_part

Bearing 2 need to go in the groove marked with arrow 1.

Assembled part is on the next page.

第 25 页中文译文

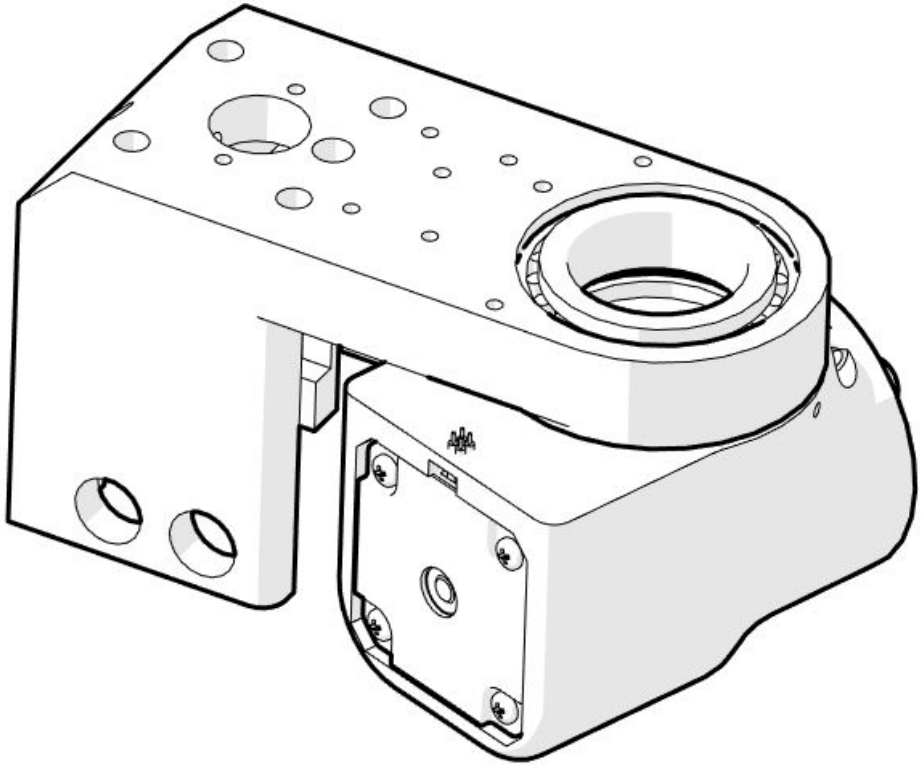
前臂组件

零件：

- Bearing 2（轴承 2）。
- J5_part。

轴承 2 需要安装到箭头 1 标示的凹槽中。组装完成后的部件见下一页。

Source Robotics



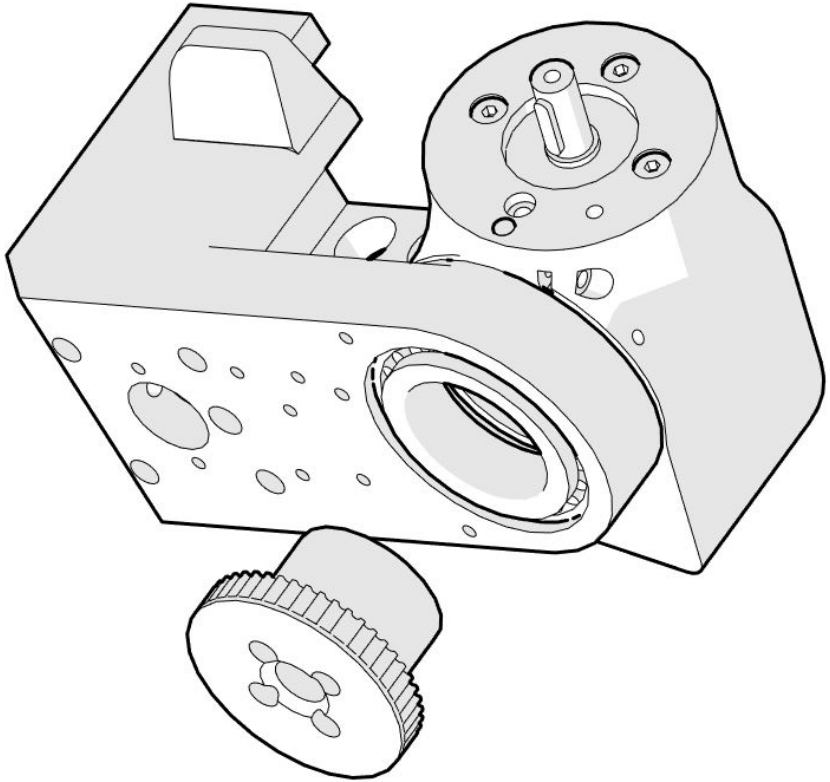
Hammer in the bearing with rubber hammer. Another option is to use screwdriver and gently tap with a hammer on the whole circumference of the bearing.

第 26 页中文译文

前臂装配（Forearm assembly）

用橡胶锤将轴承敲入。也可以使用螺丝刀和锤子，沿轴承整个圆周轻轻敲入。

Forearm assembly



HTD3_48_J5_pulley

Pull all then wires from J6 thru the pulley and the bearing! Assembled part is on the next page.

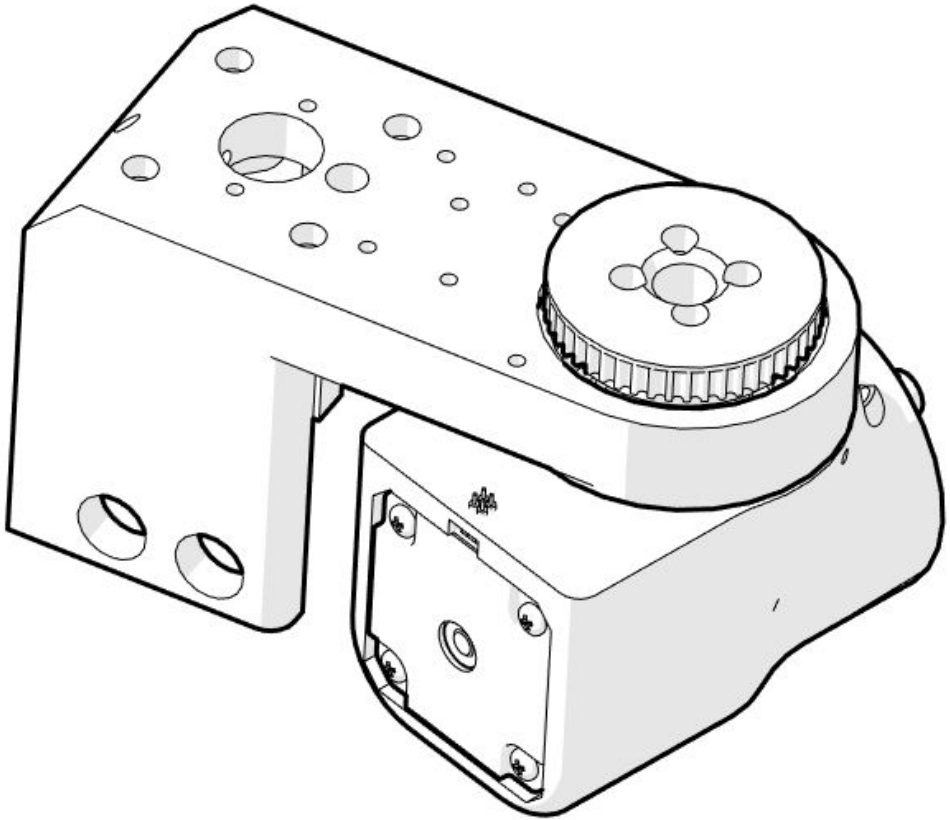
第 27 页中文译文

前臂装配（Forearm assembly）

零件：HTD3_48_J5_pulley。

将 J6 的所有线缆穿过滑轮和轴承！装配完成后的部件见下一页。

Forearm assembly

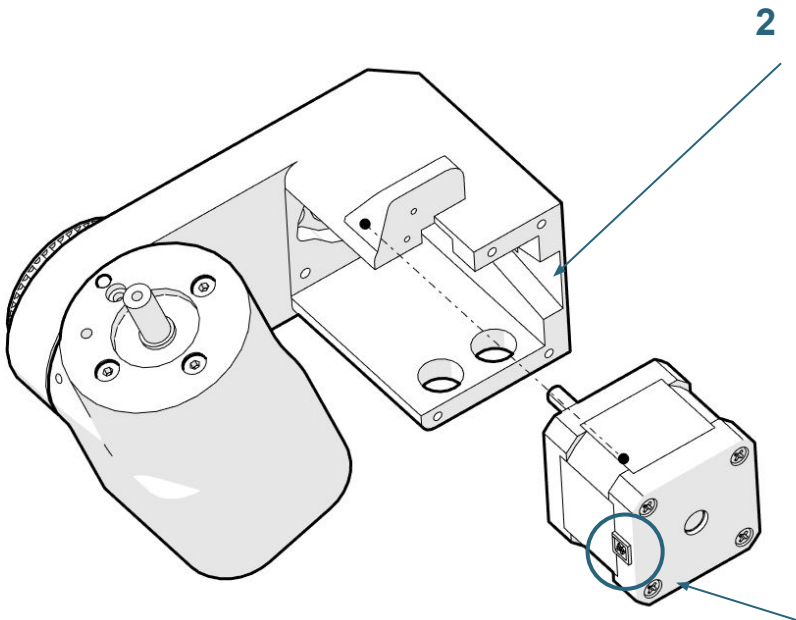


第 28 页中文译文

前臂装配（Forearm assembly）

本页为装配示意图，无可提取的正文。

Forearm assembly



Stepper 2

Stepper wire, all other wires and pipes need to go thru the wires guide marked with arrow 1 and then thru a hole on J5 part.

Stepper orientation is wrong on this image. Flip it 180 degree so that wires go thru the hole where arrow 2 points

第 29 页中文译文

前臂装配（Forearm assembly）

零件：步进电机 2（Stepper 2）。

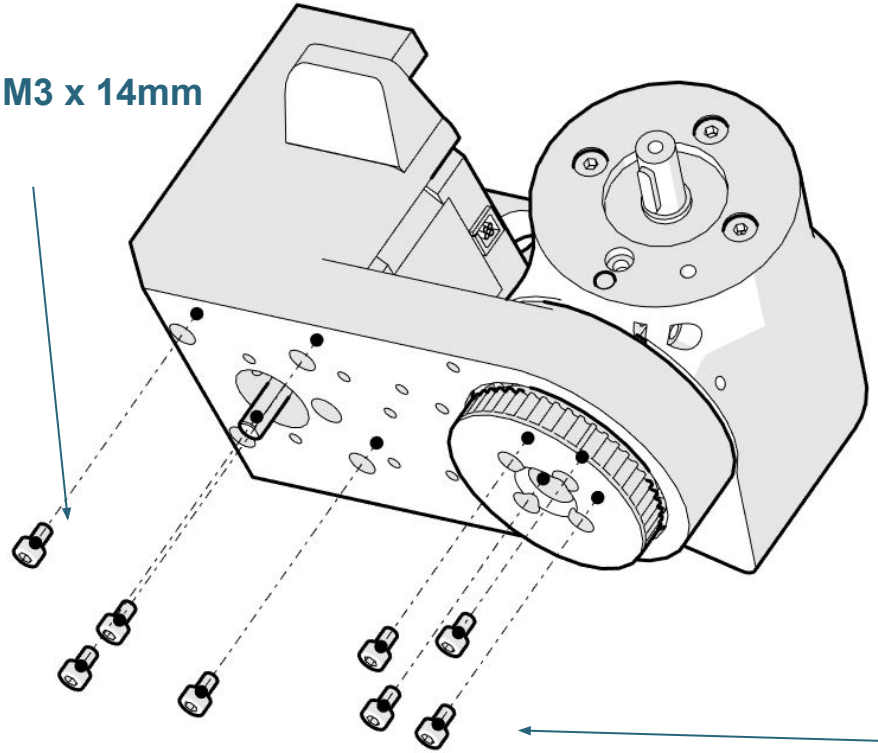
步进电机线、其他所有线缆及管路都要穿过箭头 1 标示的线缆导向槽，再穿过 J5 零件上的孔。

注意：图中步进电机的朝向错误。请将电机旋转 180°，使线缆穿过箭头 2 所指的孔。

Forearm assembly

Source Robotics

4 x M3 x 14mm



4 x M3 x 25 mm

4 x M3 x 25 mm
4 x M3 x 14mm

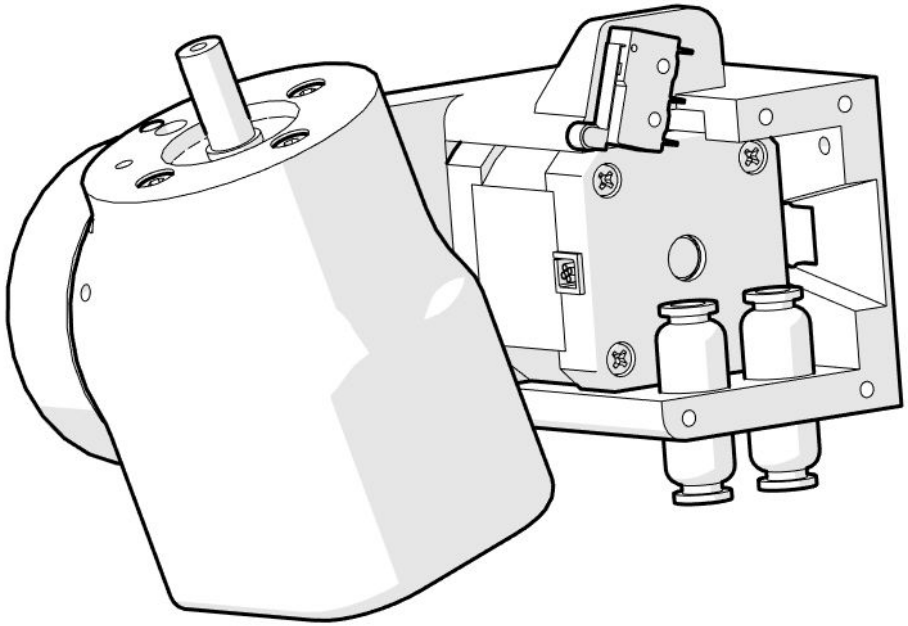
第 30 页中文译文

前臂装配（Forearm assembly）

- 4 × M3 × 25 mm
- 4 × M3 × 14 mm

图中标注的两种螺钉各需 4 颗。

Forearm assembly



2 x Pneumatic connector 1
Limit switch
2 X M2 x 10mm

Limit switch needs to be secured with M2 screws. Wires need to go thru a hole in J5 part. Pneumatic connectors need to be secured with provided nuts. Pneumatic tubes go thru a hole on J5 part.

第 31 页中文译文

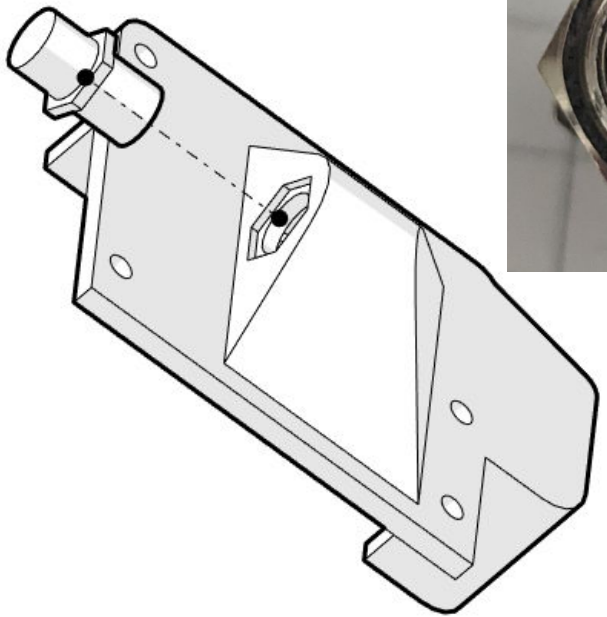
前臂装配（Forearm assembly）

零件：2 个气动接头 1（Pneumatic connector 1）；限位开关（Limit switch）。

- 2 × M2 × 10 mm

用 M2 螺钉固定限位开关。线缆需要穿过 J5 零件上的孔。气动接头用随附螺母固定。气管穿过 J5 零件上的孔。

Forearm assembly



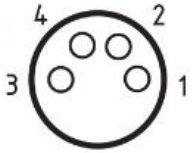


J5_electronics_lid
Gripper connector electric

On the gripper connector solder 4 wires of length 1m.
2 wires need to be twisted pair for the can bus.
Use heat shrink on solder joints. After placing gripper connector in lid use provided nut to tighten connector to the lid.

Solder 4 wires to 4 pin connector. Use this pinout:

1- 24V
2- CANH
3 - GND
4 -CANL



第 32 页中文译文

前臂装配（Forearm assembly）

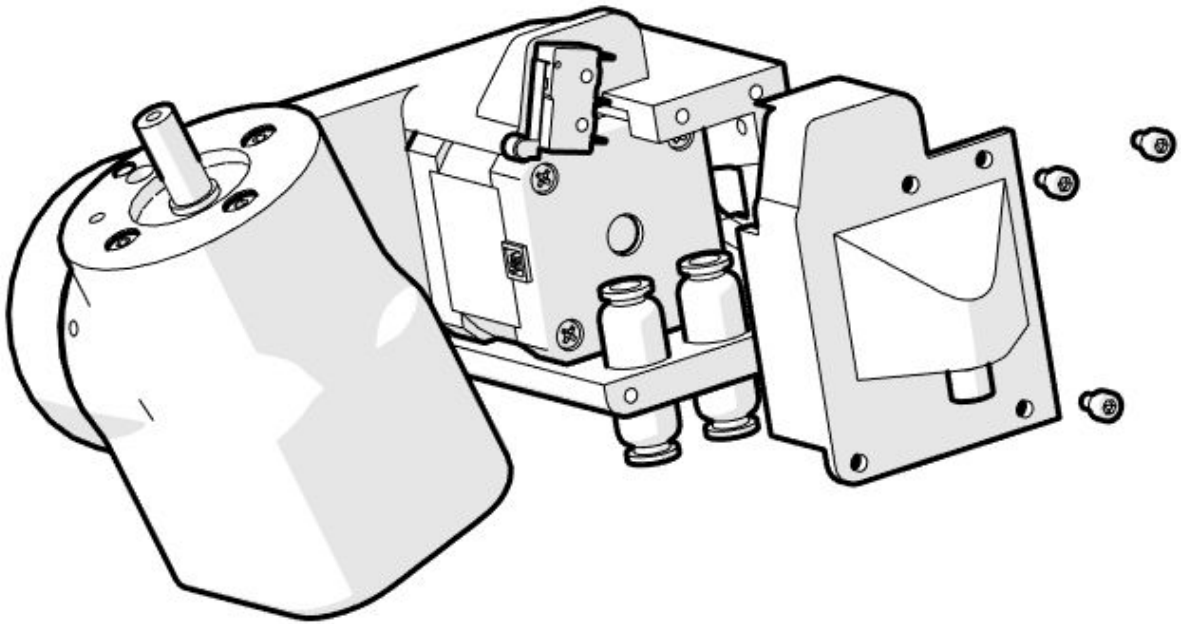
零件：J5_electronics_lid；电动夹爪连接器（Gripper connector electric）。

在夹爪连接器上焊接 4 根长度为 1 m 的导线。其中 2 根导线需要绞合成 CAN 总线双绞线。焊点使用热缩管绝缘。将夹爪连接器放入盖板后，用随附螺母将连接器固定在盖板上。

将 4 根导线焊接到 4 针连接器，针脚定义如下：

1. 24 V
2. CANH
3. GND
4. CANL

Forearm assembly



4x M3 8 mm

Assembled part is on the next page.

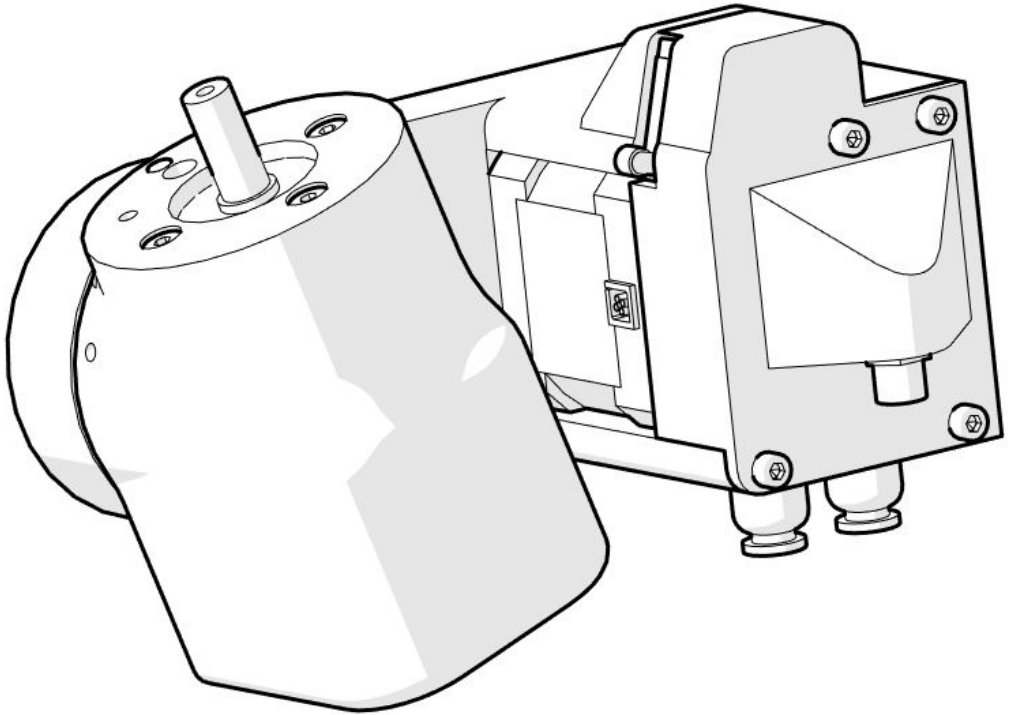
第 33 页中文译文

前臂装配（Forearm assembly）

- 4 × M3 × 8 mm

装配完成后的部件见下一页。

Forearm assembly



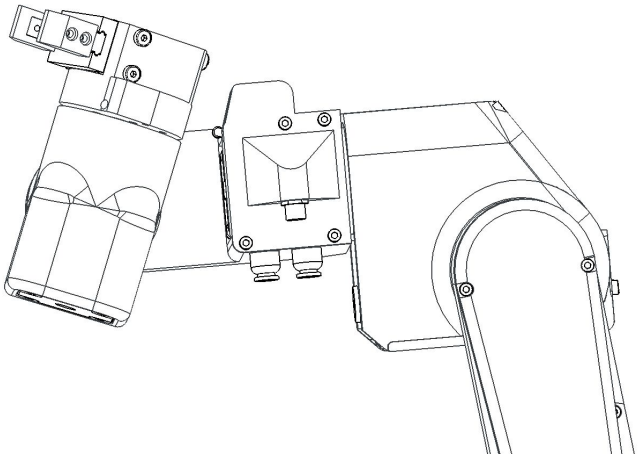
第 34 页中文译文

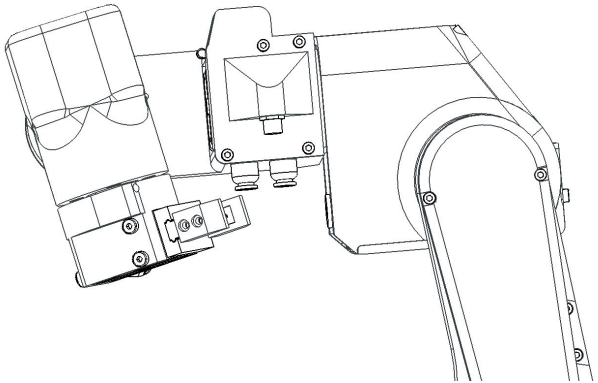
前臂装配（Forearm assembly）

本页为装配示意图，无可提取的正文。

Source Robotics

IF THERE IS NO GRIPPER ON WRIST PART IT CAN APPEAR IT CAN SPIN INDEFINITELY. DO NOT DO THAT! LIMITS OF J5 ARE SHOWN ON THE IMAGES BELOW!





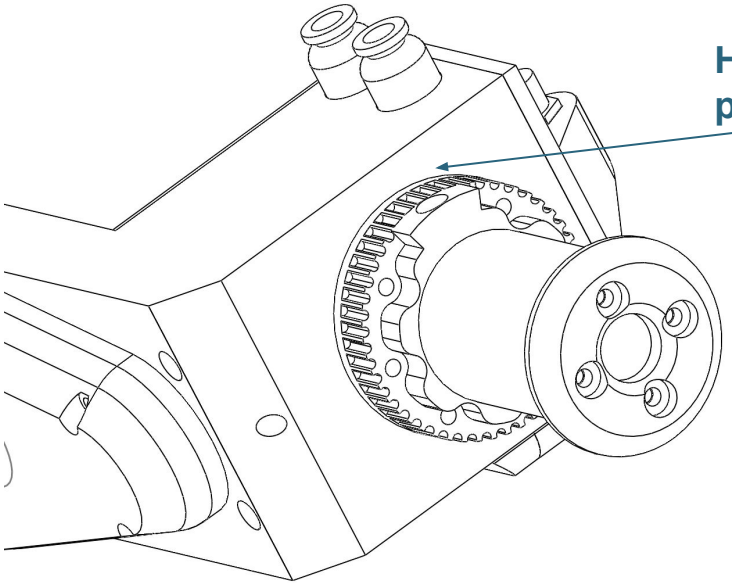
第 35 页中文译文

前臂装配（Forearm assembly）

警告：如果腕部零件上没有安装夹爪，J5 看起来可能可以无限旋转。不要这样做！J5 的运动限位见下方图片。

Forearm assembly

J4_output_pulley
6 x M3 x 16mm



All wires and tubes need to go thru J4 output pulley. Use M3 16 mm screws to attach pulley to body of lower arm.

第 36 页中文译文

前臂装配（Forearm assembly）

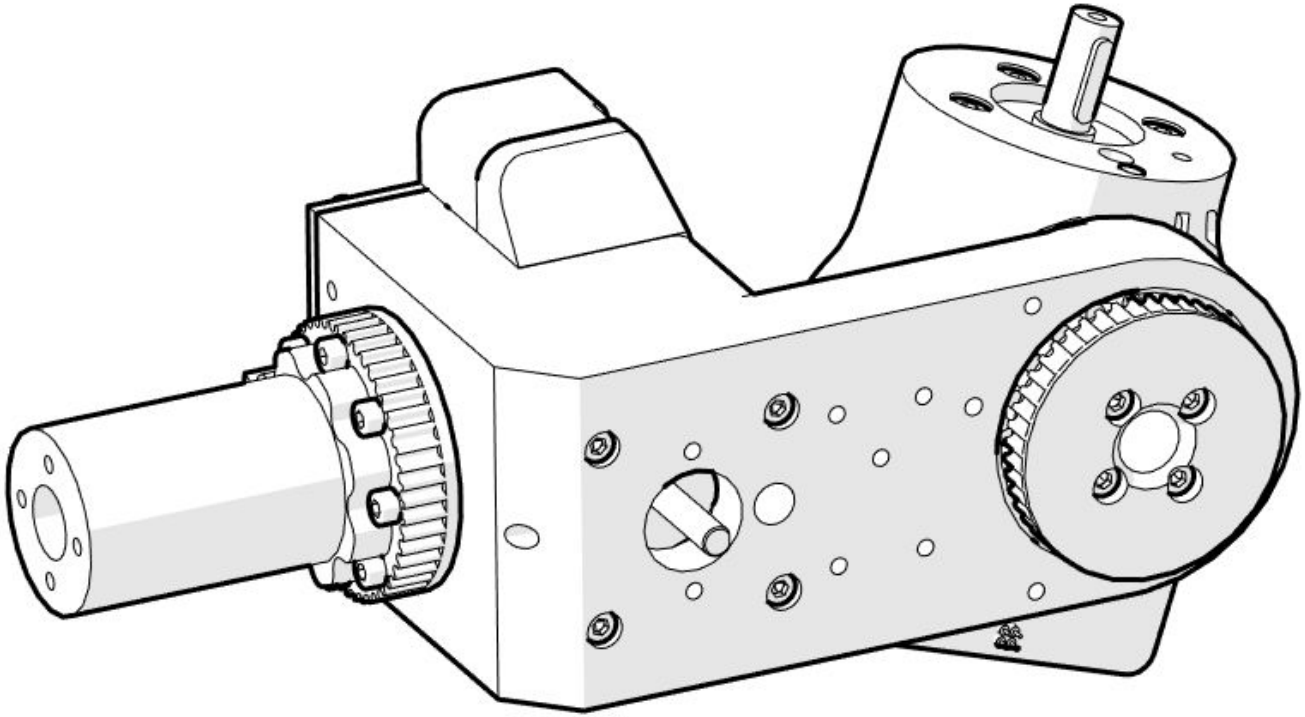
零件：J4_output_pulley。

- 6 × M3 × 16 mm

所有线缆和管路都需要穿过 J4_output_pulley。使用 M3 × 16 mm 螺钉将滑轮固定到下臂主体上。图中标注：归零销（Homing pin）。

Forearm assembly

Source Robotics

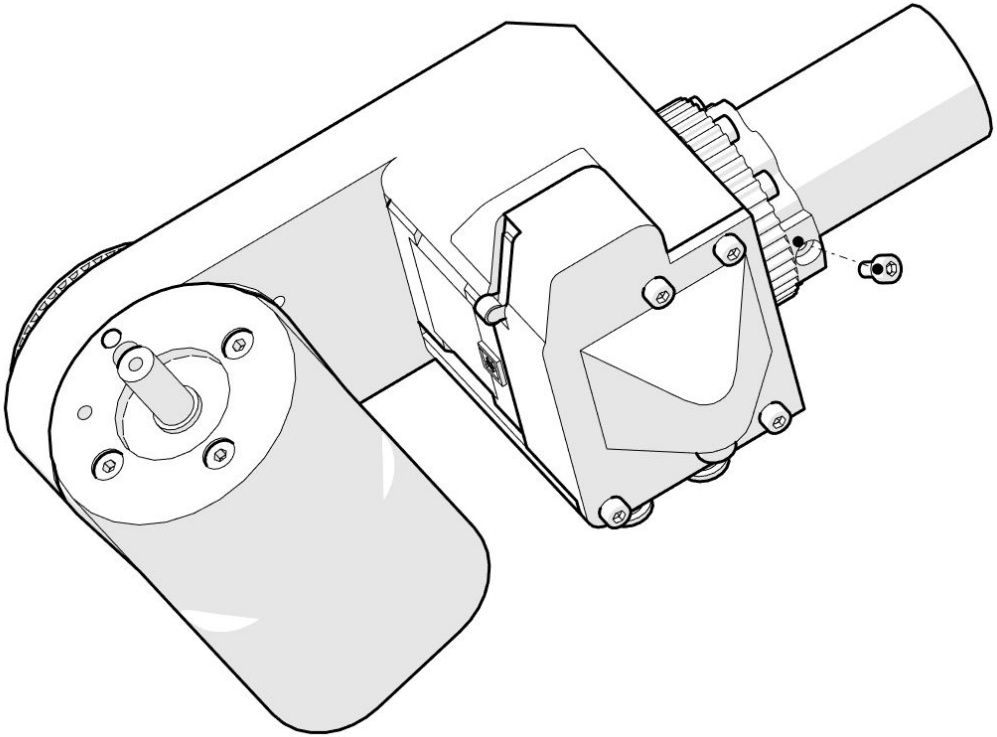


第 37 页中文译文

前臂装配（Forearm assembly）

本页为装配示意图，无可提取的正文。

Forearm assembly



1 x M3 x 6mm

This screw is used to trigger J4 sensor. Please use provided screw. If different screw is used sensor will trigger at wrong position and mess with robot kinematics.

第 38 页中文译文

前臂装配（Forearm assembly）

- 1 × M3 × 6 mm

此螺钉用于触发 J4 传感器。请使用随附螺钉。若使用其他螺钉，传感器会在错误的位置触发，从而导致机械臂运动学参数错误。

Source Robotics

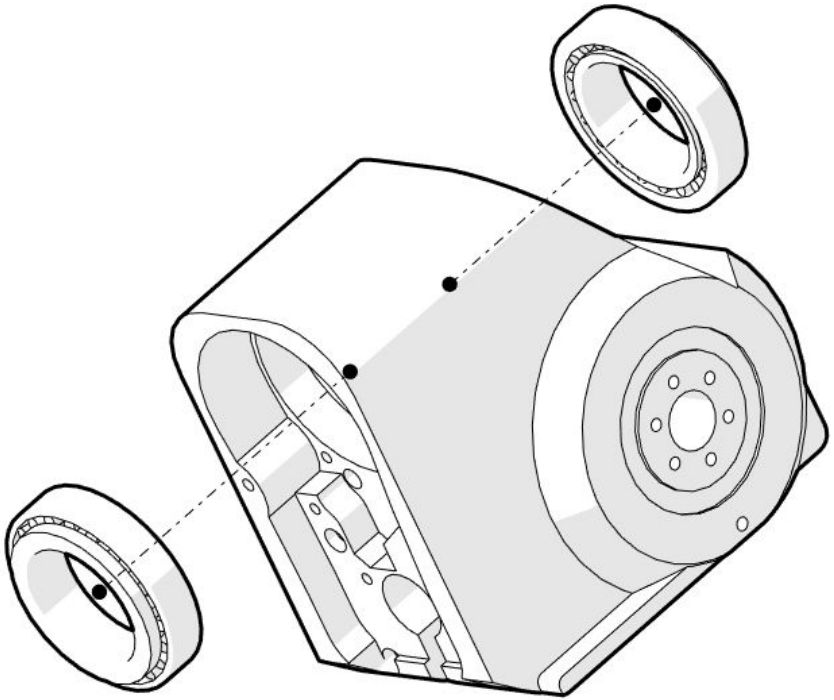
Elbow assembly

第 39 页中文译文

肘部装配（Elbow assembly）

本页为装配示意图。

Elbow assembly



Elbow_part
2 x Bearing 2

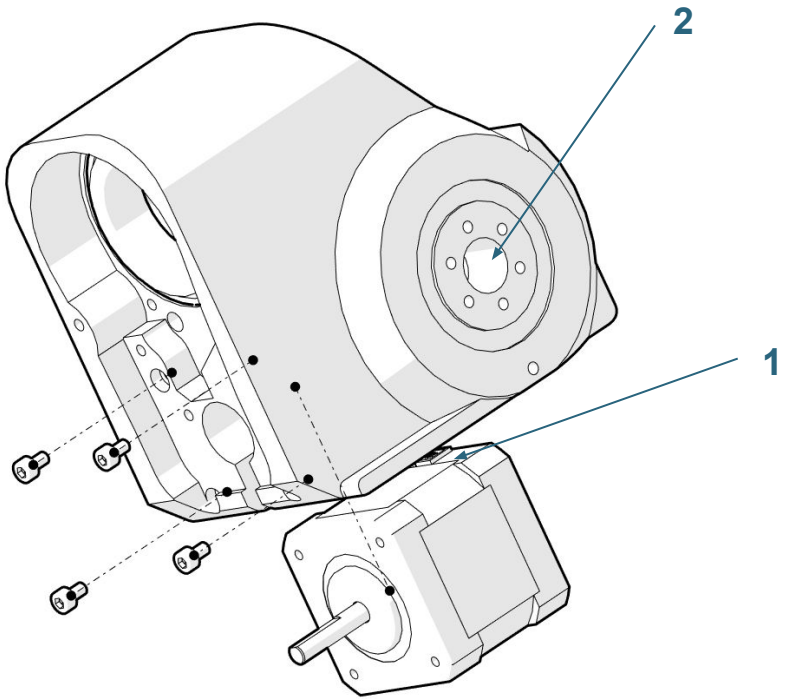
第 40 页中文译文

肘部装配（Elbow assembly）

零件：Elbow_part。

- 2 个轴承 2（Bearing 2）

Elbow assembly



Stepper 2
4 x M3 x 14 mm

Secure the stepper with 4 screws. Note the position of stepper wires with arrow 1. All wires from previous steps and from subsequent steps need to go thru hole 2.

第 41 页中文译文

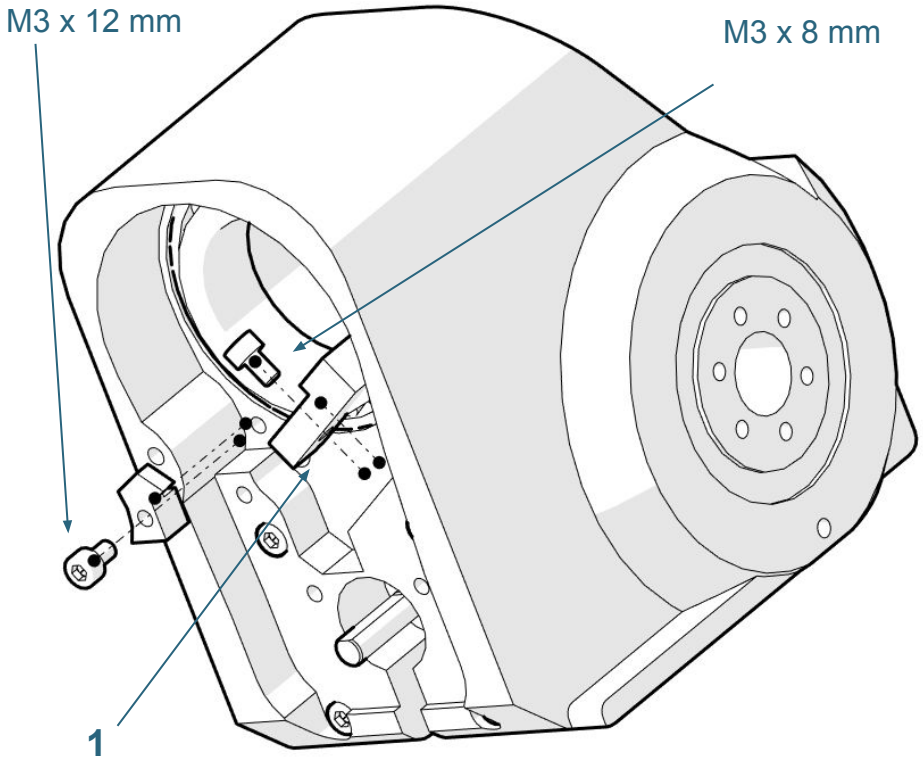
肘部装配（Elbow assembly）

零件：步进电机 2（Stepper 2）。

- 4 × M3 × 14 mm

用 4 颗螺钉固定步进电机。注意箭头 1 所示的电机线方向。前序步骤和后续步骤中的所有线缆都需要穿过孔 2。

Elbow assembly



J4_limiter
Sensor 2
M3 x 12 mm
M3 x 8 mm

Wires of the sensor 2 need to go thru a hole marked with arrow 1. Assembled part is on the next page.

第 42 页中文译文

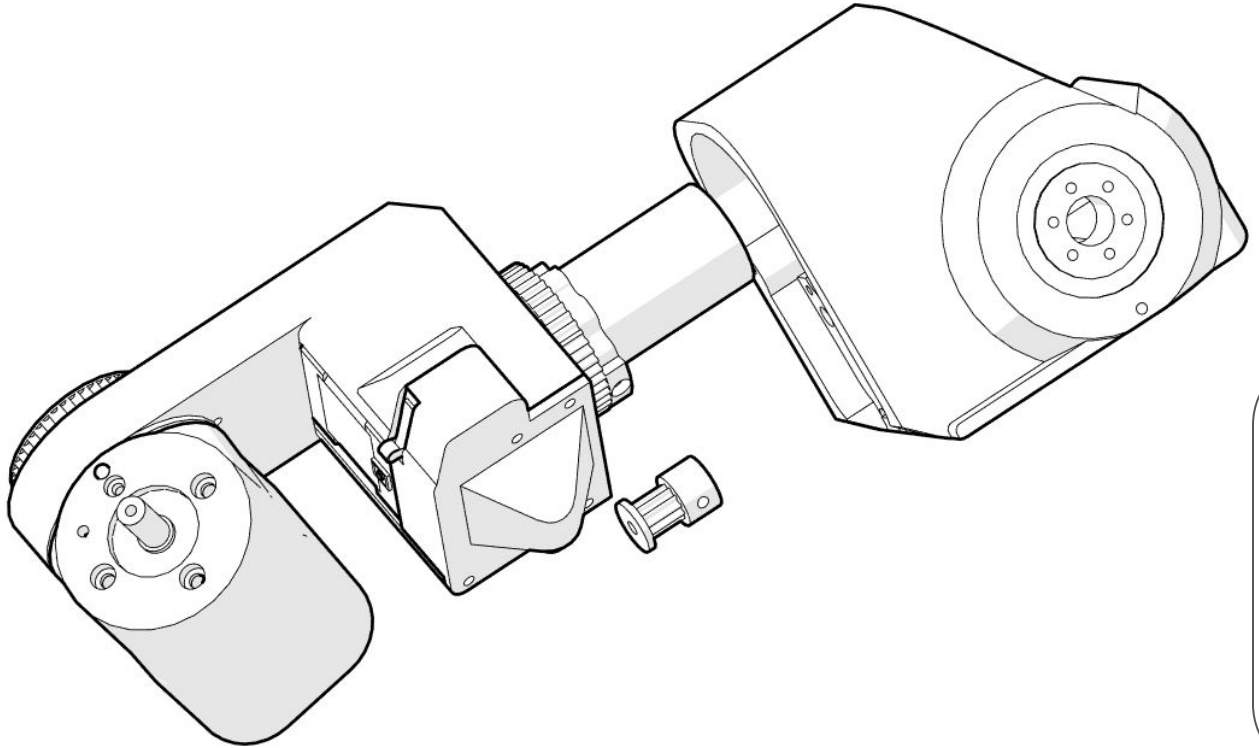
肘部装配（Elbow assembly）

零件：J4_limiter；传感器 2（Sensor 2）。

- M3 × 12 mm
- M3 × 8 mm

传感器 2 的导线需要穿过箭头 1 所标示的孔。装配完成后的部件见下一页。图中两种螺钉分别标注为 M3 × 8 mm 和 M3 × 12 mm。

Elbow assembly



J4 pulley
J4 belt

First pull all wires and tubes thru the elbow part! When J4 pulley and J4 output pulley are in illustrated position put the belt on.

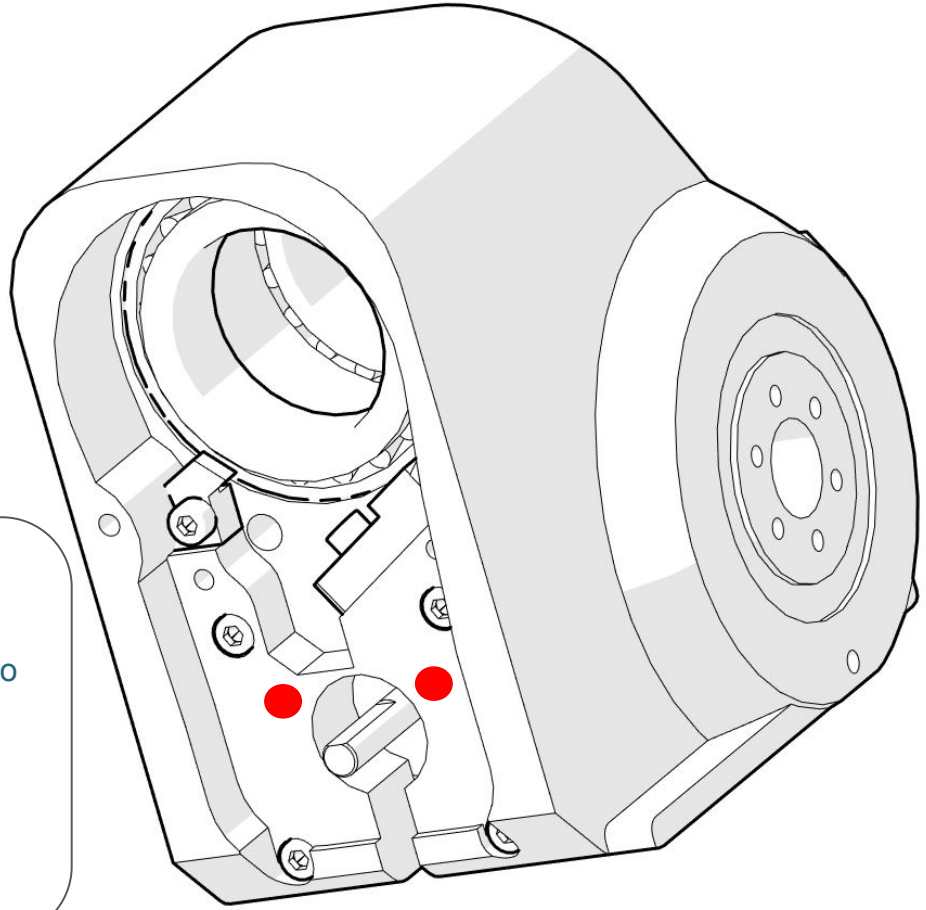
第 43 页中文译文

肘部装配（Elbow assembly）

零件：J4 pulley；J4 belt。

先将所有线缆和管路穿过肘部零件！当 J4 pulley 与 J4 output pulley 处于图示位置时，装上皮带。

Elbow assembly



Belt tension bearings go to the red marked spots!
USE M3x15 mm screws with tapered head!

Parol6 robotic arm needs homing since it uses open loop-stepper drivers. Joints 1,4 and 6 are homed with inductive sensors. Homing is done with small screws that trigger the sensor when it gets close to it. You need to find “sweet” spot where the sensors trigger.

第 44 页中文译文

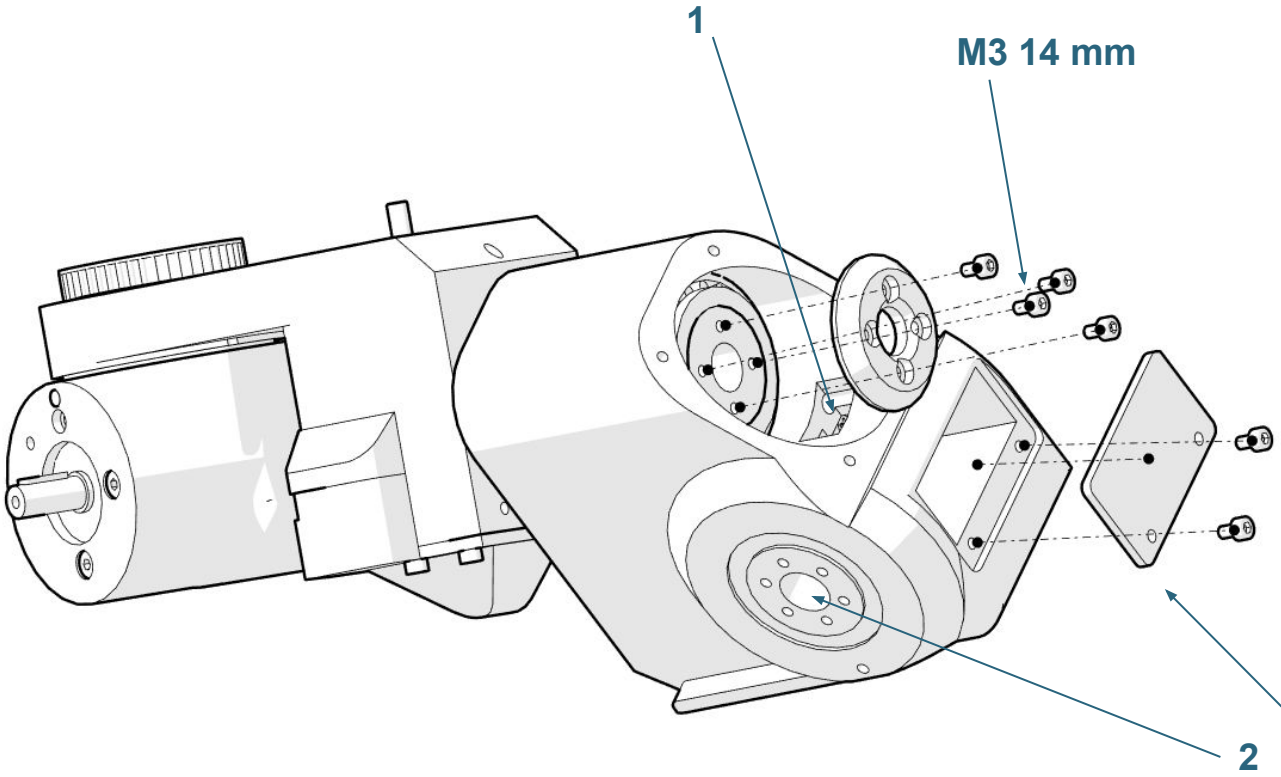
肘部装配（Elbow assembly）

皮带张紧轴承安装在红色标记处！

使用 M3 × 15 mm 锥头螺钉！

PAROL6 机械臂需要执行归零，因为它使用开环步进驱动器。关节 1、4 和 6 使用电感式传感器归零。归零由小螺钉触发：螺钉靠近传感器时会触发传感器。需要找到传感器能够触发的“合适位置（sweet spot）”。

Elbow assembly



Elbow_lid_2
J4_bearing_backplate
2 x M3 8 mm
4 x M3 14 mm

All wires and tubes need to go where arrow 1 is pointing and exit where arrow 2 is pointing.

第 45 页中文译文

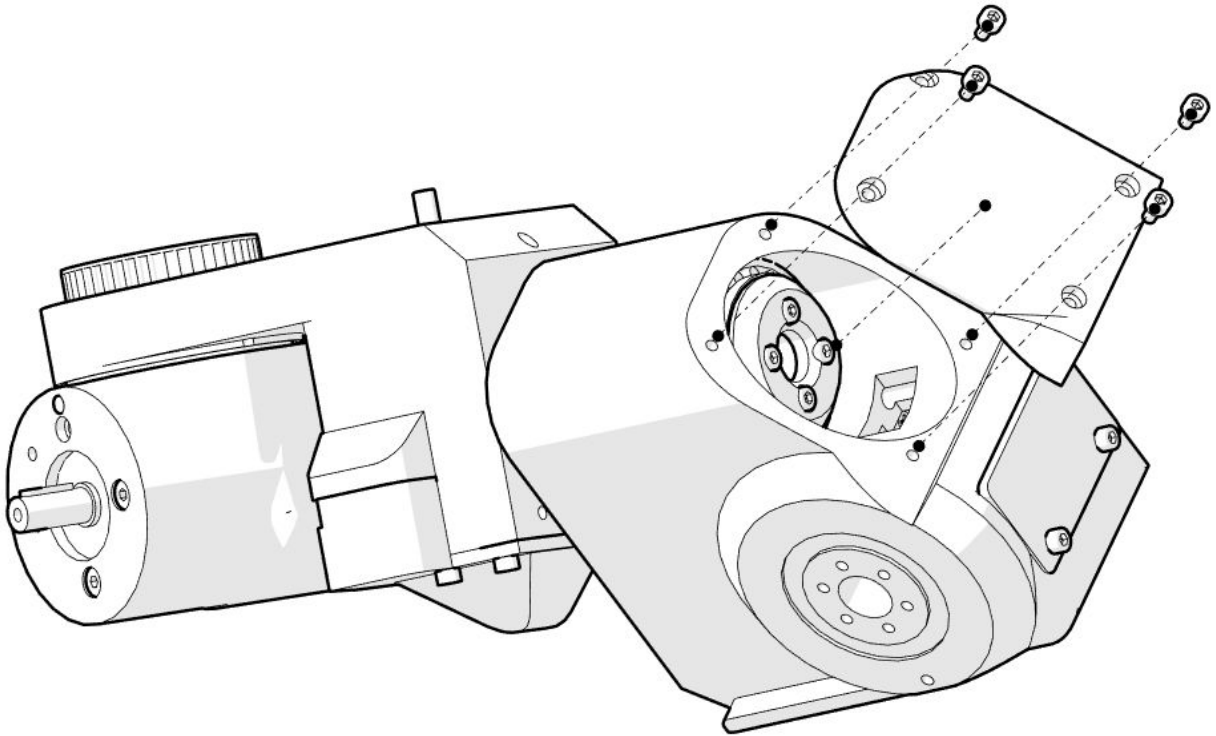
肘部装配（Elbow assembly）

零件：Elbow_lid_2；J4_bearing_backplate。

- 2 × M3 × 8 mm
- 4 × M3 × 14 mm

所有线缆和管路需要沿箭头 1 所示方向布置，并从箭头 2 所示位置引出。

Elbow assembly



Elbow_lid_1
4 x M3 IMBUS 8 mm

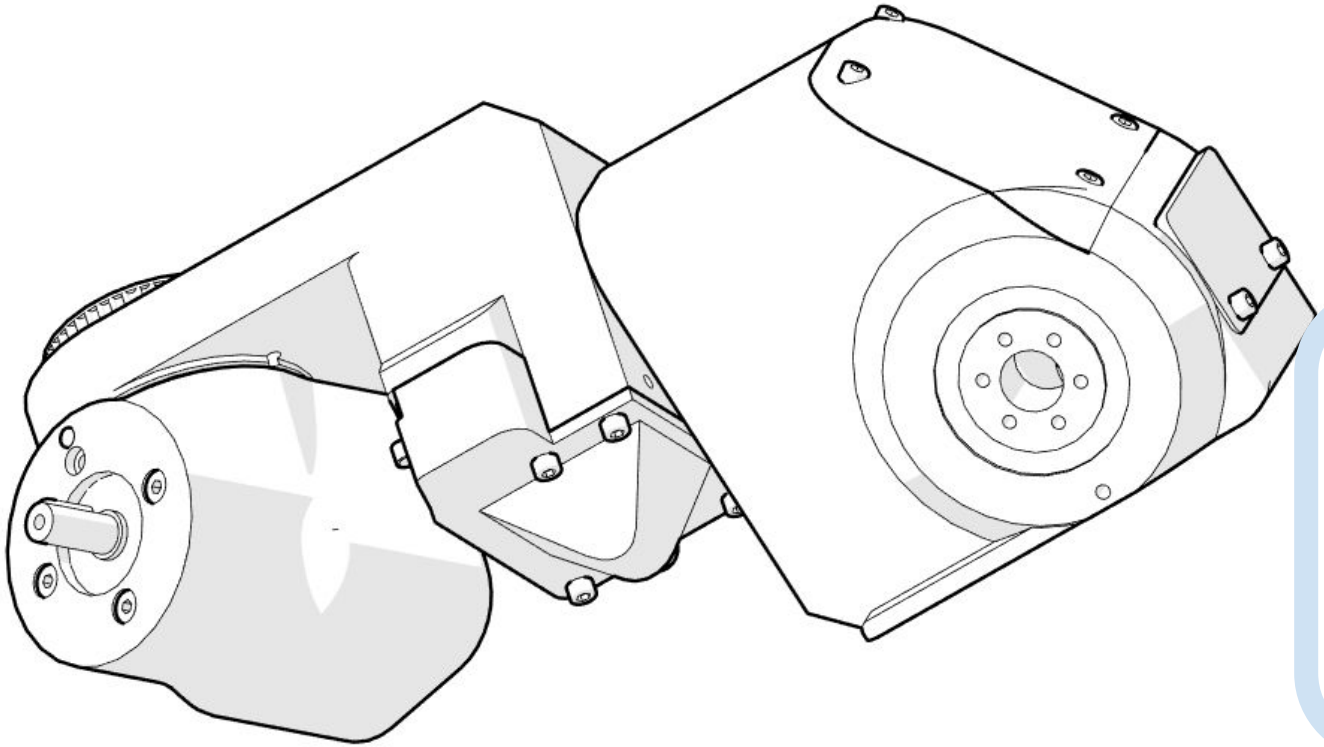
第 46 页中文译文

肘部装配（Elbow assembly）

零件：Elbow_lid_1。

- 4 × M3 内六角螺钉（imbus）× 8 mm

Elbow assembly



Make sure that all joints can move and that there is not much friction on specific spots. If there is you have:

1. Misaligned bearings
2. Used too much torque on bearing retainer screws.
3. Tangled wires.

第 47 页中文译文

肘部装配（Elbow assembly）

确认所有关节都可以活动，并且没有某些位置摩擦过大。如果有，请检查：

1. 轴承是否未对准。

2. 轴承压板螺钉是否拧得过紧。

3. 线缆是否缠绕。

Source Robotics

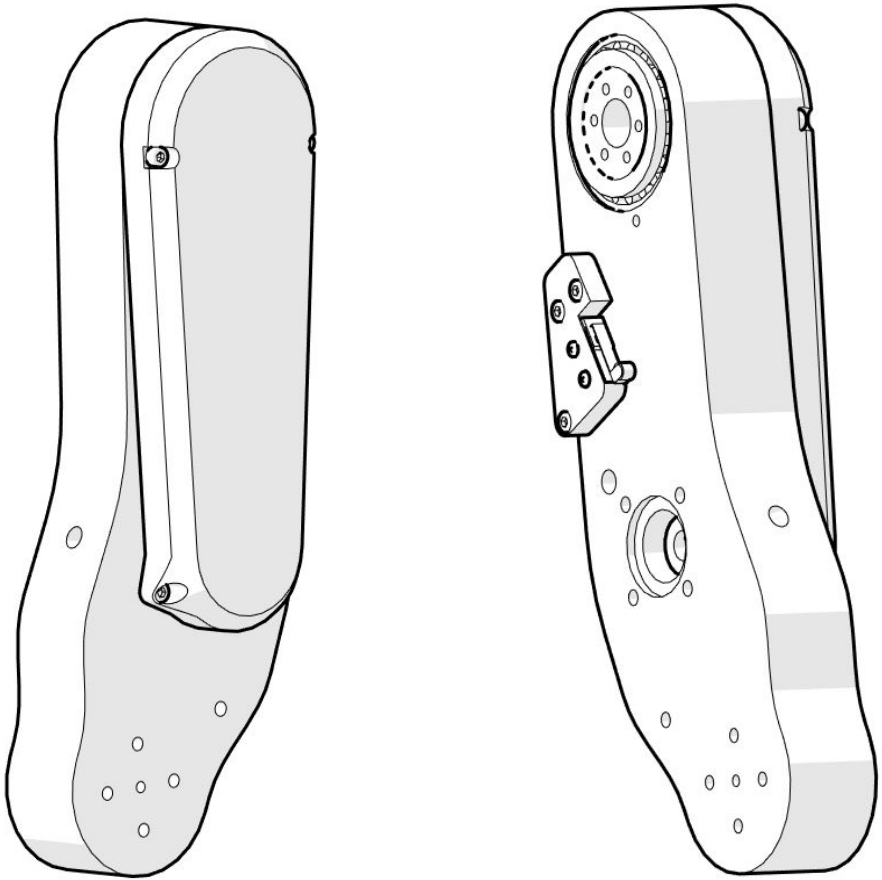
Upper arm assembly

第 48 页中文译文

上臂装配（Upper arm assembly）

本页为装配示意图。

Upper arm assembly



第 49 页中文译文

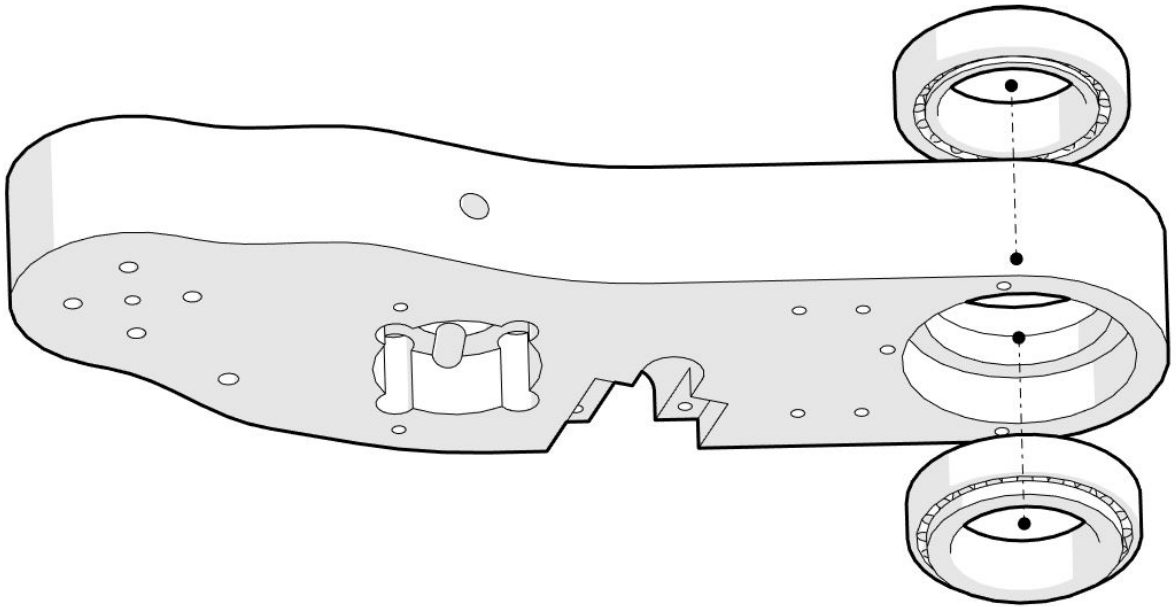
上臂装配（Upper arm assembly）

本页为装配示意图。

Upper arm assembly

2 x Bearing 2
Upper arm

Assembled part in the next page.



第 50 页中文译文

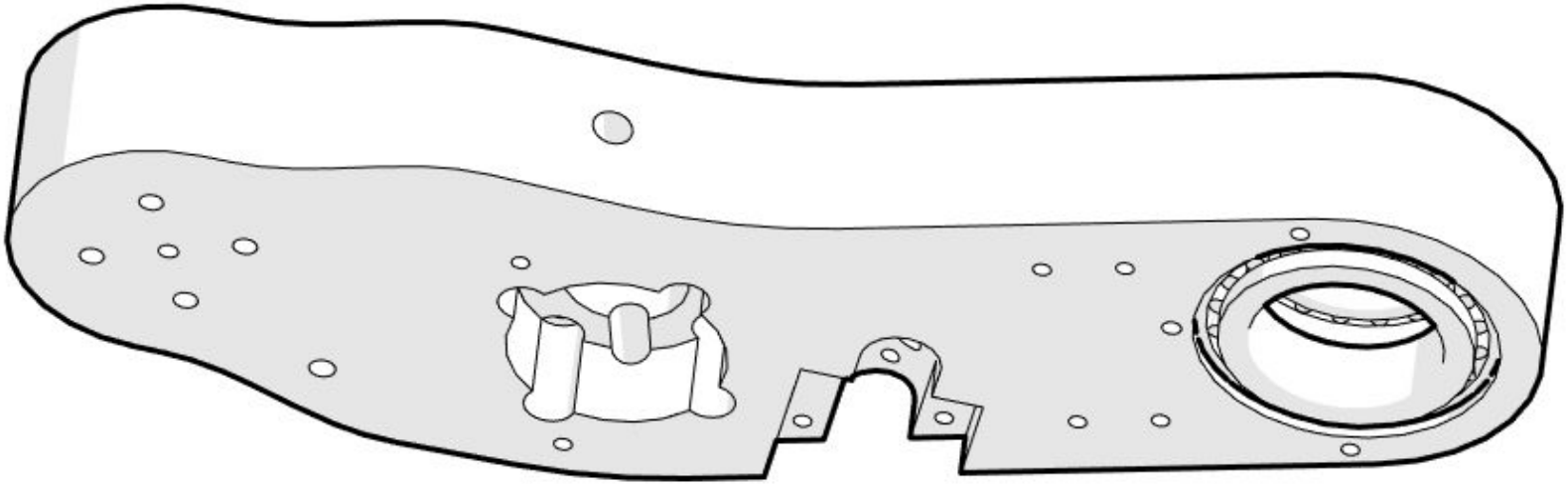
上臂装配（Upper arm assembly）

- 2 个轴承 2（Bearing 2）
- 上臂（Upper arm）

装配完成后的部件见下一页。

Upper arm assembly

Source Robotics

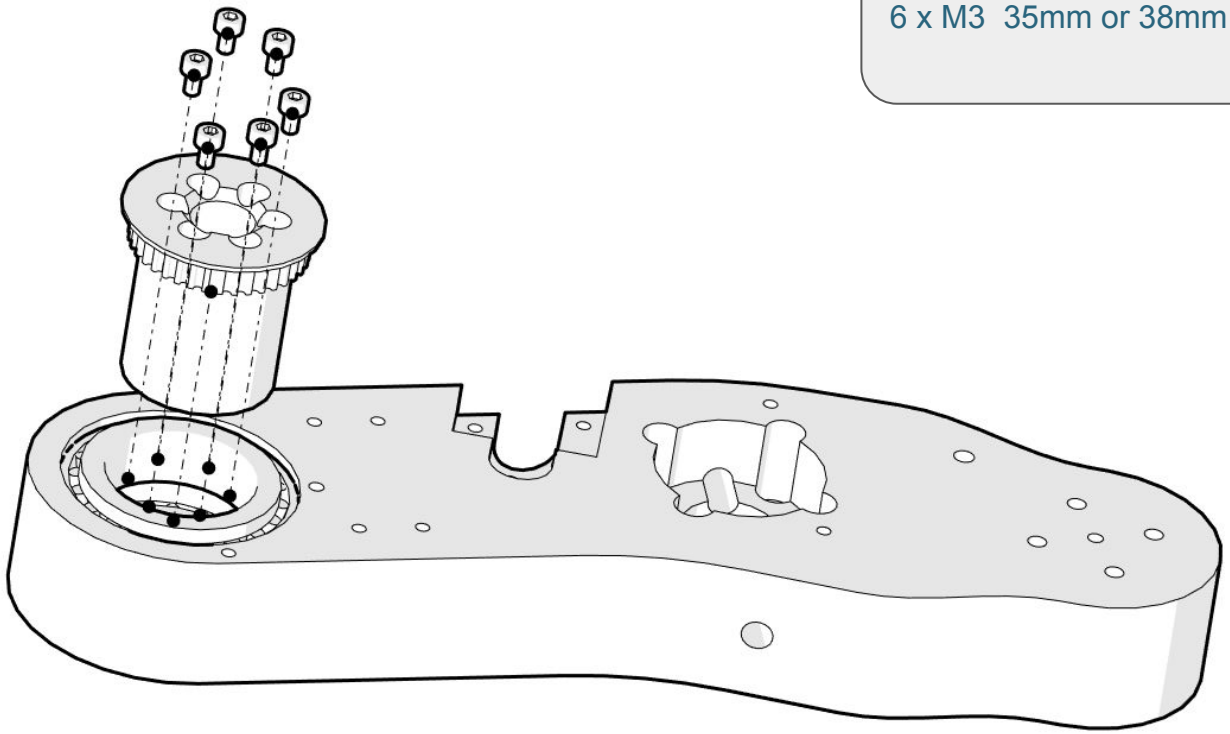


第 51 页中文译文

上臂装配（Upper arm assembly）。本页为章节图示页，请对照原 PDF 图片。

Upper arm assembly

38_pulley_J3
6 x M3 35mm or 38mm

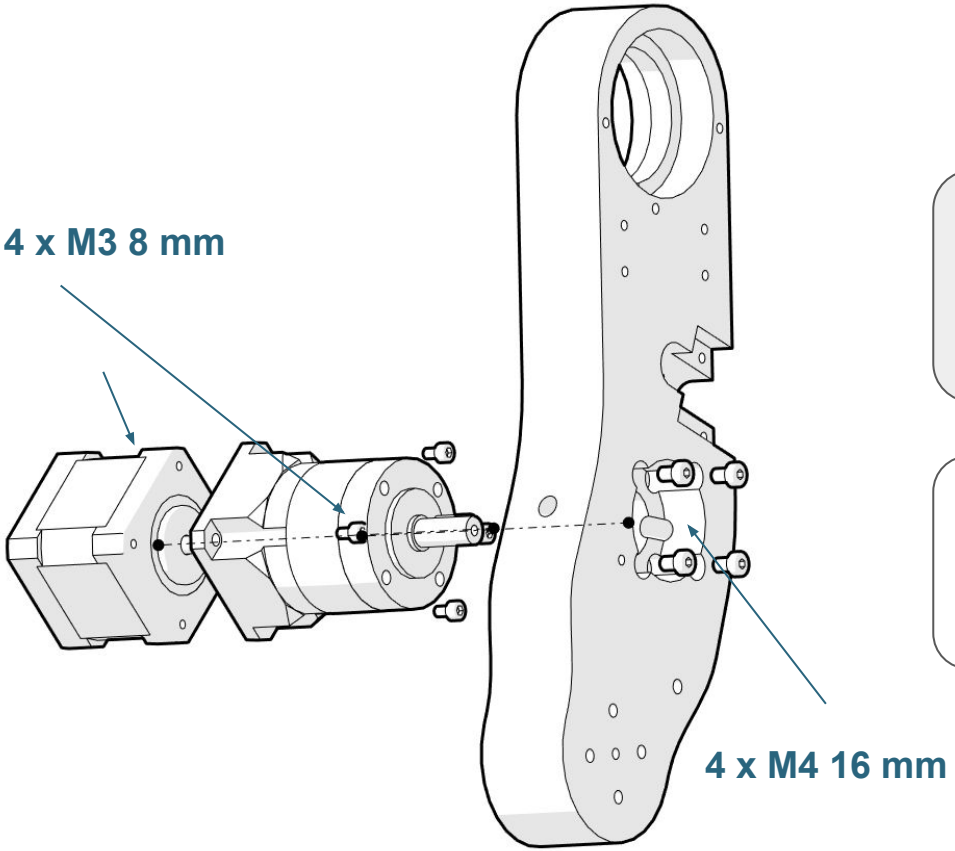


第 52 页中文译文

上臂装配（Upper arm assembly）

- 零件：38_pulley_J3
- 螺钉：M3 × 35 mm 或 M3 × 38 mm，共 6 颗。

Upper arm assembly



Stepper 2
Gearbox 20:1
4 x M3 8 mm
4 x M4 16 mm

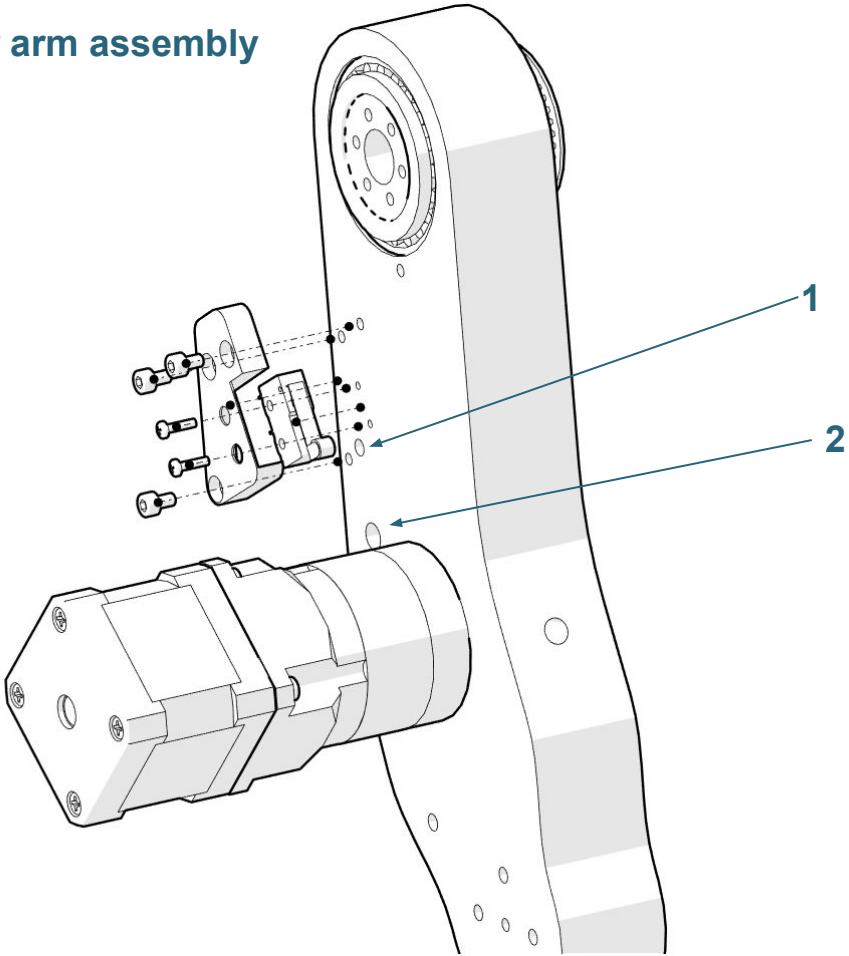
Make sure that the location of the stepper side with wires is as pointed by the arrow!

第 53 页中文译文

上臂装配（Upper arm assembly）

- 步进电机 2（Stepper 2）
- 20:1 减速器（Gearbox 20:1）
- M3 × 8 mm 螺钉 4 颗；M4 × 16 mm 螺钉 4 颗。
- 确保步进电机带电线的一侧朝向箭头所示方向。
- 原图两处均标有 M3 × 8 mm 4 颗、M4 × 16 mm 4 颗；请结合装配位置核对，避免重复计数。

Upper arm assembly



J3_limit_switch_cover
Limit switch
3x M3 8 mm
2x M2 x 10 mm

Limit switch wires go thru hole market with arrow 1. Stepper wires go thru a hole marked with 2. Finished part on next page.

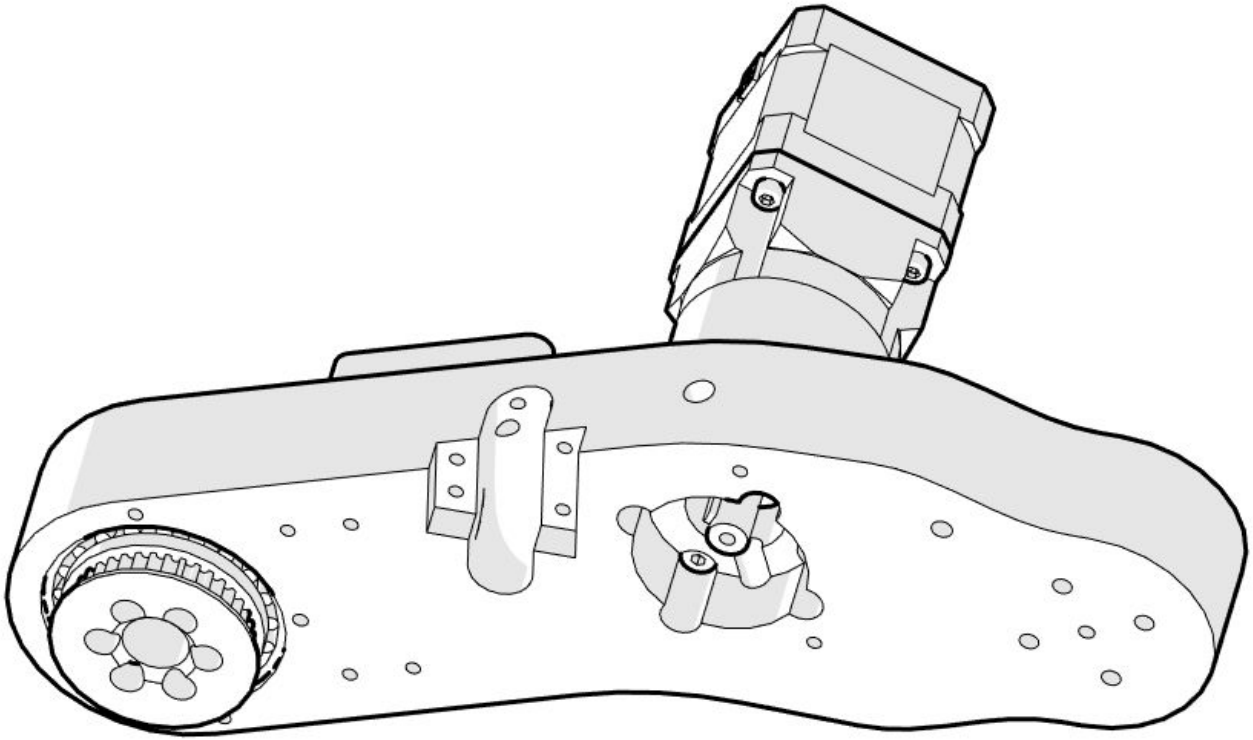
第 54 页中文译文

上臂装配（Upper arm assembly）

- 零件：J3_limit_switch_cover
- 限位开关（Limit switch）
- M3 × 8 mm 螺钉 3 颗；M2 × 10 mm 螺钉 2 颗。
- 限位开关导线穿过箭头 1 标记的孔。步进电机导线穿过标记 2 的孔。完成后的部件见下一页。

Upper arm assembly

Source Robotics

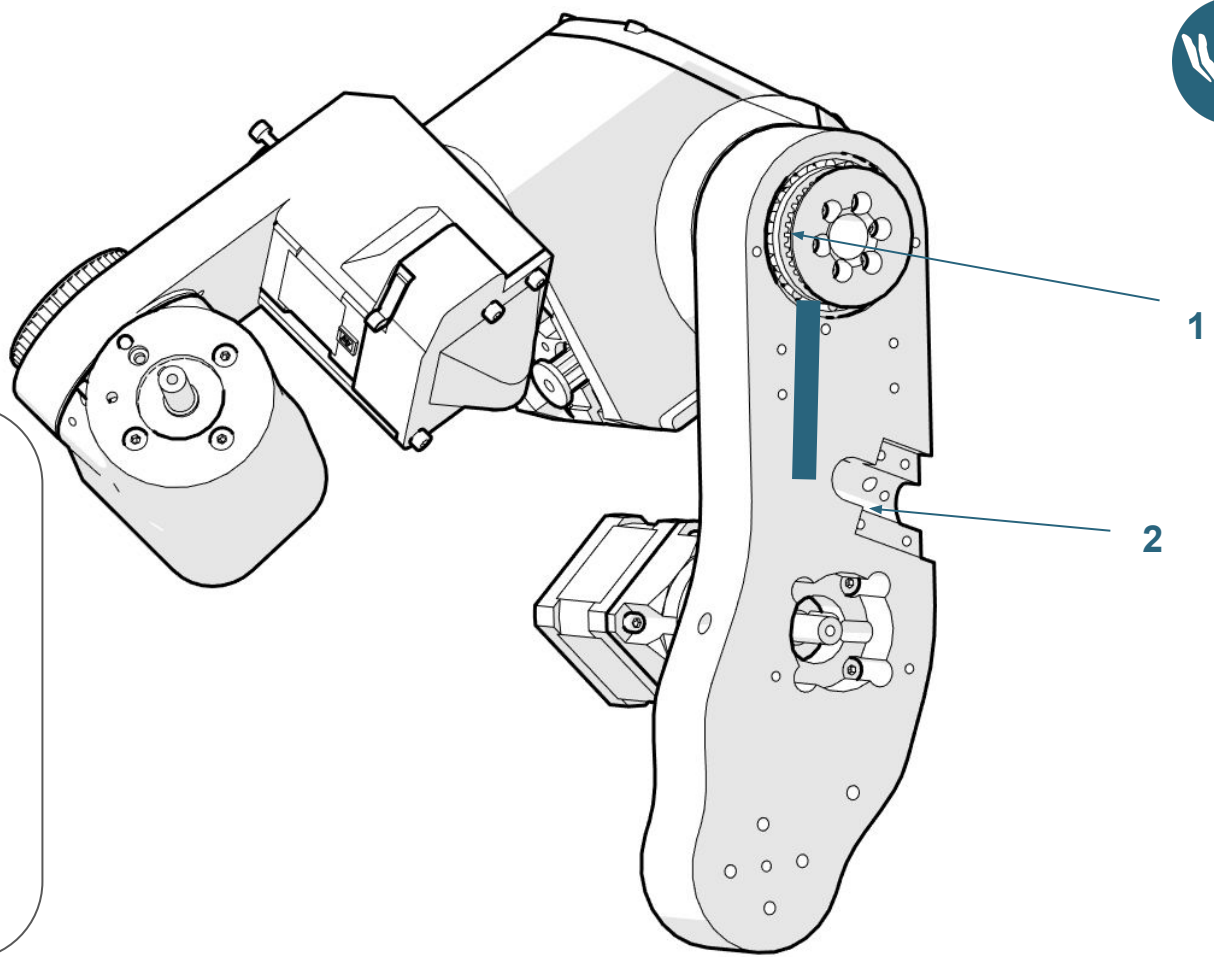


第 55 页中文译文

上臂装配（Upper arm assembly）。本页为装配结果图示，请对照原 PDF 图片。

Upper arm assembly

Source Robotics



Attach Upper arm assembly to the Forearm assembly. Wrap the wires with cloth fabric. Wires need to go thru the hole pointed by arrow 1 to the slot pointed by arrow 2. Wires need to follow the blue line.
*Note that you need to pull wires from J3 limit switch and J3 stepper thru the same slot!

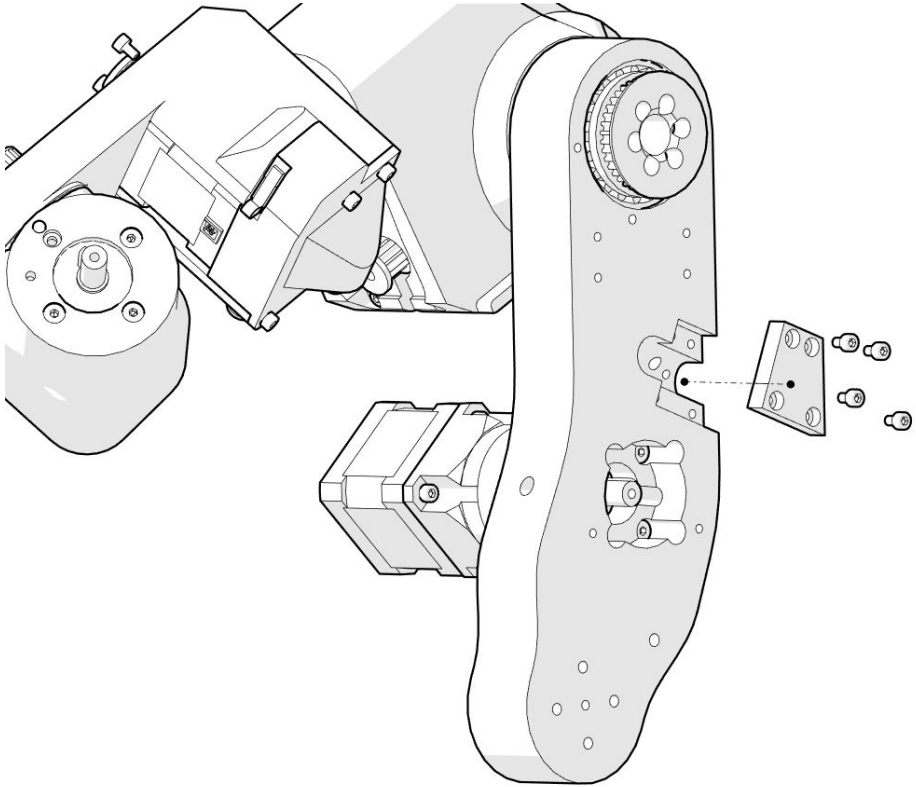
第 56 页中文译文

上臂装配（Upper arm assembly）

将上臂组件连接到前臂组件。用布基胶带包束导线。导线需从箭头 1 所指的孔穿入，再经过箭头 2 所指的槽。导线应沿蓝线所示路径布置。

注意：J3 限位开关导线与 J3 步进电机导线必须穿过同一个槽。

Upper arm assembly



Upper_arm_wires_cover
4 x M3 12 mm

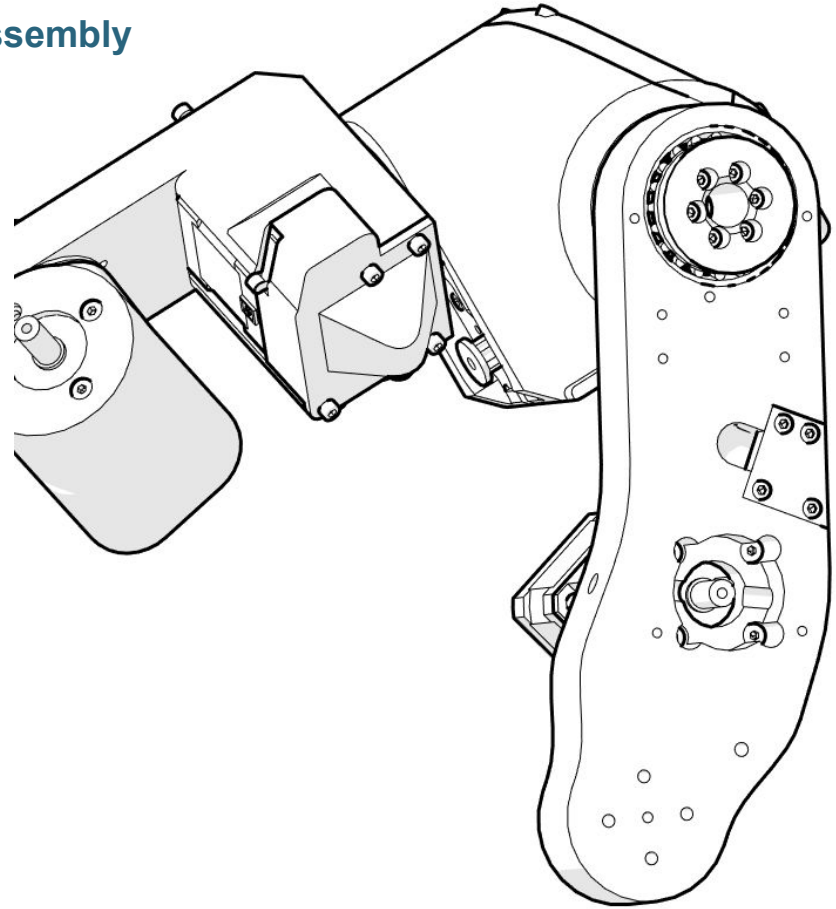
Enclose the wires by upper arm wires cover. Make sure the wires are tight.
Assembled part on next page

第 57 页中文译文

上臂装配（Upper arm assembly）

- 零件：Upper_arm_wires_cover
- M3 × 12 mm 螺钉 4 颗。
- 用上臂线缆盖板包住导线，并确保导线收紧、不松散。装配完成后的部件见下一页。

Upper arm assembly



We will attach the belt in later steps!

Once the wires are enclosed try to move joint 3.

第 58 页中文译文

上臂装配（Upper arm assembly）

导线收纳好后，尝试转动关节 3。皮带将在后续步骤中安装。

Source Robotics

Shoulder assembly

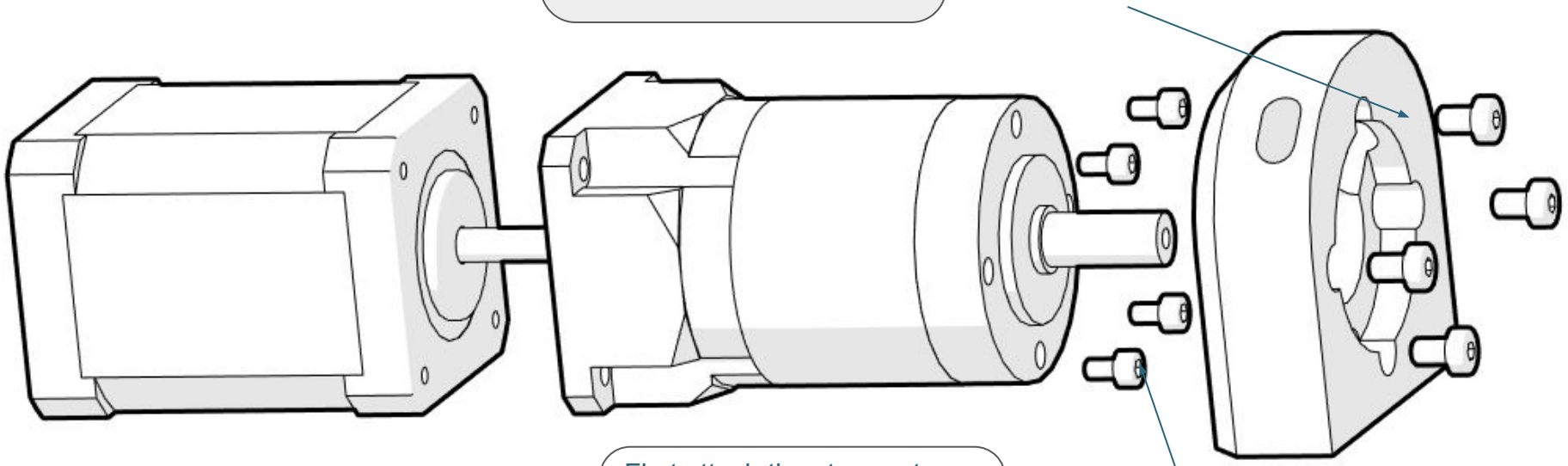
第 59 页中文译文

肩部装配（Shoulder assembly）。本页为章节图示页，请对照原 PDF 图片。

Shoulder assembly

J1_turret_motor_holder
Stepper 3
Gearbox 20:1
4 x M3 8 mm
4 x M4 16 mm

4 x M4 16 mm



First attach the stepper to the gearbox. After that attach turret holder to the gearbox. Assembled part is on the next page.

4 x M3 8 mm

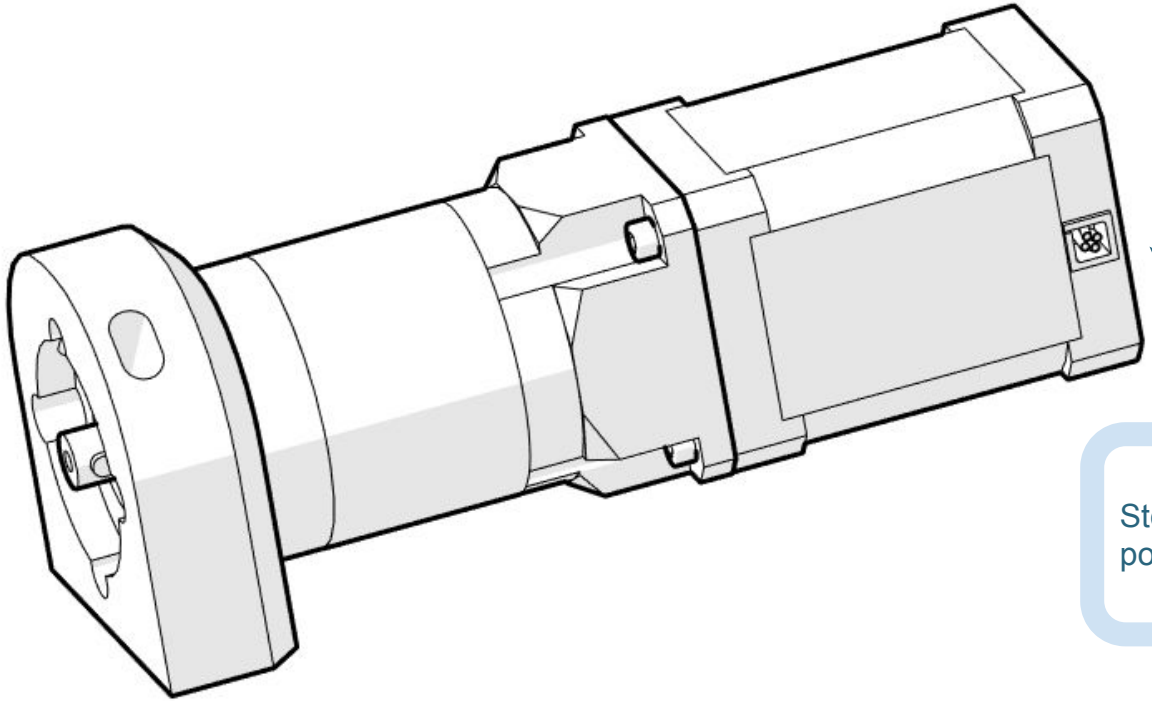
第 60 页中文译文

肩部装配（Shoulder assembly）

- 零件：J1_turret_motor_holder
- 步进电机 3（Stepper 3）
- 20:1 减速器（Gearbox 20:1）
- M3 × 8 mm 螺钉 4 颗；M4 × 16 mm 螺钉 4 颗。
- 先将步进电机安装到减速器上，再将转台电机支架安装到减速器上。装配完成后的部件见下一页。
- 原图两处均标有 M3 × 8 mm 4 颗、M4 × 16 mm 4 颗；请结合图示核对紧固件数量。

Shoulder assembly

Source Robotics



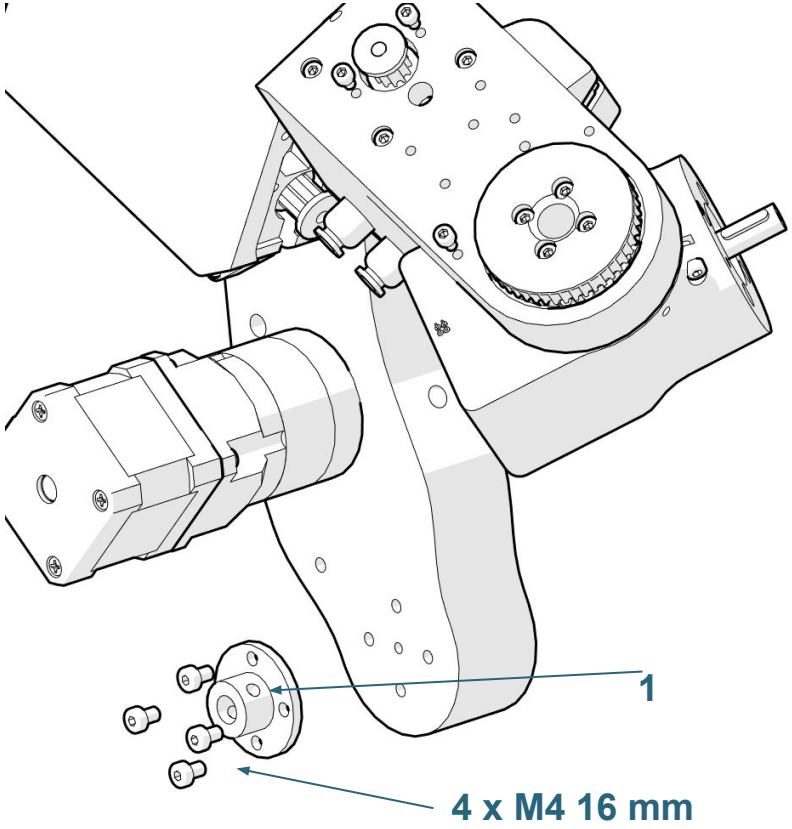
Stepper wires need to be positioned like this

第 61 页中文译文

肩部装配（Shoulder assembly）

步进电机导线应按图示方向和位置布置。请对照原 PDF 图片。

Shoulder assembly



Shaft coupler
4 x M4 16 mm

Note the position of the hole marked with 1. It is important for it to be positioned this way for easier screw tightening.

第 62 页中文译文

肩部装配（Shoulder assembly）

- 法兰联轴器（Shaft coupler）
- M4 × 16 mm 螺钉 4 颗（原图标注两处）。
- 注意标记 1 的孔位。该孔按图示方向定位，便于后续拧紧螺钉。

Source Robotics

shaft couplers need to be aligned with the hole, you must use blue threadlocker and it is best to let it cure for few hours



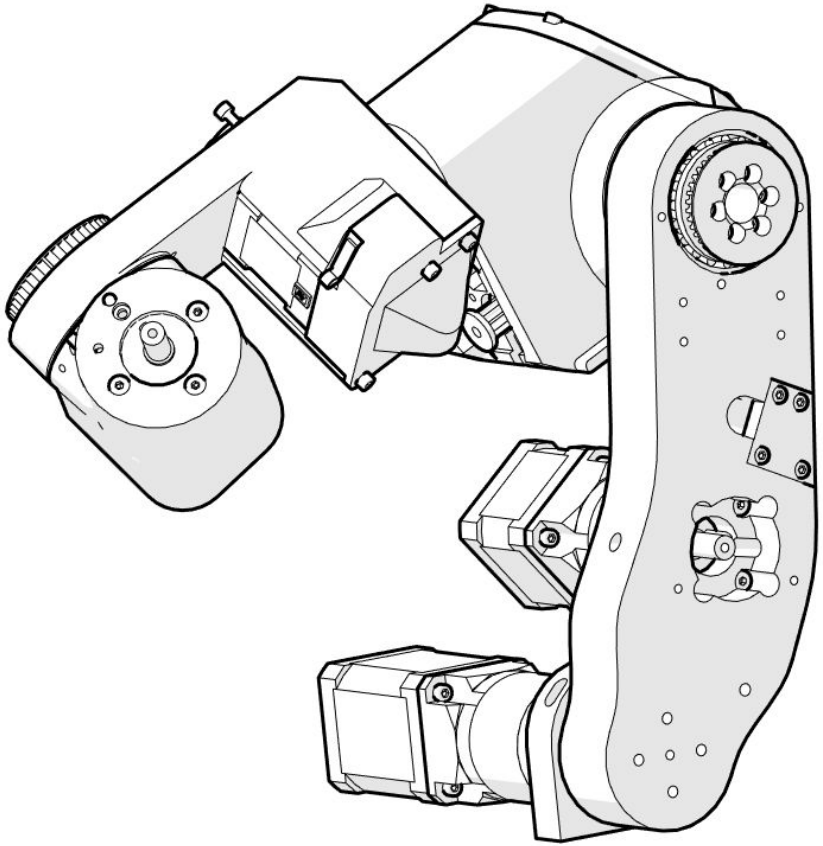


第 63 页中文译文

肩部装配（Shoulder assembly）

法兰联轴器应与孔对齐。此处需使用蓝色螺纹锁固剂（blue threadlocker），最好静置数小时使其固化。

Shoulder assembly



Gearbox has a hole (keyway) where key should be going. Since we are not using keys make sure that the screw of shaft coupler goes in the keyway! Use threadlocker here!

第 64 页中文译文

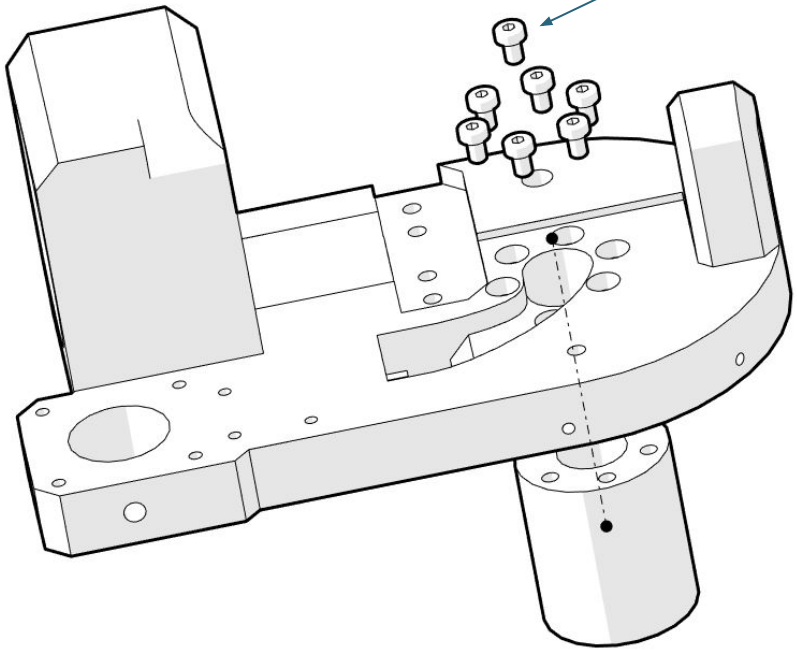
肩部装配（Shoulder assembly）

减速器有一个供键使用的槽（键槽，keyway）。由于本设计不安装键，必须确保法兰联轴器的紧定螺钉拧入该键槽位置。此处使用螺纹锁固剂。

Shoulder assembly

Source Robotics

M4 14 mm - used for homing J1



J1_turret
J1_rotation_shaft
6 x M4 16 mm
1 x M4 14 mm

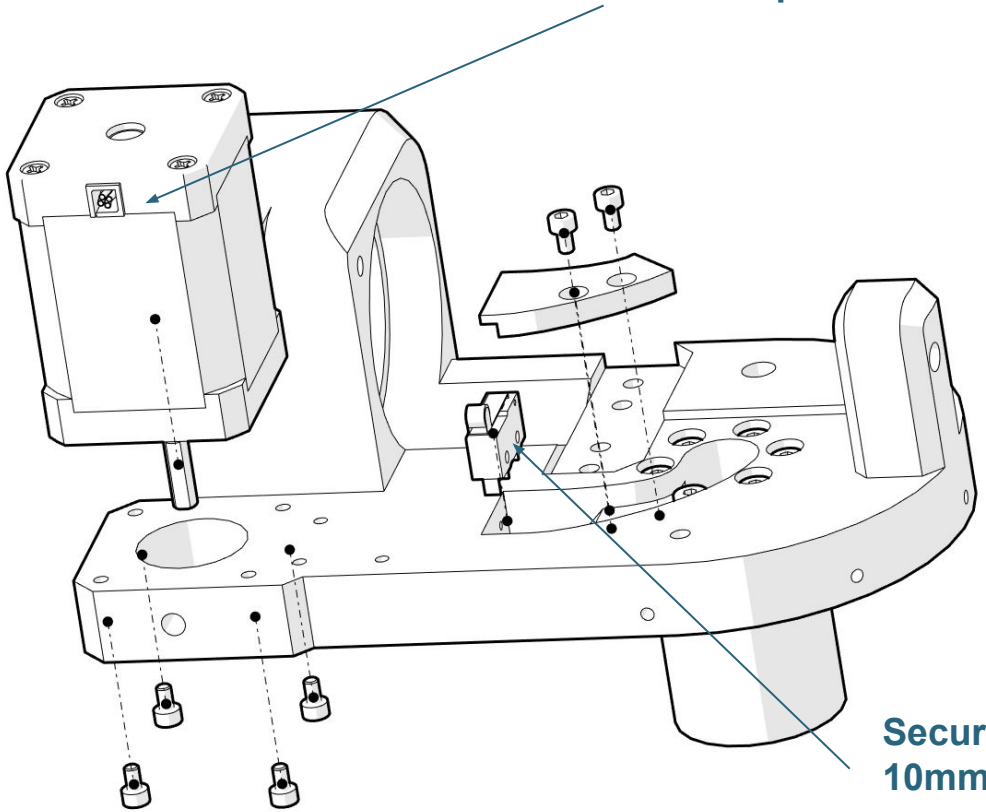
M4 14mm screw is used to home J1. It needs to be peeking out a little on the bottom side after you screw it in. Do not screw it in all the way!

第 65 页中文译文

肩部装配（Shoulder assembly）

- 零件：J1_turret、J1_rotation_shaft
- M4 × 16 mm 螺钉 6 颗；M4 × 14 mm 螺钉 1 颗。
- M4 × 14 mm 螺钉用于 J1 归零。拧入后，螺钉应在底侧略微突出；不要将螺钉完全拧到底。
- 原图注明：M4 × 14 mm 用于 J1 归零。

Shoulder assembly



Stepper 3
Limit switch
J2_limit_switch_cover
2 x M3 8 mm
4 x M3 14 mm
2 x M2 10 mm

Before placing the lid attach limit switch with 2 M2 screws. After that pull its wires thru J1 rotation shaft.

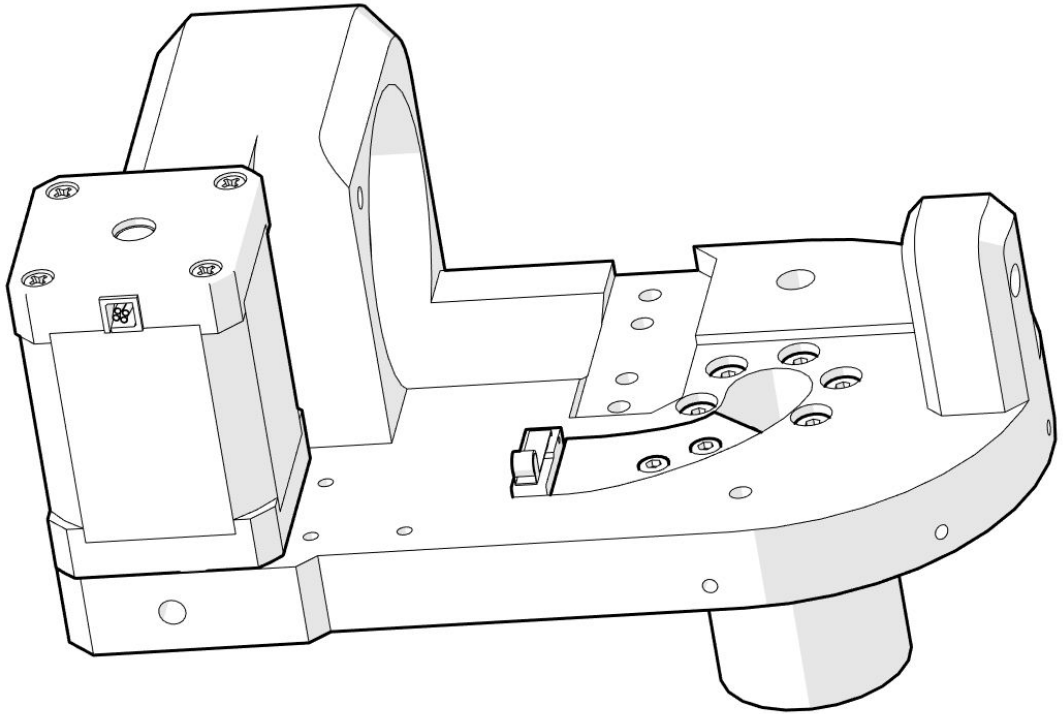
第 66 页中文译文

肩部装配（Shoulder assembly）

- 步进电机 3（Stepper 3）
- 限位开关（Limit switch）
- 零件：J2_limit_switch_cover
- M3 × 8 mm 螺钉 2 颗；M3 × 14 mm 螺钉 4 颗；M2 × 10 mm 螺钉 2 颗。
- 安装盖板前，先用 2 颗 M2 螺钉固定限位开关。随后将其导线穿过 J1_rotation_shaft。注意导线位置。
- 用 2 颗 M2 × 10 mm 螺钉固定限位开关。

Shoulder assembly

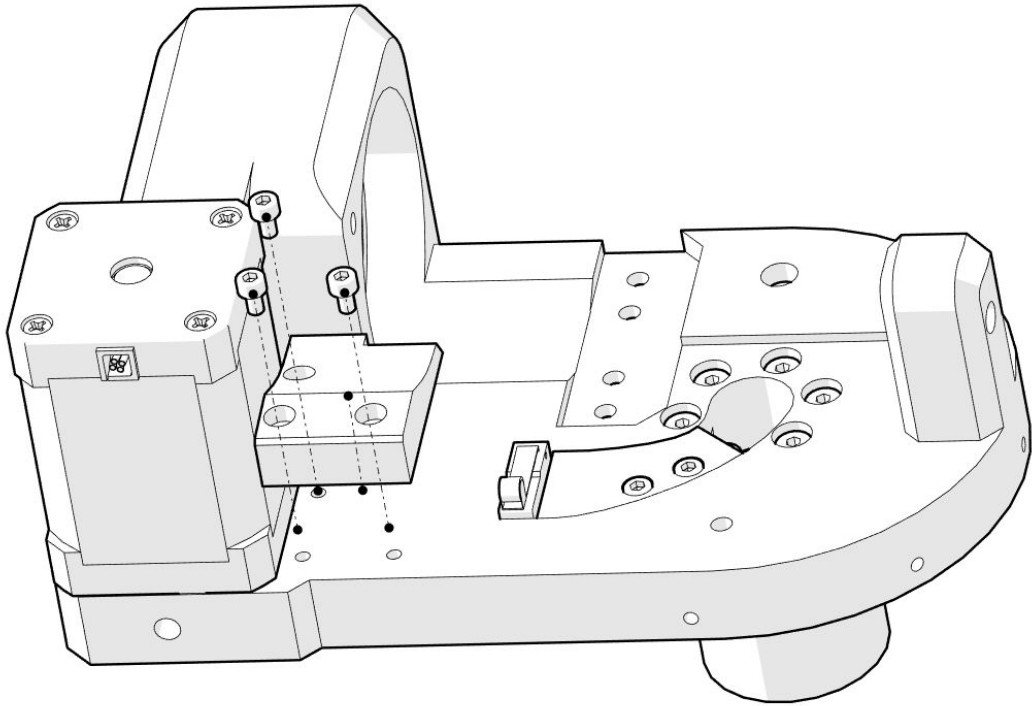
Source Robotics



第 67 页中文译文

肩部装配（Shoulder assembly）。本页为图示页，请对照原 PDF 图片。

Shoulder assembly



J2_stopper_block
1 x M3 8 mm
2 x M3 12 mm

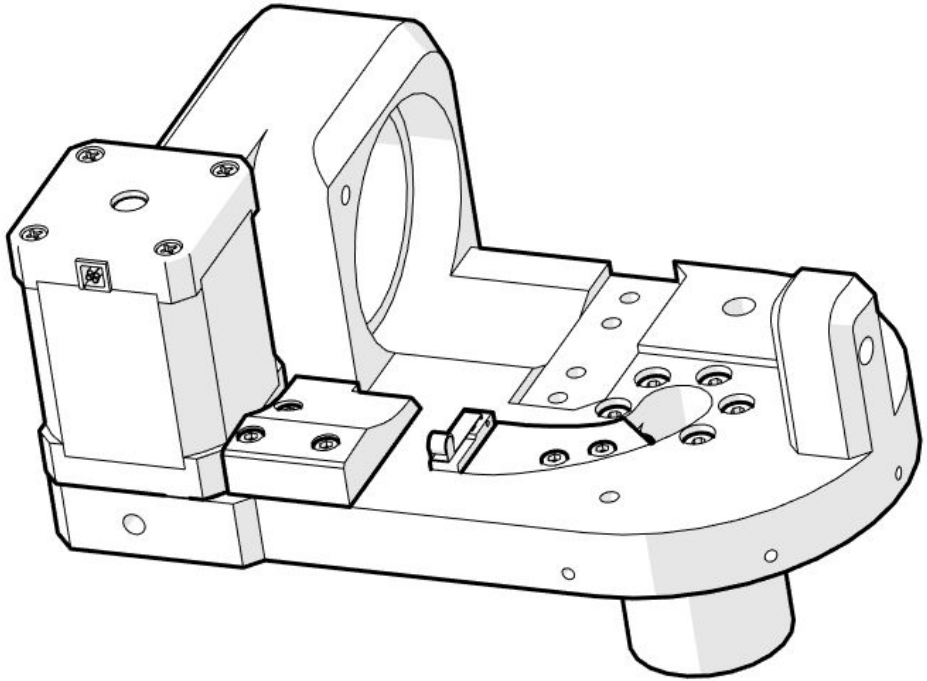
第 68 页中文译文

肩部装配（Shoulder assembly）

- 零件：J2_stopper_block
- M3 × 8 mm 螺钉 1 颗；M3 × 12 mm 螺钉 2 颗。

Shoulder assembly

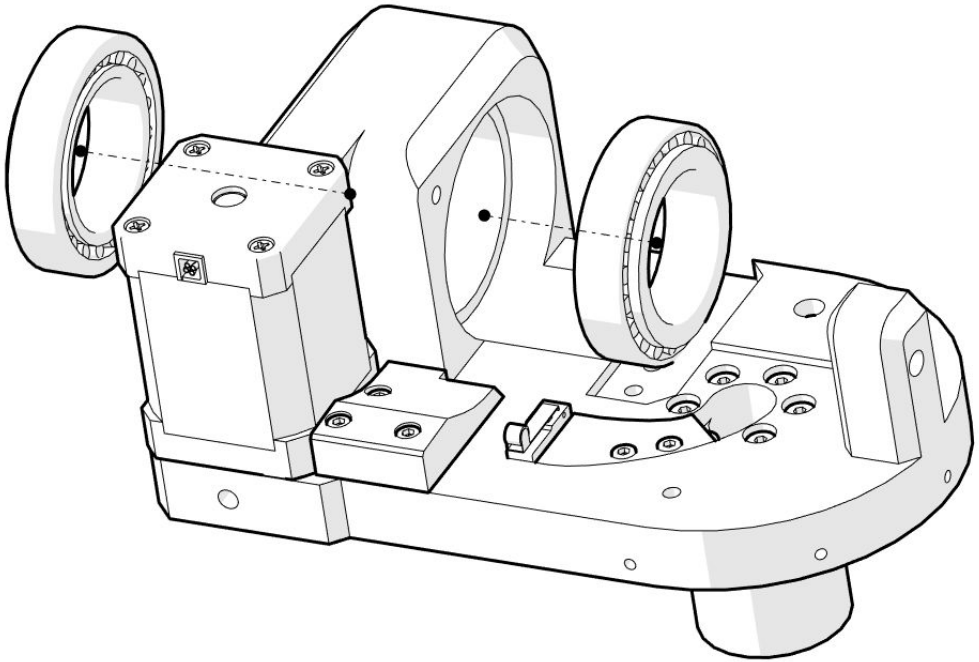
Source Robotics



第 69 页中文译文

肩部装配（Shoulder assembly）。本页为图示页，请对照原 PDF 图片。

Shoulder assembly



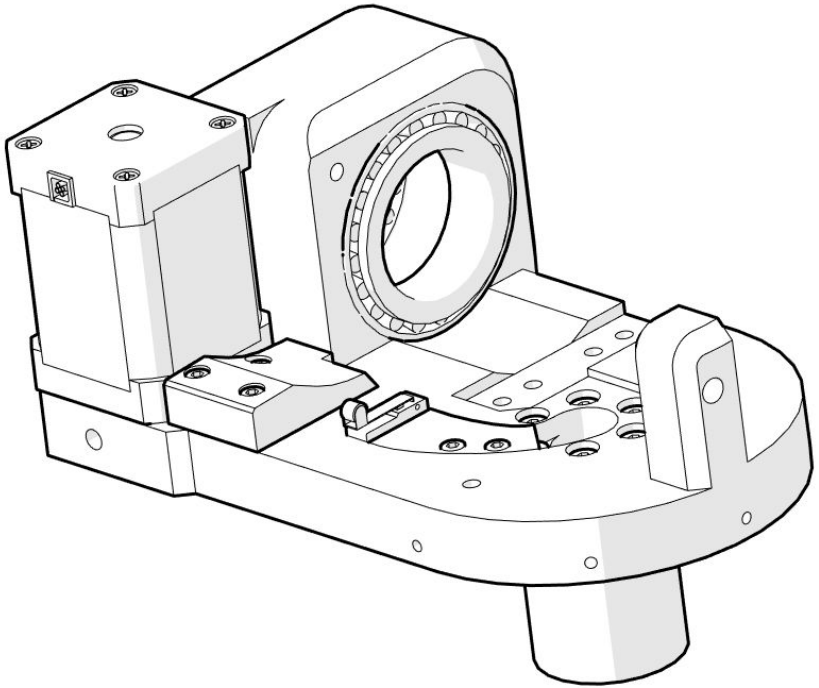
2 x Bearing 3

第 70 页中文译文

肩部装配（Shoulder assembly）

- Bearing 3（轴承 3）× 2。

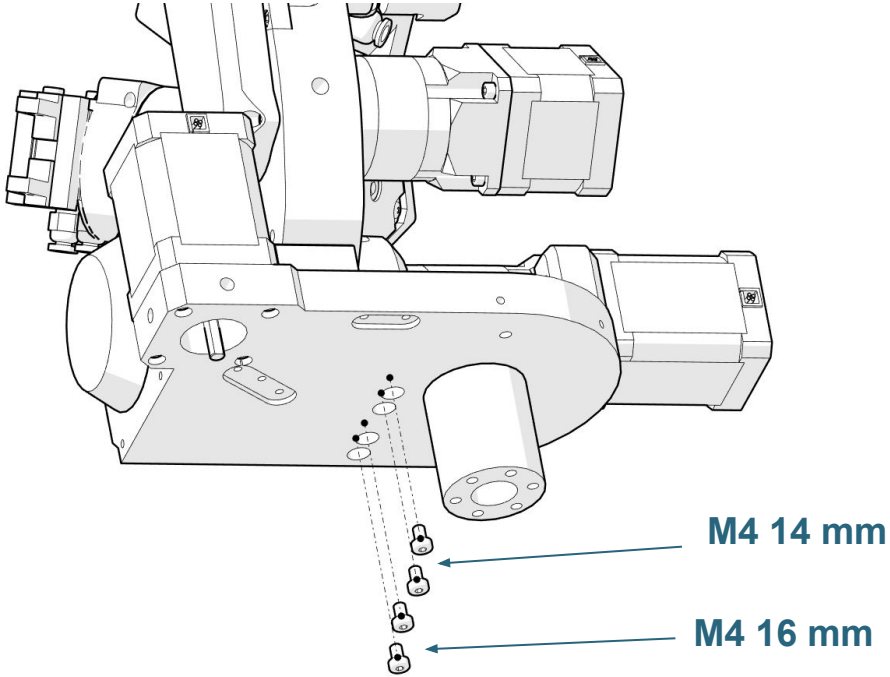
Shoulder assembly



第 71 页中文译文

肩部装配（Shoulder assembly）。本页为图示页，请对照原 PDF 图片。

Shoulder assembly



2 x M4 16 mm
2 x M4 14 mm

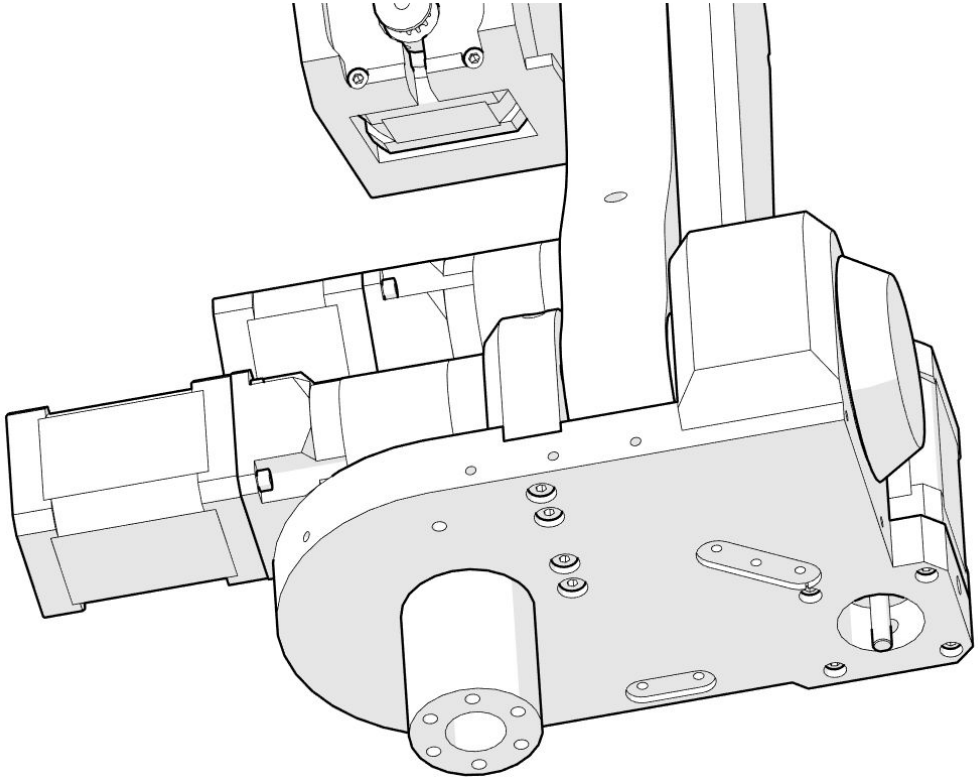
J1_turret is modified in new versions to have guide rails for stopper for J1.

第 72 页中文译文

肩部装配（Shoulder assembly）

- M4 × 16 mm 螺钉 2 颗；M4 × 14 mm 螺钉 2 颗。
- 新版本的 J1_turret 已作修改，加入了用于 J1 限位块（stopper）的导轨。
- 原图另标出 M4 × 14 mm、M4 × 16 mm；请结合图示确认每种螺钉的安装位置。

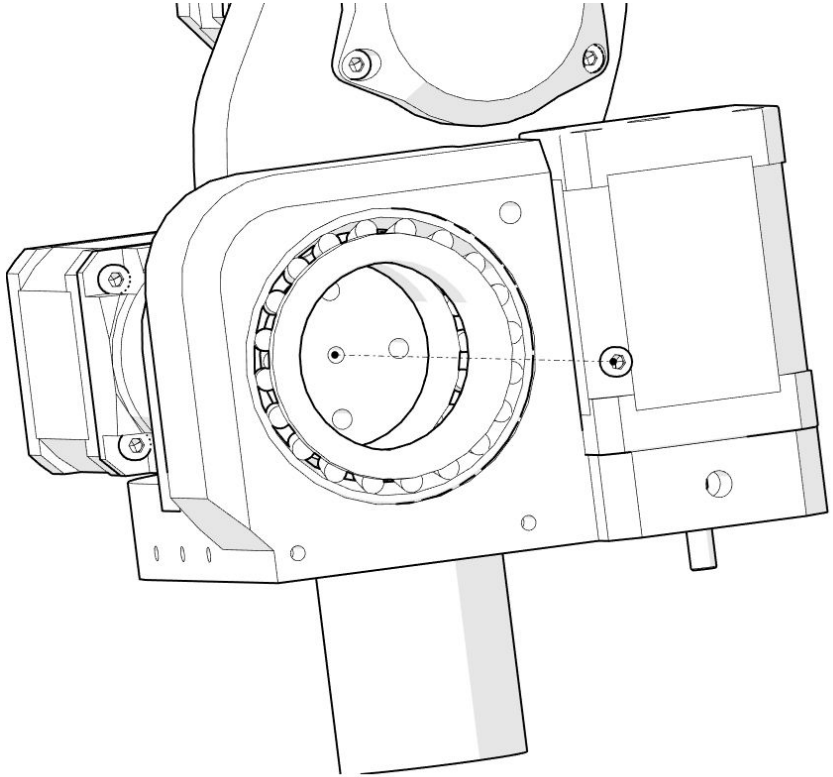
Shoulder assembly



第 73 页中文译文

肩部装配（Shoulder assembly）。本页为图示页，请对照原 PDF 图片。

Shoulder assembly



1 x M3 ?

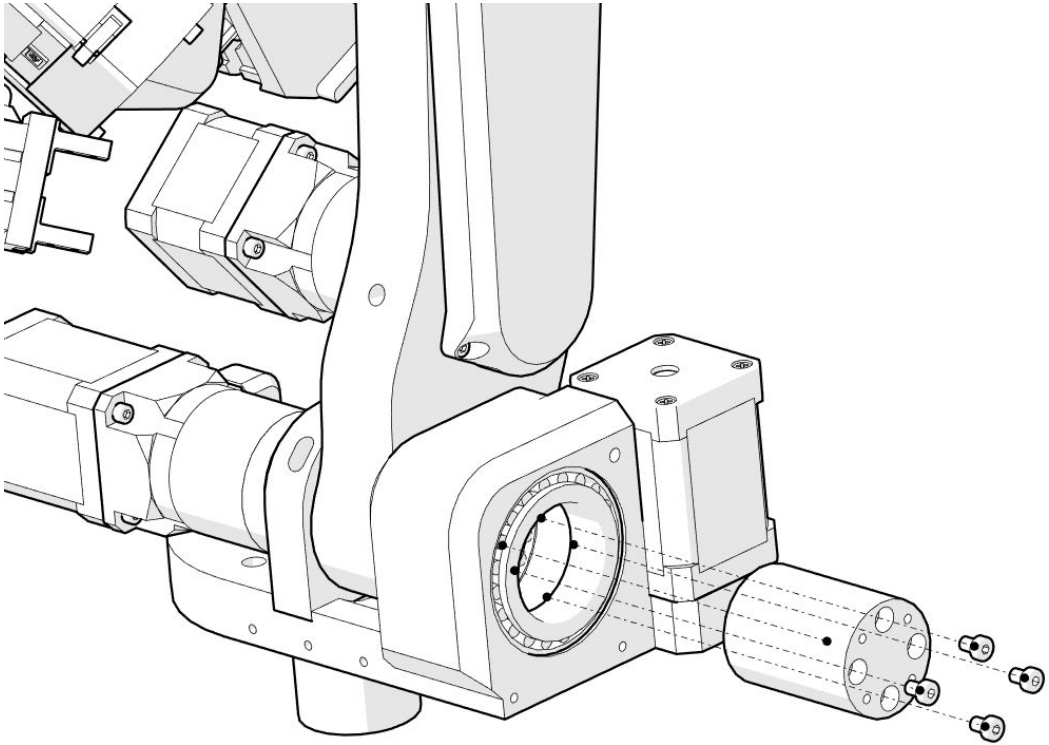
This step is not necessary!

第 74 页中文译文

肩部装配（Shoulder assembly）

- M3 螺钉 1 颗，长度原文标为“？”（1 x M3 ?），长度未注明。
- 此步骤并非必需。

Shoulder assembly



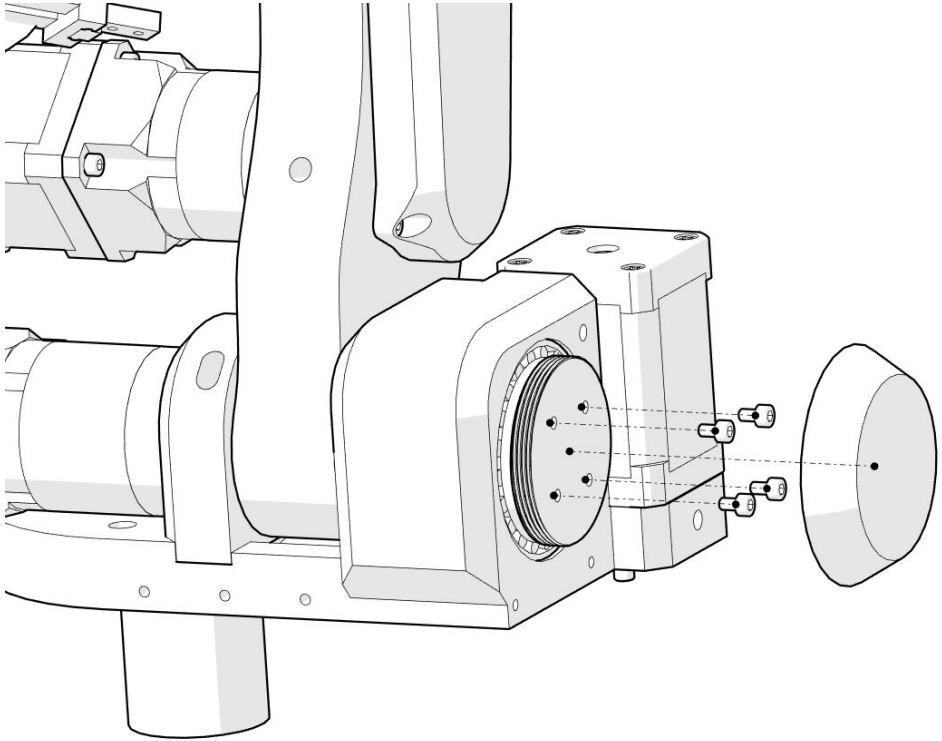
J2_shaft
4 x M4 x 50 mm

第 75 页中文译文

肩部装配（Shoulder assembly）

- 零件：J2_shaft
- M4 × 50 mm 螺钉 4 颗。

Shoulder assembly



J2_bearing_backplate
J2_cover
4 x M3 12 mm

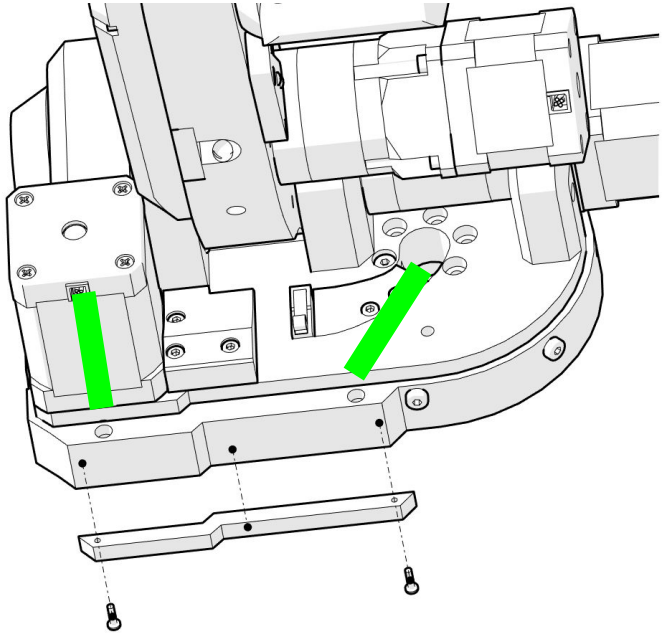
J2_cover and bearing_backplate have threads.

第 76 页中文译文

肩部装配（Shoulder assembly）

- 零件：J2_bearing_backplate、J2_cover
- 使用 4 颗 M3 × 12 mm 螺钉。
- J2_cover 和 bearing_backplate 都带有螺纹。

Shoulder assembly



J1_belt_cover
J1_wires_cover
M2 x 10 mm

Pull the wires from stepper for J1 thru the holes on the J1_belt_cover. After that pull the wires thru the J1_rotation shaft!
J1_belt_cover does not get screwed in on this stage!
Attach wires cover with 2 M2 screws.

第 77 页中文译文

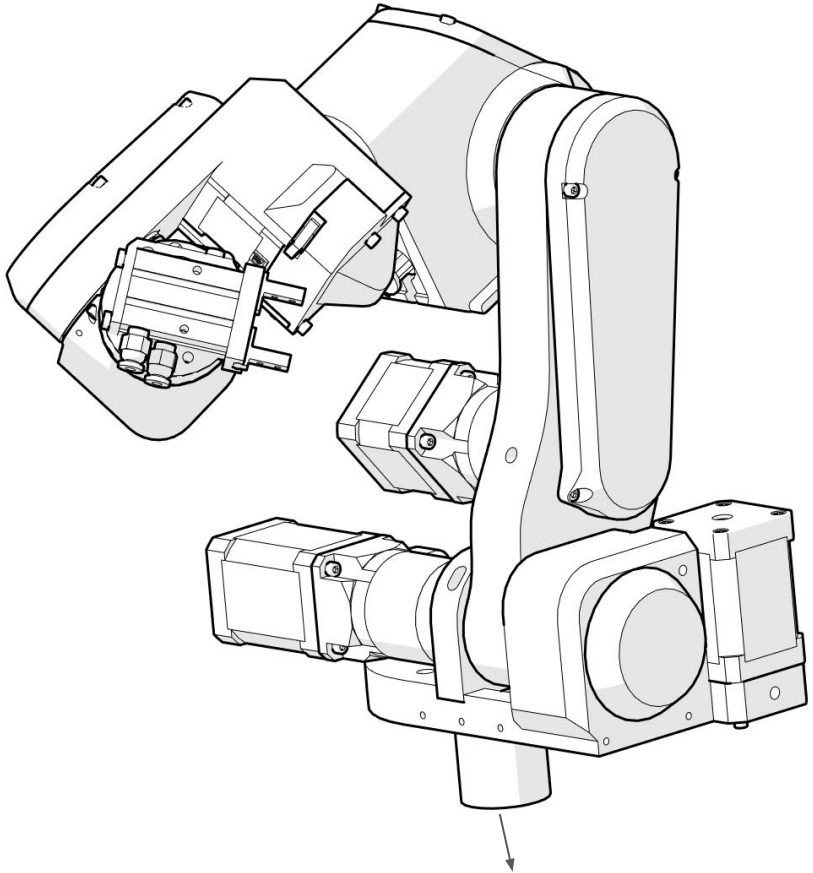
肩部装配（Shoulder assembly）

- 零件：J1_belt_cover、J1_wires_cover
- 使用 M2 × 10 mm 螺钉。
- 将 J1 步进电机的导线穿过 J1_belt_cover 上的孔，再穿过 J1_rotation_shaft。

> 此阶段不要把 J1_belt_cover 拧紧固定！

- 用 2 颗 M2 螺钉固定线缆盖（wires cover）。

Shoulder assembly



Now pull all the rest of the wires thru the J1 rotation shaft!

第 78 页中文译文

肩部装配（Shoulder assembly）

- 将其余所有导线继续穿过 J1_rotation_shaft。

Source Robotics

Base assembly

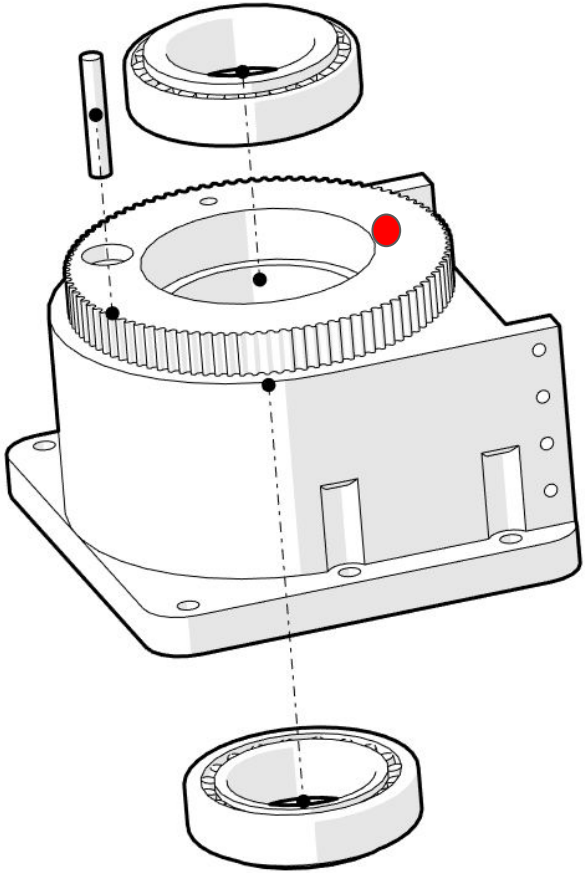
第 79 页中文译文

底座装配（Base assembly）

Base assembly

Source Robotics

Main_base
2 x bearing 3
Sensor 3
1 x M3 25 mm



Parol6 robotic arm needs homing since it uses open loop-stepper drivers. Joints 1,4 and 6 are homed with inductive sensors. Homing is done with small screws that trigger the sensor when it gets close to it. You need to find “sweet” spot where the sensors trigger.

Sensor 3 needs to be in level with the main base. Screw it in with provided nuts, one from each side. In the hole marked with red attach M3 20mm screw all the way. It is used as a rotation limiter for J1.
IF YOU DON'T HAVE ROTATION BLOCKER IN PLACE YOU RISK DAMAGING YOUR ROBOT BY DOING MORE THAN 1 ROTATION IN THE BASE OF THE ROBOT!

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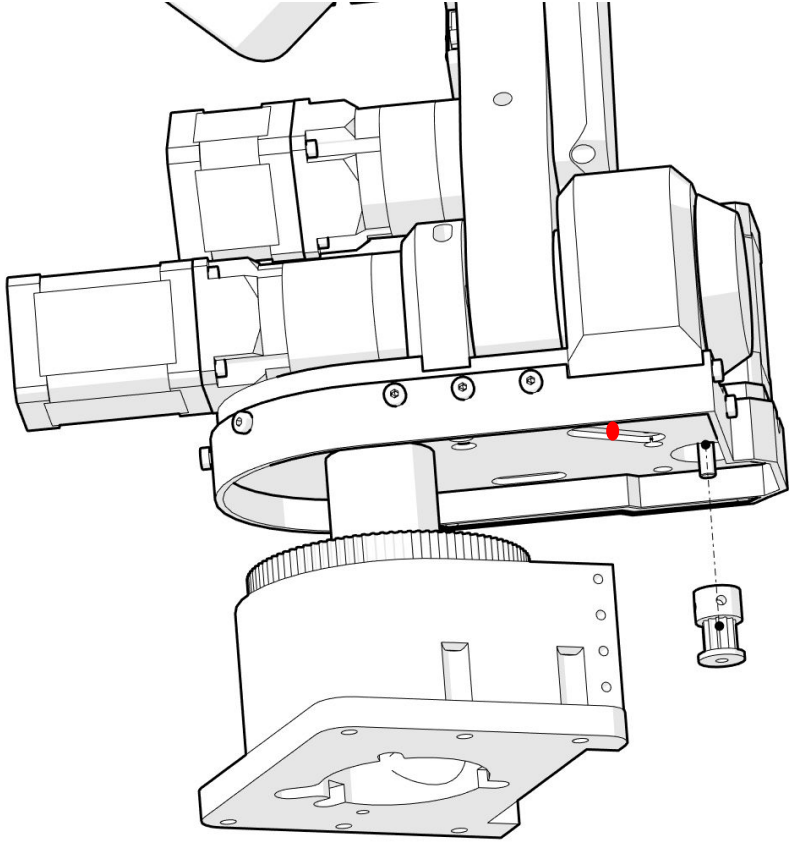
底座装配（Base assembly）

- 零件：Main_base、2 个 3 号轴承（bearing 3）、传感器 3（Sensor 3）。
- 使用 1 颗 M3 × 25 mm 螺钉。
- 传感器 3 必须与主底座表面齐平。使用随附螺母从两侧各固定一个。
- 在红色标记的孔中将 M3 × 20 mm 螺钉拧到底。它用作 J1 的旋转限位。

如果没有安装旋转限位块，底座中的 J1 转动超过一圈可能损坏机械臂！

- PAROL6 使用开环步进驱动器，需要执行回零。J1、J4、J6 使用电感式传感器回零。回零由小螺钉靠近传感器并触发完成。需要找到能可靠触发传感器的合适位置（sweet spot）。

Base assembly



J1 pulley
J1 belt

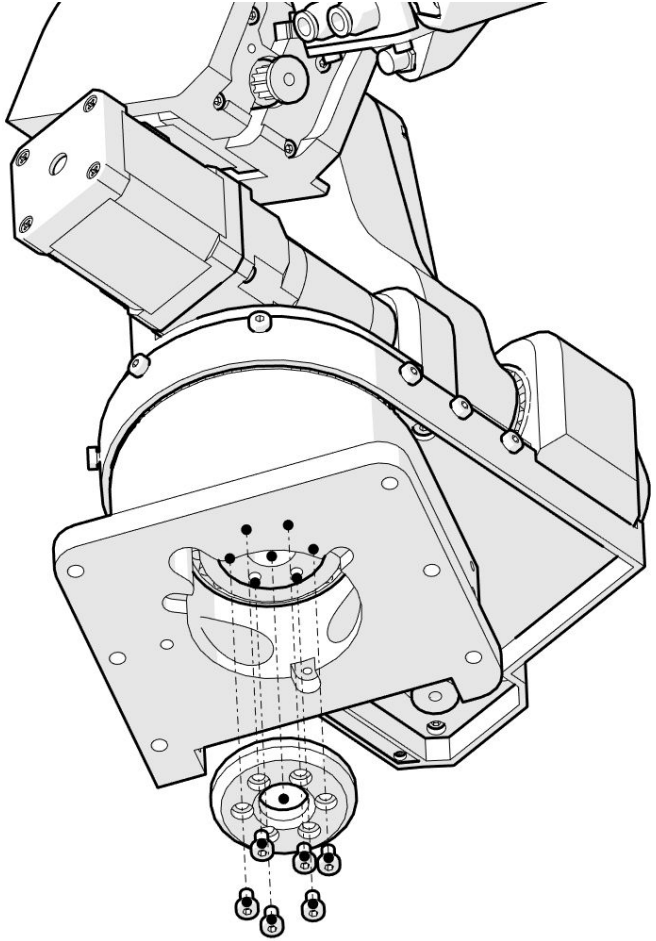
Place J1 belt in position. Pull all the wires thru the base part (except J1 sensor) . After that slowly put the base all the way up to the J1 rotation shaft. Belt at this stage needs to be around base. After that you can screw in the covers and attach the pulley. Pulley has set screw holes on its side same as the cover part. If needed add belt tension bearing on the red spot.

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底座装配（Base assembly）

- 零件：J1 带轮（J1 pulley）、J1 皮带（J1 belt）。
- 将 J1 皮带放到位。除了 J1 传感器的线以外，将所有导线穿过底座部件；随后缓慢地将底座向上套到 J1_rotation_shaft 上。
- 此时皮带应绕在底座周围。之后可以拧上盖板并安装带轮。带轮侧面有紧定螺钉孔，盖件上也有对应孔。
- 如有需要，在红色标记处增加皮带张紧轴承。

Base assembly



J1_backplate
6 x M4 16 mm

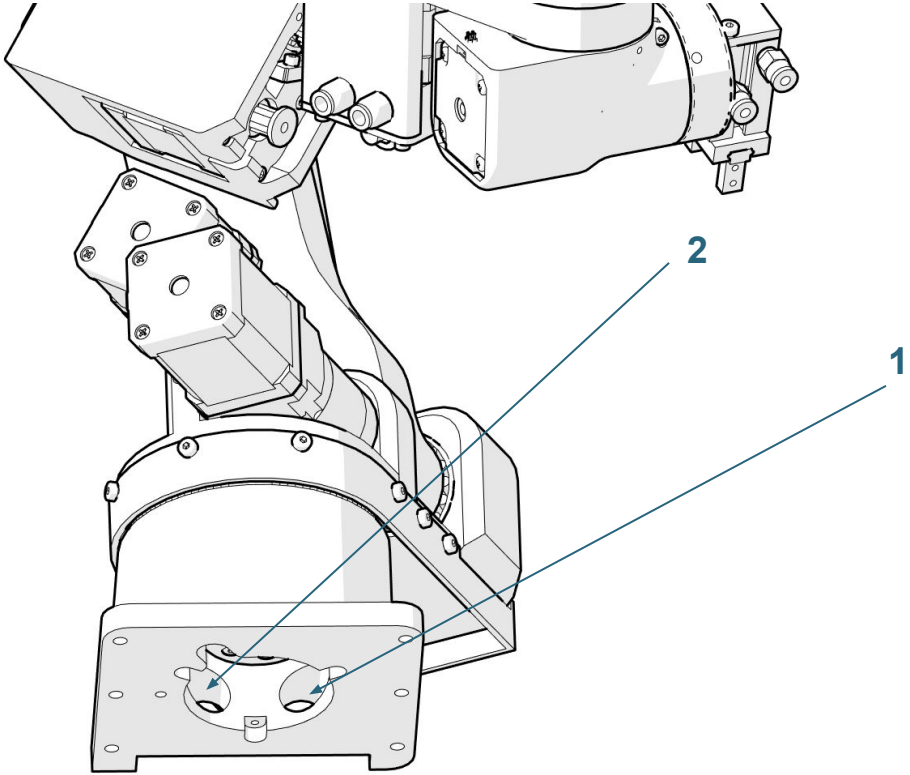
Pull all the wires (Except J1 sensor) thru the J1 backplate.

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底座装配（Base assembly）

- 零件：J1_backplate。
- 使用 6 颗 M4 × 16 mm 螺钉。
- 将除 J1 传感器线以外的所有导线穿过 J1_backplate。

Base assembly



All robot wires need to go thru hole 1 and 2. To keep wiring mess to a minimum and reduce strain on the wires use this guide:

Hole 1:

J1,J3,J5 Steppers
Limit 2,1,5

Hole 2:

J2,J4,J6 Steppers
Limit 6,3,4, CAN
Pneumatic tubes

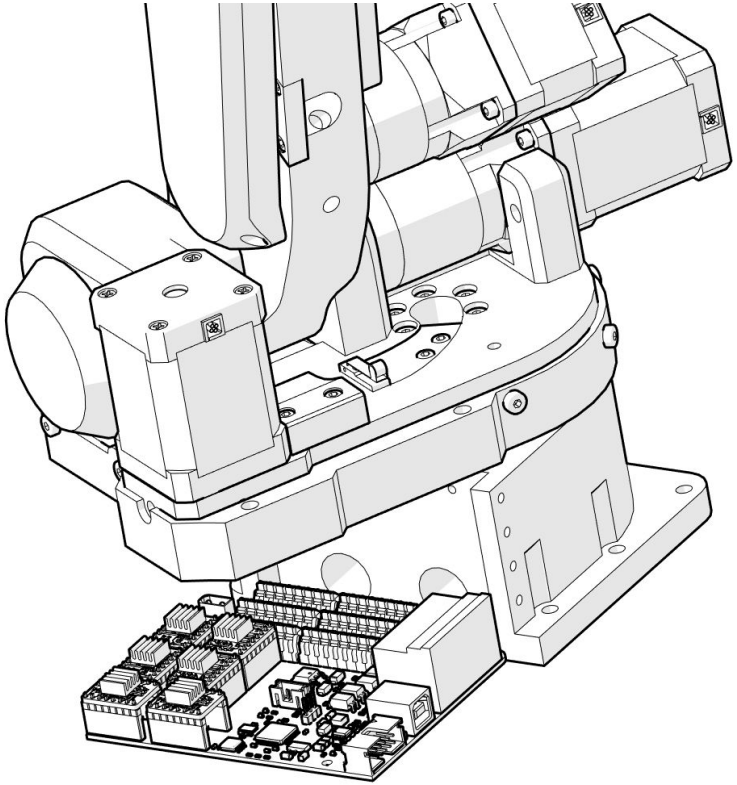
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底座装配（Base assembly）

所有机械臂线缆都必须穿过孔 1 和孔 2。为减少线束杂乱并减轻线缆受力，请按以下方式布线：

- 孔 1：J1、J3、J5 步进电机线；限位开关 2、1、5 的线。
- 孔 2：J2、J4、J6 步进电机线；限位开关 6、3、4 的线；CAN 线；气动管。

Base assembly



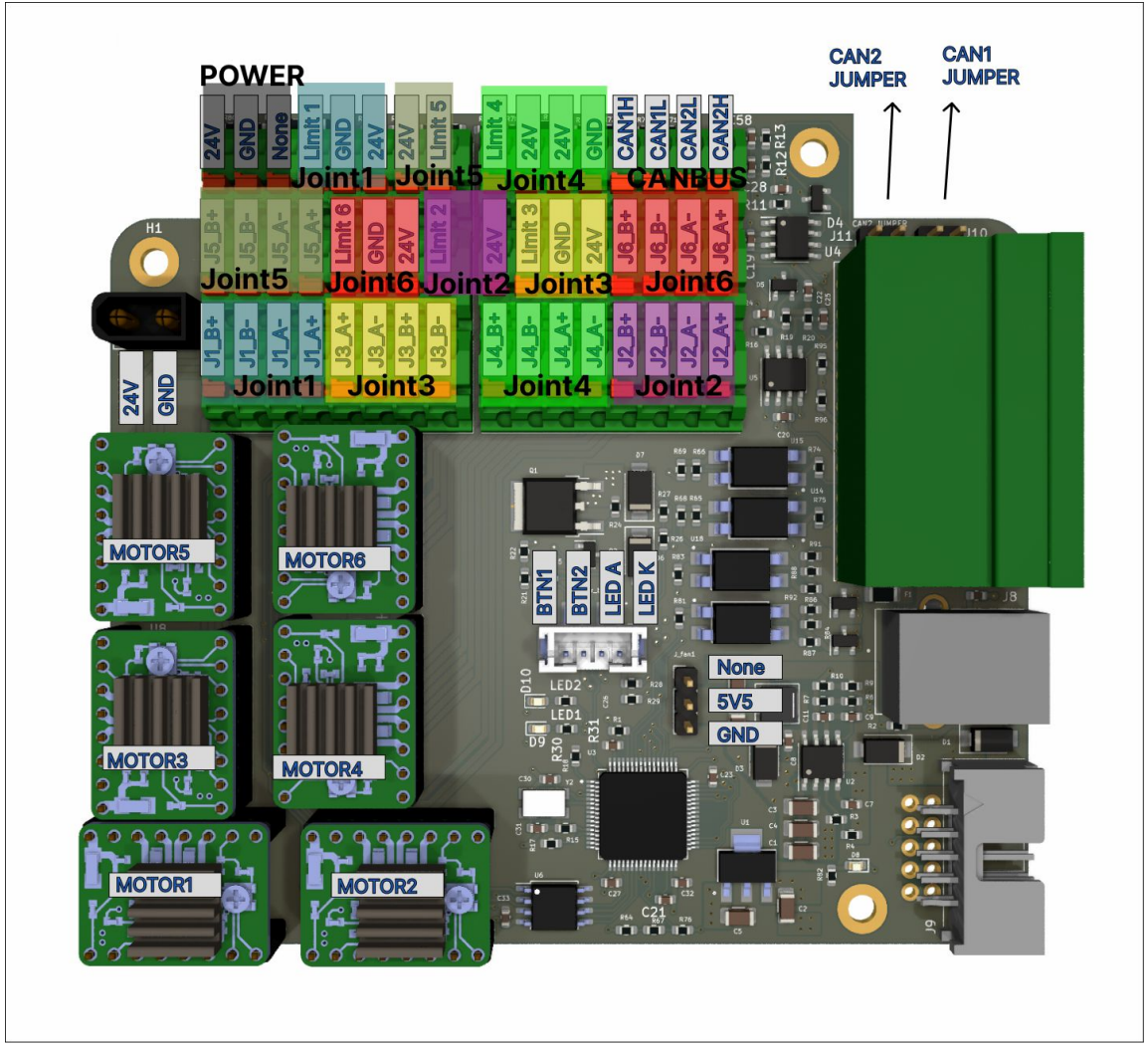
PAROL6 control board
You can buy PAROL6 control board at:
https://source-robotics.com/products/parol6-control-board
Or send us an email at
info@source-robotics.com

Position the board like shown in the picture. Wire the robot by following these wiring instructions or ones in the docs!

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底座装配（Base assembly）

- PAROL6 主控板（control board）。原手册提供购买链接：<https://source-robotics.com/products/parol6-control-board>，也可联系 info@source-robotics.com。
- 按图示方向放置控制板。按照本手册或项目文档中的接线说明连接机械臂。



Source Robotics

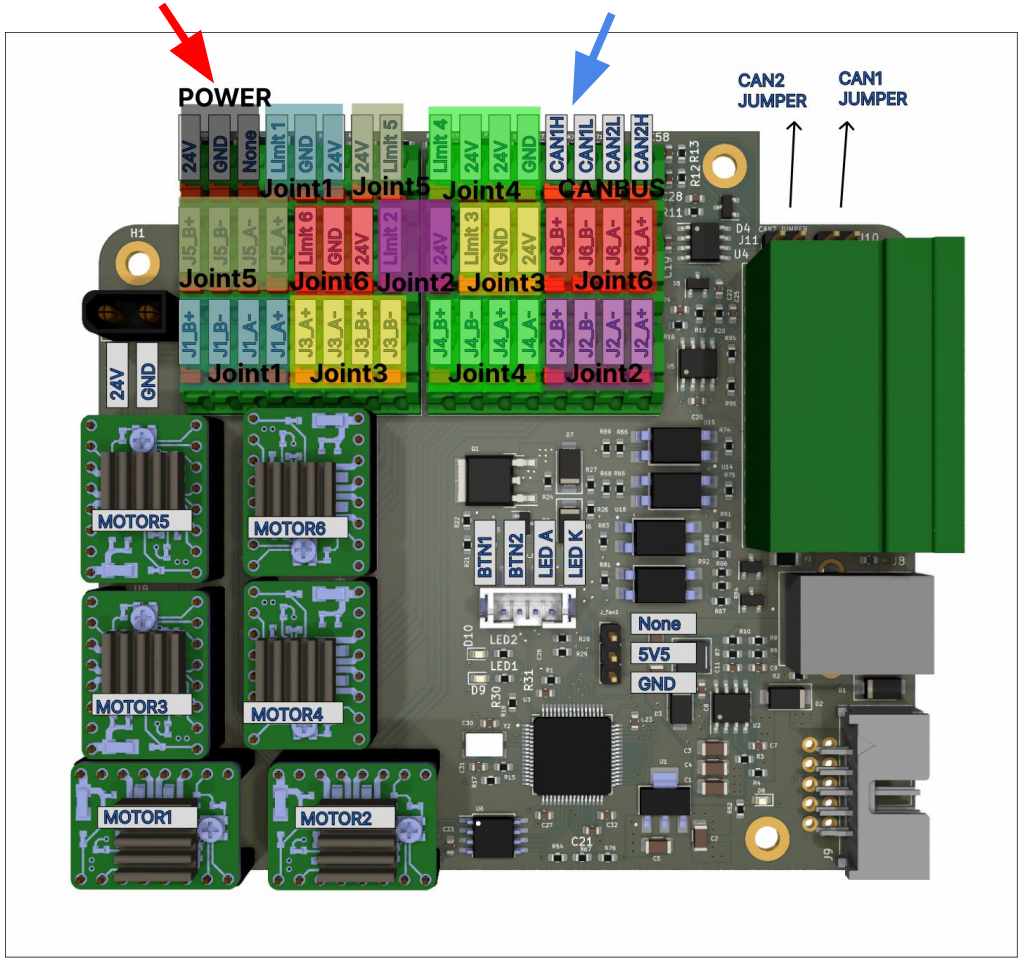
Wire the robot using the image on the left. For the stepper motors it is important to wire phases A and B correctly but if you switch the wires on one phase ex. phase A+ and A- it is not important since this will be taken care in the software. For limit switches orientation is also not important (one lead to 24V other to signal. For Inductive sensors.

- Black - signal
- Blue - negative
- Brown - positive

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按照左侧图片为机械臂接线。

- 步进电机必须正确连接 A、B 两相。若只交换其中一相的两根线（例如 A+ 和 A−），方向可在软件中处理。
- 限位开关不区分方向：一根线接 24 V，另一根线接信号端。
- 电感式传感器线色：
- 黑色（Black）：信号
- 蓝色（Blue）：负极
- 棕色（Brown）：正极

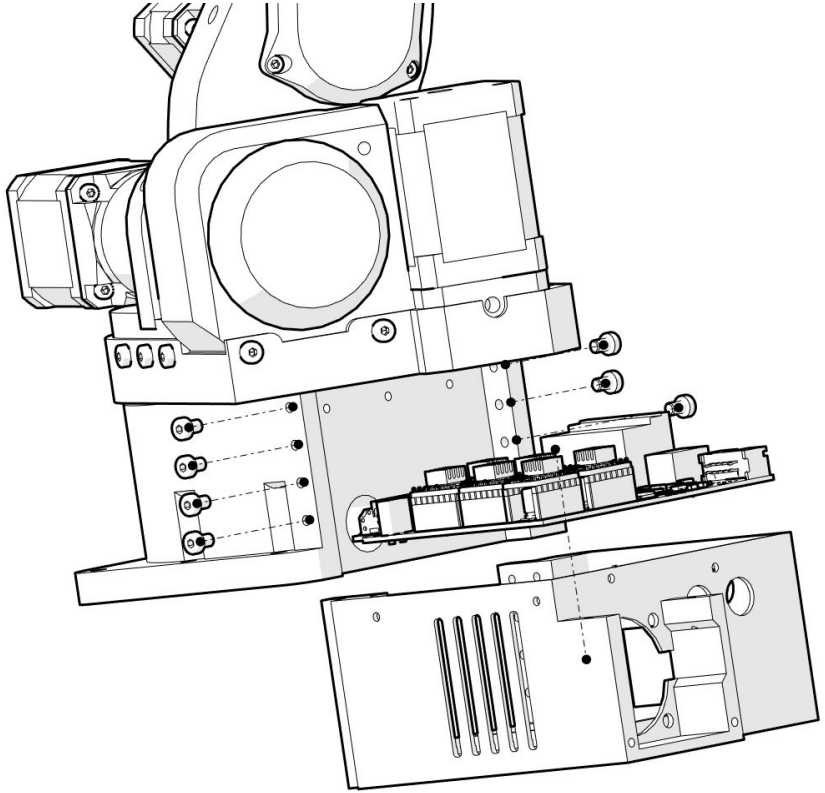


Connect CAN gripper connection wires to the 24V and GND of the POWER connection (marked with red arrow). Connect CANH and CANL gripper connection wires to the CAN1H and CAN1L of the CANBUS connector marked with blue arrow.

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将夹爪 CAN 连接线接到 POWER 电源接口的 24 V 和 GND（图中红色箭头）。将夹爪的 CANH、CANL 线分别接到 CANBUS 接口的 CAN1H、CAN1L（图中蓝色箭头）。

Base assembly



Electronics case
8 x M3 12 mm

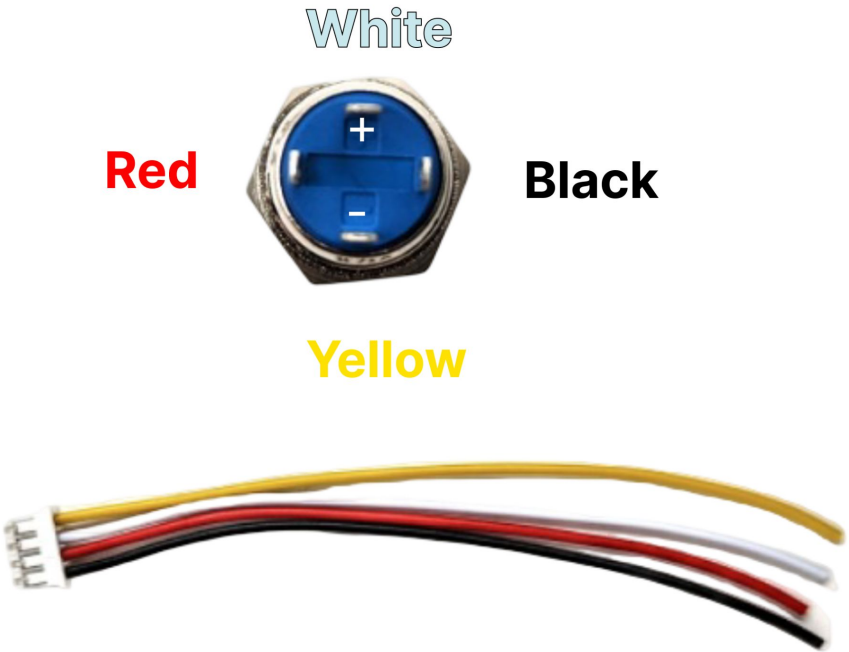
Slide the electronics case from the bottom of the board. Secure it to the robot with 8 x M3 screws. Remove the terminals from the green connector next to the USB.

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底座装配（Base assembly）

- 零件：电子设备外壳（Electronics case）。
- 使用 8 颗 M3 × 12 mm 螺钉。
- 从控制板下方滑入电子设备外壳，并用 8 颗 M3 螺钉固定到机械臂上。拆下 USB 旁绿色接线端子上的端子部件。

Prepare the wires



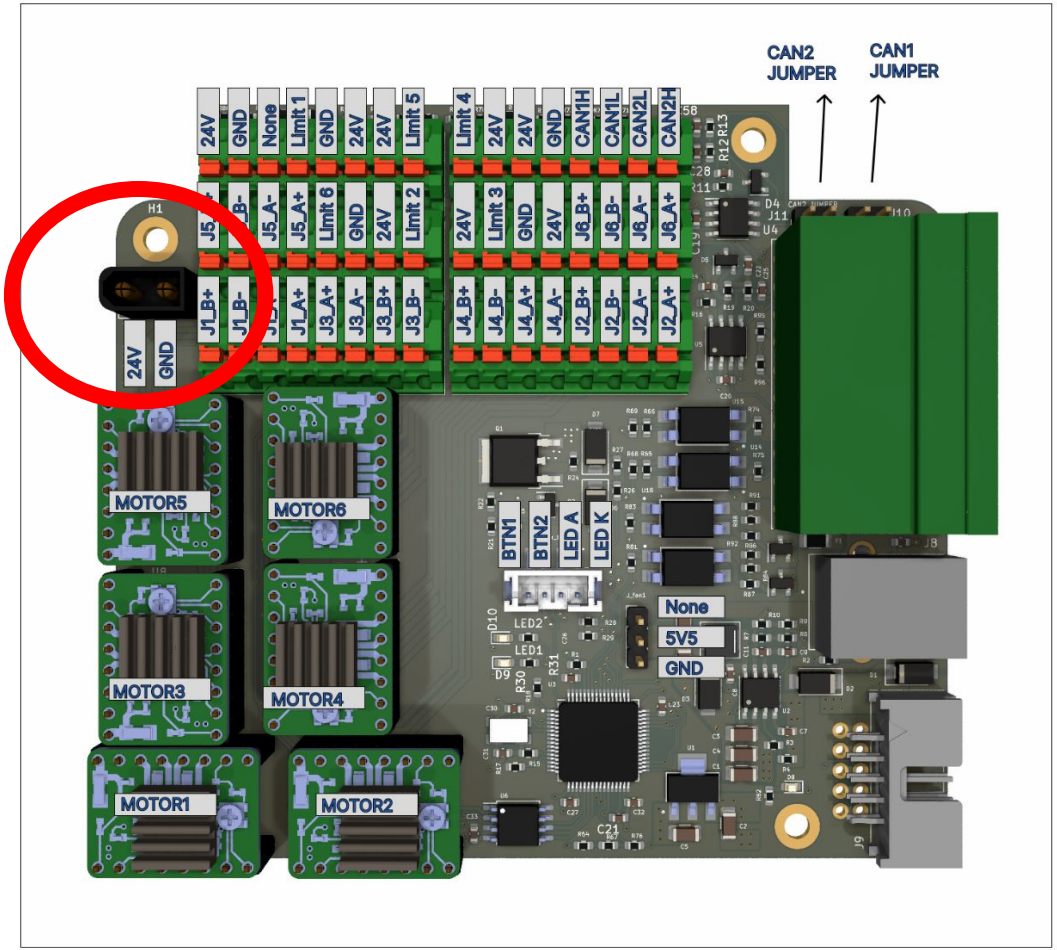
On/off button
JST PH 2.0mm 4pin wire

Attach JST wire to the button following the colour markings. Only follow it if you have exact same wire and button. Length of the JST cable needs to be 80 mm!

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准备线缆（Prepare the wires）

- 电源开关（On/off button）
- JST PH 2.0 mm、4 针导线
- 按导线颜色标记将 JST 导线接到按钮上。只有在导线和按钮型号完全相同时，才照颜色对应连接。
- JST 线长应为 80 mm。



GX16 2PIN
xt30 female connector

Attach GX16 2PIN male connector with at least 110 mm wire to the xt30 female connector! MAKE SURE YOU FOLLOW THE POLARITY AS SHOW ON THE IMAGE!

Source Robotics

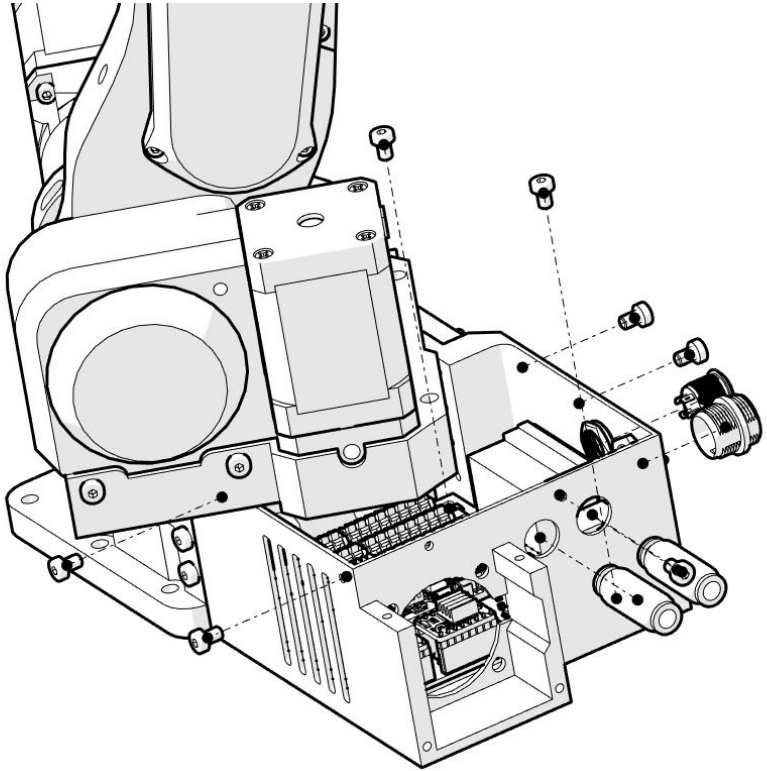
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准备线缆（Prepare the wires）

- GX16 2 针插头（GX16 2PIN）
- XT30 母头（XT30 female connector）
- 在至少 110 mm 长的导线上连接 GX16 2 针公头与 XT30 母头。

务必按图片所示核对极性！

Base assembly



2 x Pneumatic connector 1
On/Off button
Power connector
8 x M3 8 mm

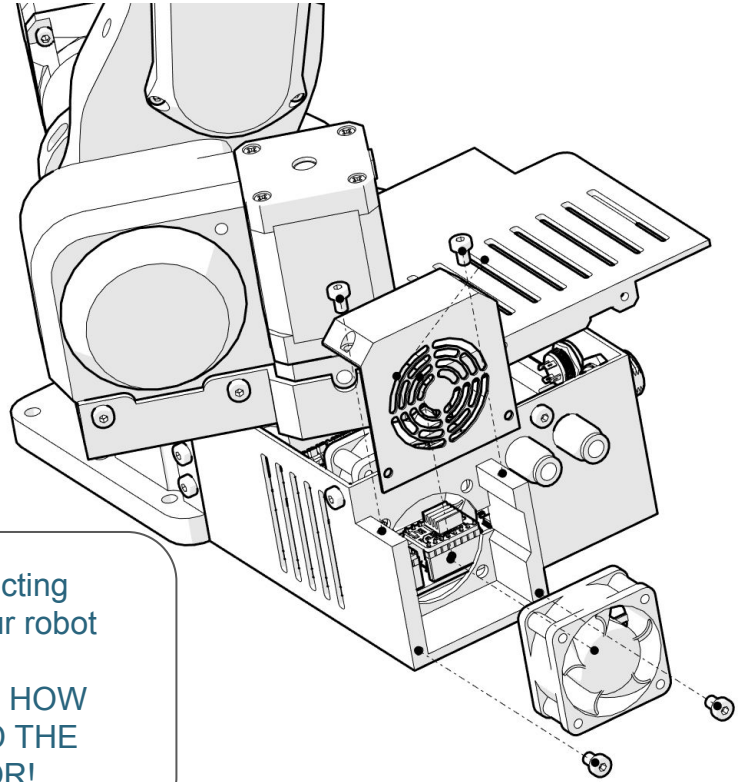
Pneumatic tubes after they exit the base of the robot need to be at least 150mm long! If needed shorten cooling fan wires. Also make note of the polarity of the fan!

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底座装配（Base assembly）

- 2 个气动接头（Pneumatic connector 1）、电源开关、电源连接器。
- 使用 8 颗 M3 × 8 mm 螺钉。
- 气动管从机械臂底座穿出后，长度至少应为 150 mm。
- 如有需要，缩短冷却风扇导线。注意风扇的正负极性。

Base assembly



Lid_electronics
Fan cover
Cooling fan
2 x M3 8 mm

Attach cooling fan using provided screws. Fans wires need to shortened if they are too long. Depending what fan you use; it may come with different set of screws.

When connecting power to your robot (24V) PAY ATTENTION HOW YOU WIRED THE CONNECTOR!

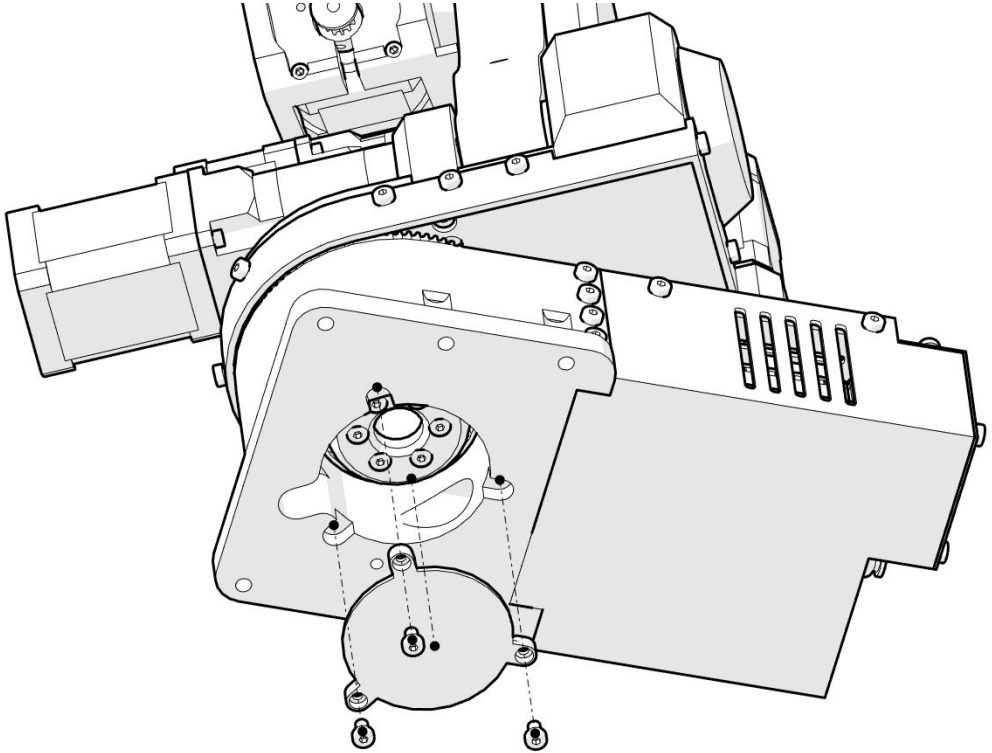
第 91 页中文译文

底座装配（Base assembly）

- 零件：Lid_electronics、风扇盖（Fan cover）、冷却风扇（Cooling fan）。
- 使用 2 颗 M3 × 8 mm 螺钉。
- 使用随附螺钉安装冷却风扇。若风扇线过长则缩短。不同风扇可能需要不同规格的螺钉。

将 24 V 电源接入机械臂时，务必确认连接器接线极性正确！

Base assembly



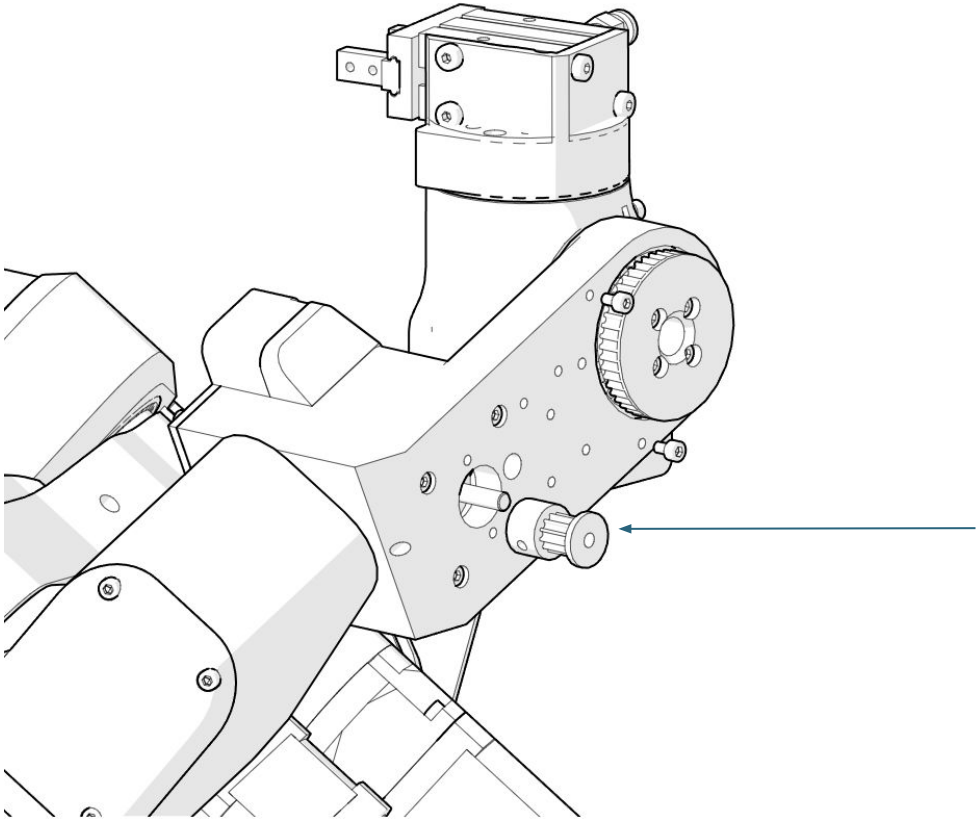
Bottom lid
3 x M3 8 mm; 2 mm head height

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底座装配（Base assembly）

- 零件：底盖（Bottom lid）。
- 使用 3 颗 M3 × 8 mm 螺钉；螺钉头高度为 2 mm。

Attaching J5 belt



J5 Pulley
J5 belt

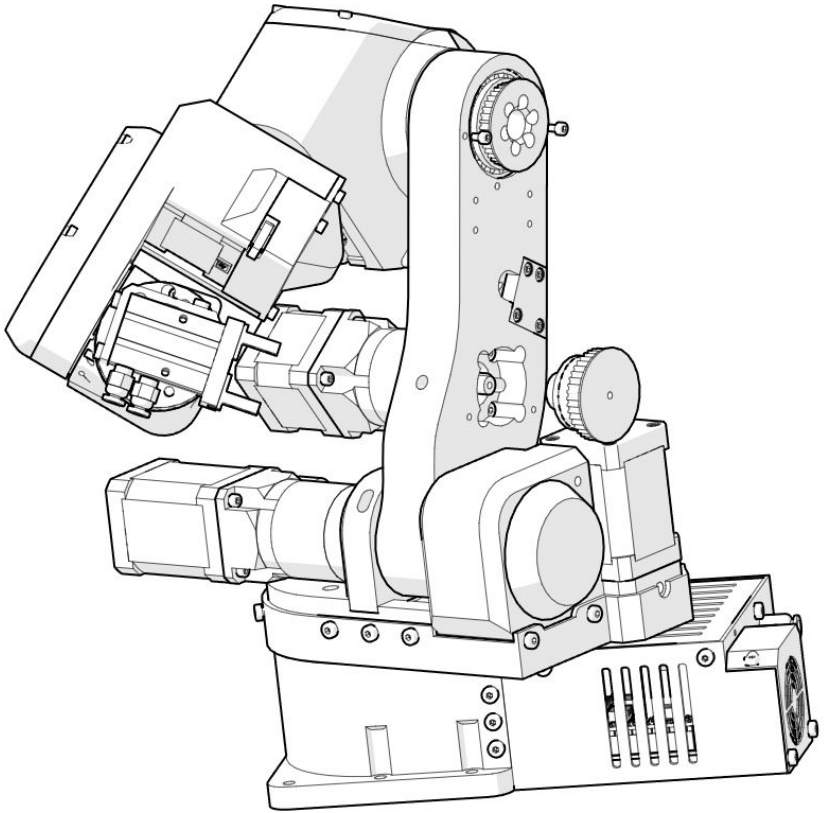
Attach the belt between J5 pulley and HTD3_48_J5_pulley

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安装 J5 皮带（Attaching J5 belt）

- 零件：J5 带轮（J5 Pulley）、J5 皮带（J5 belt）。
- 将皮带安装在 J5 带轮与 HTD3_48_J5_pulley 之间。

Attaching J3 belt



4 x 3x8 mm bearing
42_pulley
Shaft coupler
4 x M4 10mm

Attach shaft coupler to the pulley with 4 M4 10mm screws.

Gearbox has a hole (keyway) where key should be going. Since we are not using keys make sure that the screw of shaft coupler goes in the keyway! Use threadlocker here!

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安装 J3 皮带（Attaching J3 belt）

- 4 个 3 × 8 mm 轴承、42_pulley、法兰联轴器（shaft coupler）。
- 使用 4 颗 M4 × 10 mm 螺钉将法兰联轴器安装到带轮上。
- 减速器上有一个供键使用的键槽。本方案不使用键，因此确保法兰联轴器的紧定螺钉对准键槽！此处使用螺纹锁固剂（threadlocker）。

shaft couplers!





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法兰联轴器（shaft couplers）！

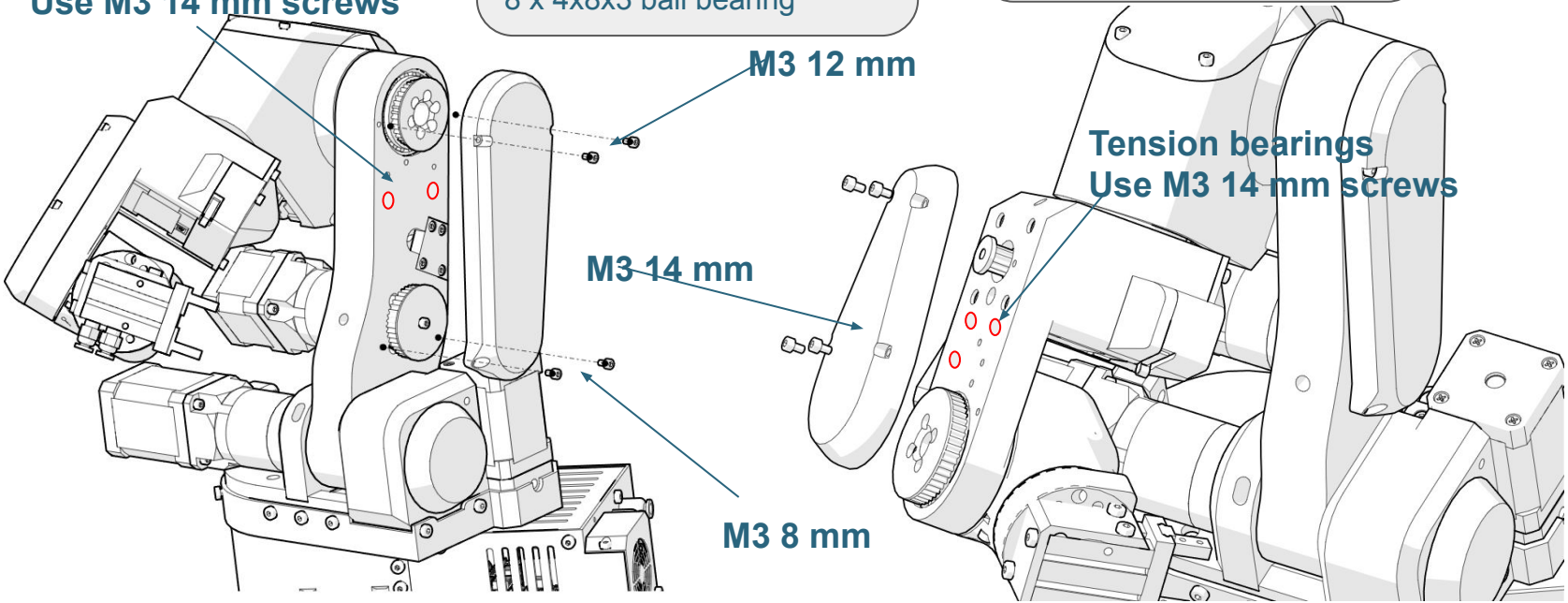
> 本页为图示/提醒页；文字层仅提及法兰联轴器。

Attaching joint covers

2 x M3 12 mm
2 x M3 8 mm
8 x M3 14 mm
8 x 4x8x3 ball bearing

For belt tension use 2 bearings on M3 14 mm screw. Do that on each side of the belt

Tension bearings
Use M3 14 mm screws

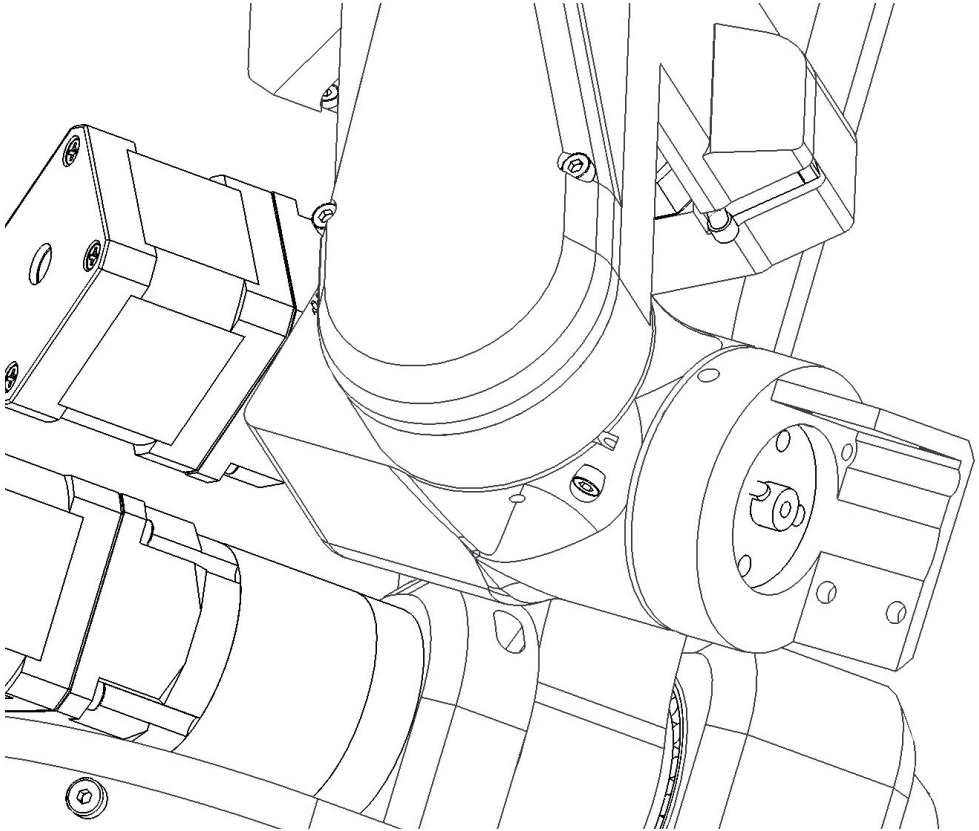


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安装关节盖（Attaching joint covers）

- 使用：2 颗 M3 × 12 mm、2 颗 M3 × 8 mm、8 颗 M3 × 14 mm 螺钉，以及 8 个 4 × 8 × 3 mm 球轴承。
- 每侧皮带各安装 2 个轴承作为张紧轴承。使用 M3 × 14 mm 螺钉安装。
- 图中还标注 M3 × 12 mm、M3 × 8 mm、M3 × 14 mm 及张紧轴承（tension bearings）。

Pneumatic gripper



Shaft coupler
4 x M4 10mm
Pneumatic gripper holder
M3x 14mm screw

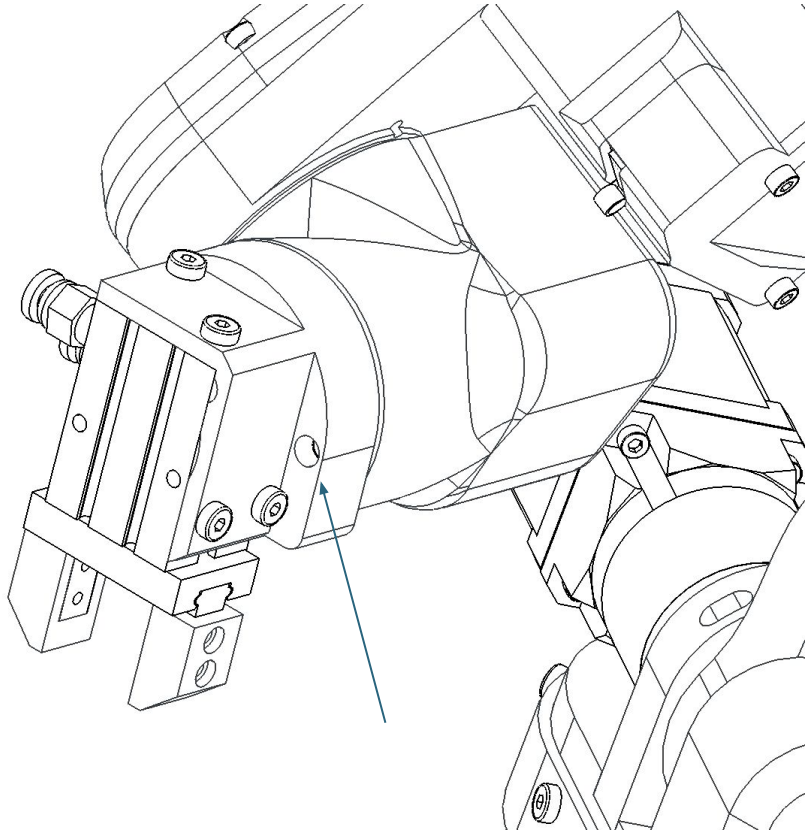
Attach shaft coupler to the pneumatic gripper holder. Make sure to align holes of the shaft coupler for set screws with the ones on the pneumatic gripper holder.

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气动夹爪（Pneumatic gripper）

- 法兰联轴器（shaft coupler）、4 颗 M4 × 10 mm 螺钉、气动夹爪支架（Pneumatic gripper holder）、M3 × 14 mm 螺钉。
- 将法兰联轴器安装到气动夹爪支架上。确保法兰联轴器紧定螺钉孔与支架上的孔对齐。

Pneumatic gripper



M3x 14mm screw
Pneumatic gripper
4 x M4 10mm

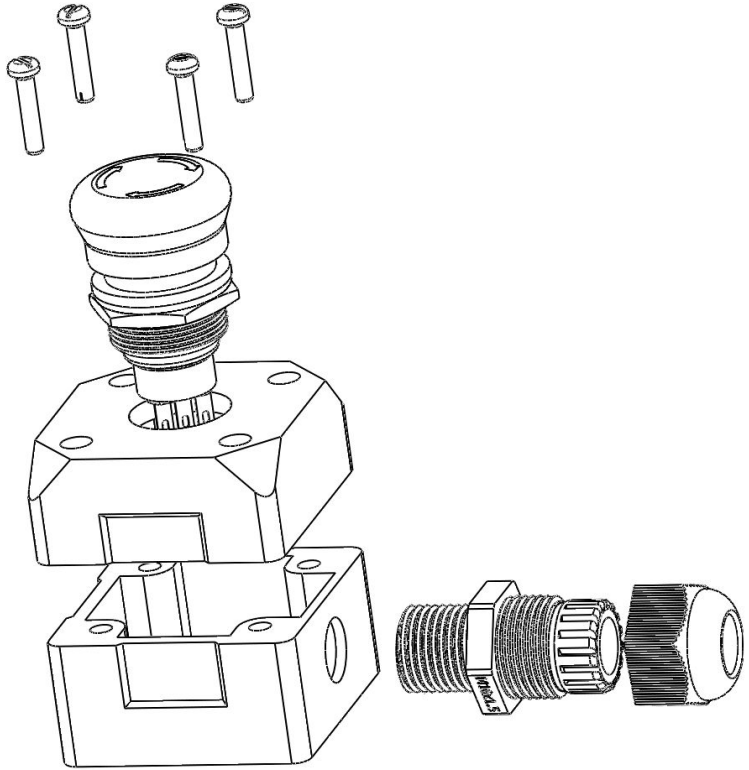
Attach gripper to the holder with 4 M4 screws. Add pneumatic connectors to the gripper if they are not installed. Install homing screw (M3 x 14mm) to the hole marked with arrow. Make sure that homing screw triggers the sensor. Screw in the set screws on the shaft coupler.

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气动夹爪（Pneumatic gripper）

- M3 × 14 mm 螺钉、气动夹爪、4 颗 M4 × 10 mm 螺钉。
- 用 4 颗 M4 螺钉将夹爪安装到支架上。若夹爪尚未安装气动接头，则加装接头。
- 将回零螺钉（M3 × 14 mm）安装到箭头所示的孔中，确认该螺钉能触发传感器。
- 拧紧法兰联轴器上的紧定螺钉。

Estop assembly



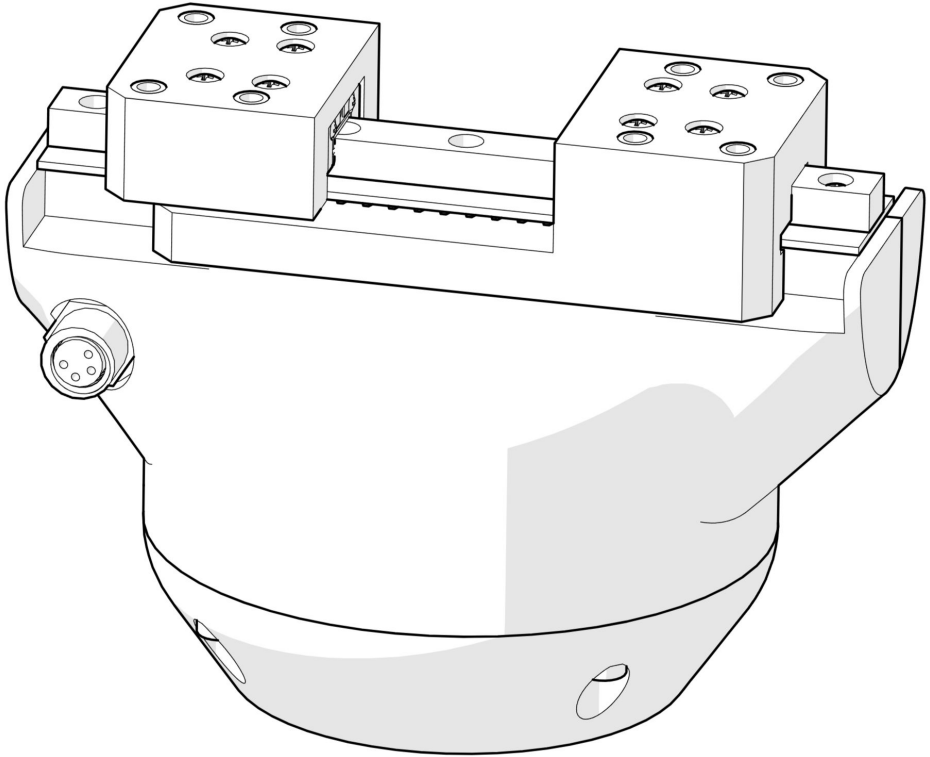
4x M3x 25mm screw
Estop
Estop bottom
Estop top
Cable gland PG7
4 x M3 brass inserts

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急停装配（Estop assembly）

- 急停按钮（Estop）、急停盒底部（Estop bottom）、急停盒顶部（Estop top）、PG7 电缆密封接头（Cable gland PG7）。
- 使用 4 颗 M3 × 25 mm 螺钉及 4 个 M3 黄铜热熔嵌件。

SSG48 gripper



If you want to build SSG electric gripper to use with your PAROL6 check this guide: https://github.com/PCrnjak/SSG-48-adaptive-electric-gripper

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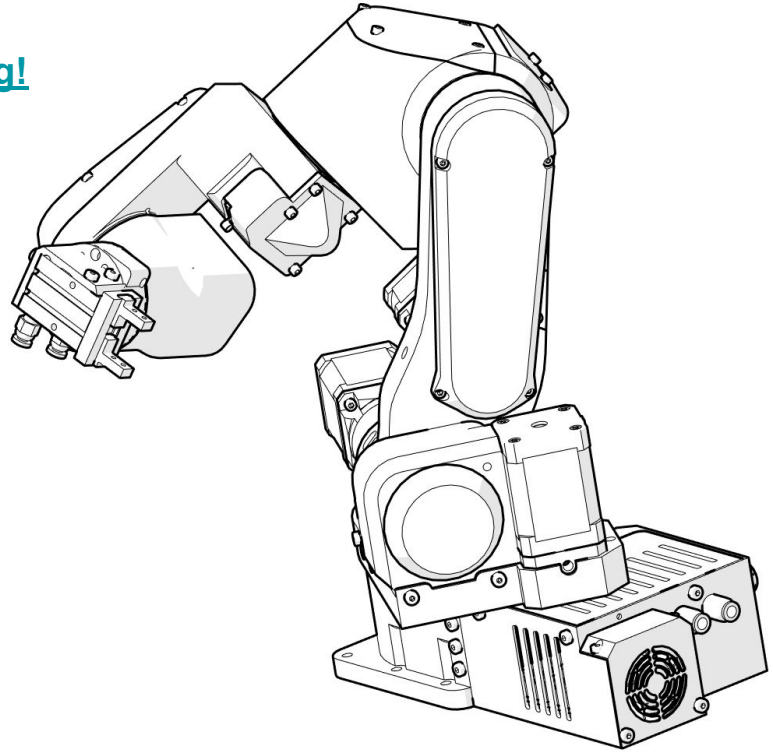
SSG48 夹爪（SSG48 gripper）

如果你想为 PAROL6 制作 SSG 电动夹爪，请参考指南：<https://github.com/PCrnjak/SSG-48-adaptive-electric-gripper>。

Great job!

Source Robotics

[Go to the docs to get your robot up and running!
https://source-robotics.github.io/PAROL-docs/](https://source-robotics.github.io/PAROL-docs/)



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恭喜完成！

请前往项目文档，了解如何启动并使用机械臂：<https://source-robotics.github.io/PAROL-docs/>。